



EUROPEAN **MiCROSOFT** **FABRiC** Community Conference

VIENNA 15-18 SEPTEMBER 2025

JOIN THE CONVERSATION

#FABCONEUROPE25



Mastering the Essentials of Data Factory



Eran Benayun

Principal Product Manager
Microsoft



Wilson Lee

Principal Product Manager
Microsoft

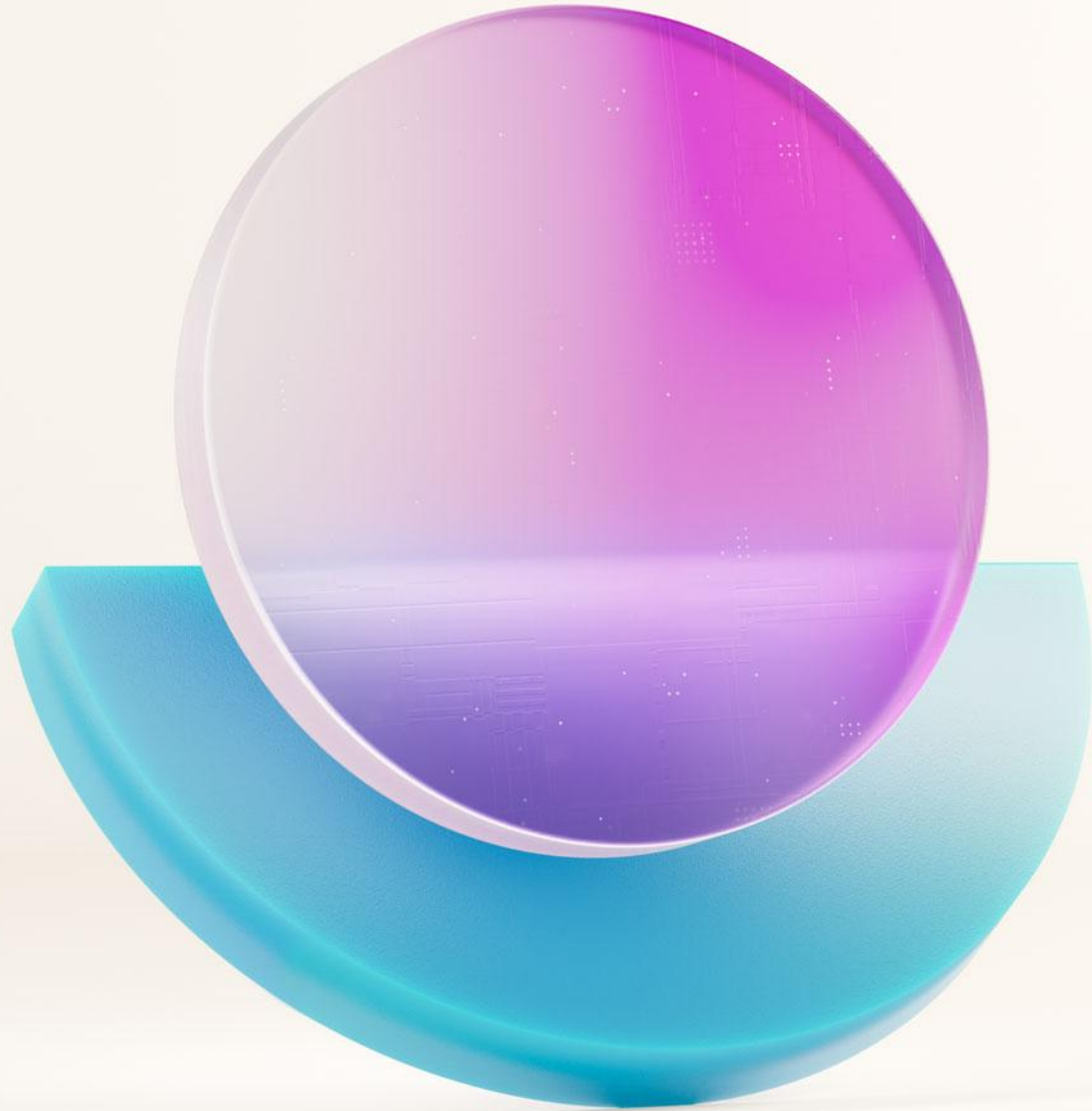


Mark Kromer

Principal PM Manager
Microsoft

Agenda (times are approximate and will be fluid with the event)

Morning		
9:00 AM – 9:15 AM		Introduction & setup
9:15 AM – 10:30 AM	Eran, Wilson, Mark	Microsoft Fabric and Data Factory overview
10:30 AM – 11:00 AM		 Morning Break
11:00 AM – 1:00 PM	Eran, Wilson, Mark	Orchestrating data movement
1:00 PM – 2:00 PM		 Break for lunch
Afternoon		
2:00 PM – 3:15 PM	Eran, Wilson, Mark	Transforming and copying data
3:15 PM – 3:45 PM		 Afternoon Break
3:45 PM – 5:00 PM	Eran, Wilson, Mark	Transforming and copying data



**Let's get some
prep work out of
the way**

Workshop content

Slide content

Designed to provide overviews and highlight key concepts



Demo videos

Pre-recorded insights into how these concepts and features function within Fabric Data Factory that are out of the scope for a hands-on lab in this workshop



Lab experience: Hands-on demos

Experience Microsoft Fabric **for free** with our demo environment and guided tutorials on Fabric Data Factory's key concepts and features

Everyone learns differently



Visual

Reads diagrams
Takes detailed notes
Studies notes on slides
Great sense of direction

+



Auditory

Participates vocally
Enjoys story-telling
Engages in open discussions
Understands changes in tone

+



Kinesthetic

Likes to explore
Learns by doing
Gets joy from building
Enjoys the “clicks” and “keys”

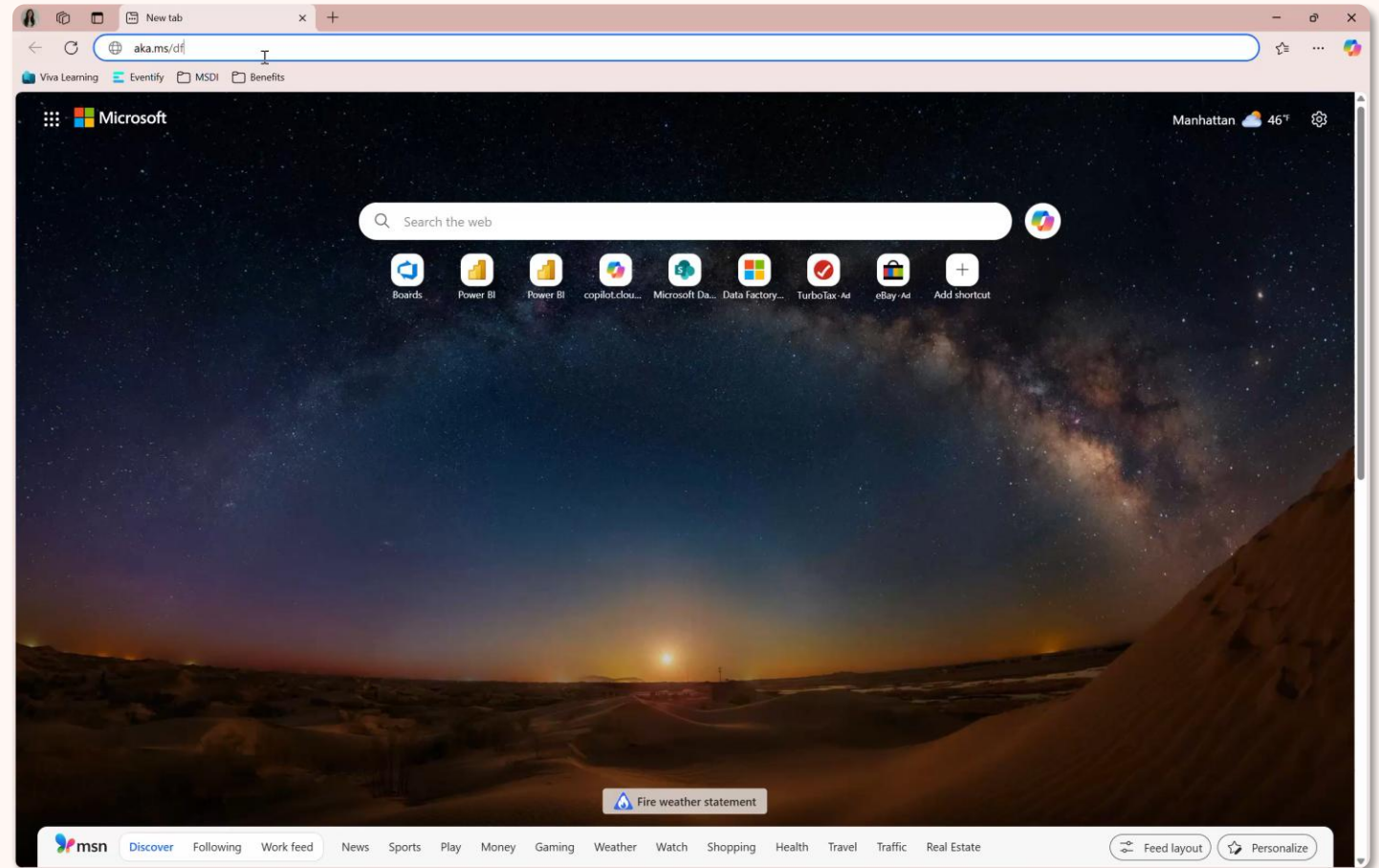
Today's workshop material + How to access lab instructions

All materials are available at:

<https://aka.ms/dfiad>

Start the lab with:

Getting started



Logging in with your free Fabric Demo User Profile

Open a new incognito browser:

Ctrl + Shift + N

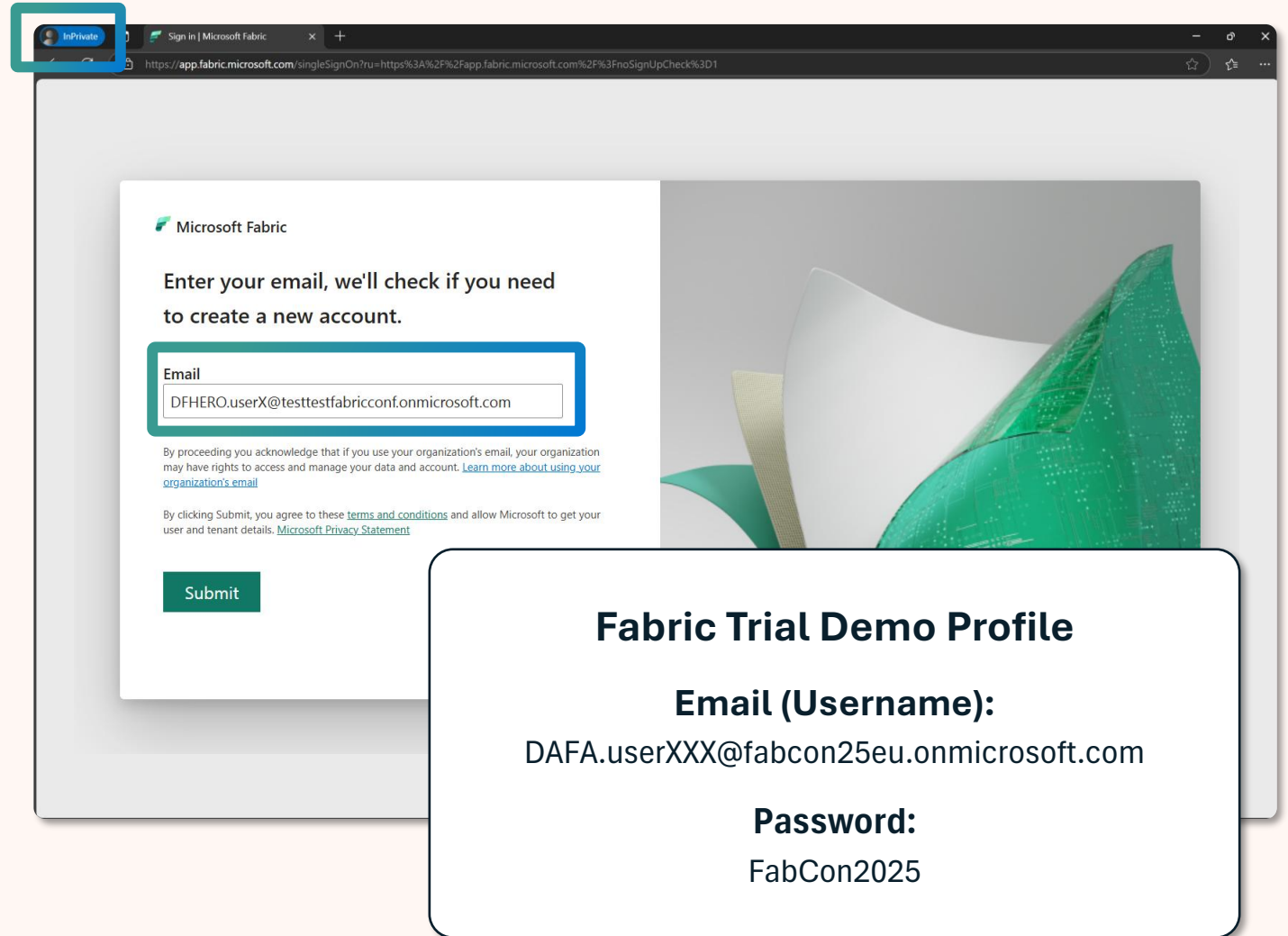
Login with demo user profile on:

<https://fabric.microsoft.com>

*Do not use multi-factor authentication

Username: on your paper handout

Password: FabCon2025



Fabric Trial Demo Profile

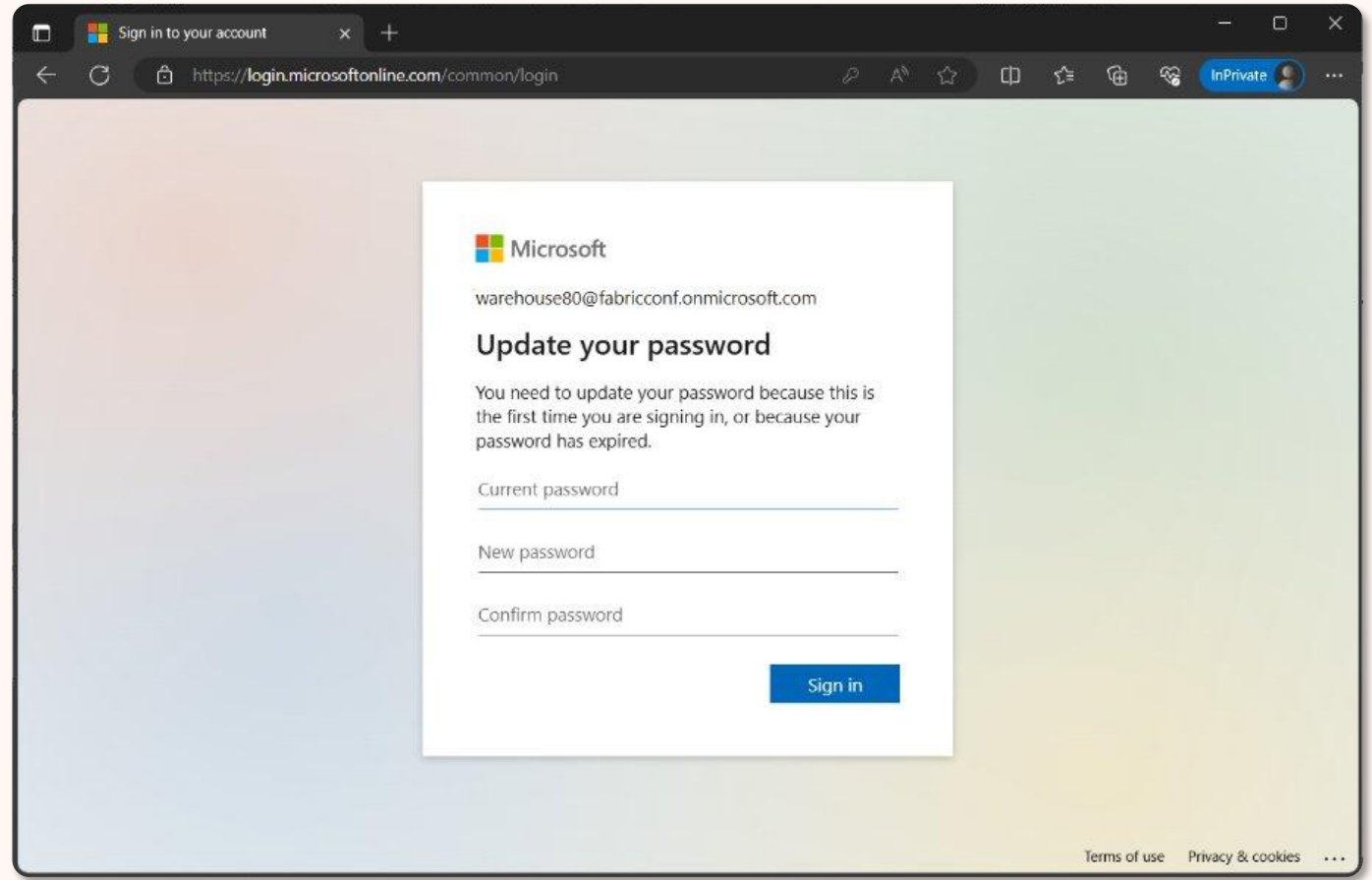
Email (Username):
DAFA.userXXX@fabcon25eu.onmicrosoft.com

Password:
FabCon2025

Logging in with your free Fabric Demo User Profile

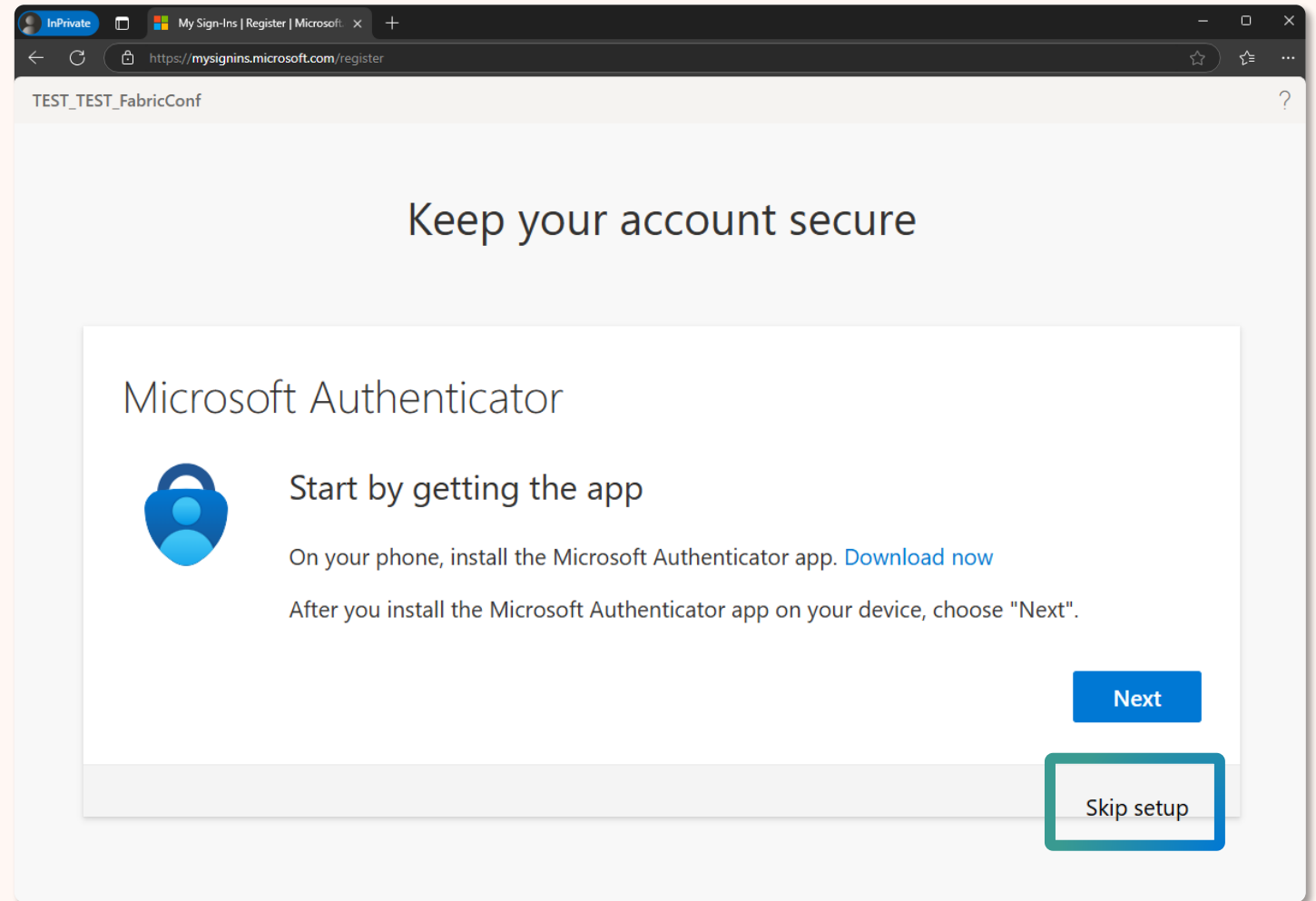
The first time you login,
you will be prompted to
change the password

*make sure you remember this
new password



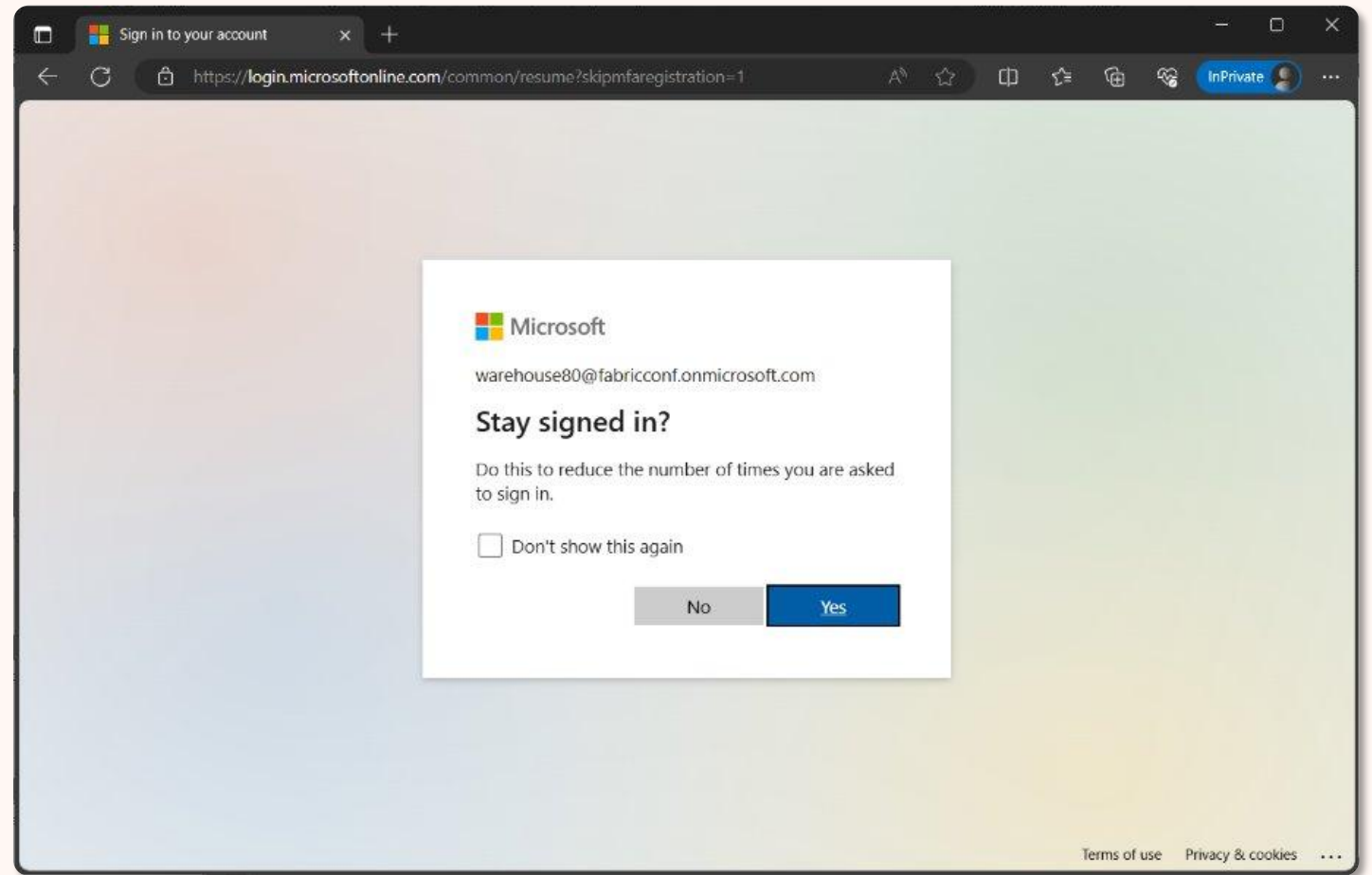
Logging in with your free Fabric Demo User Profile

Do *not* use multi-factor authentication by making sure to click **“Skip setup”**



Logging in with your free Fabric Demo User Profile

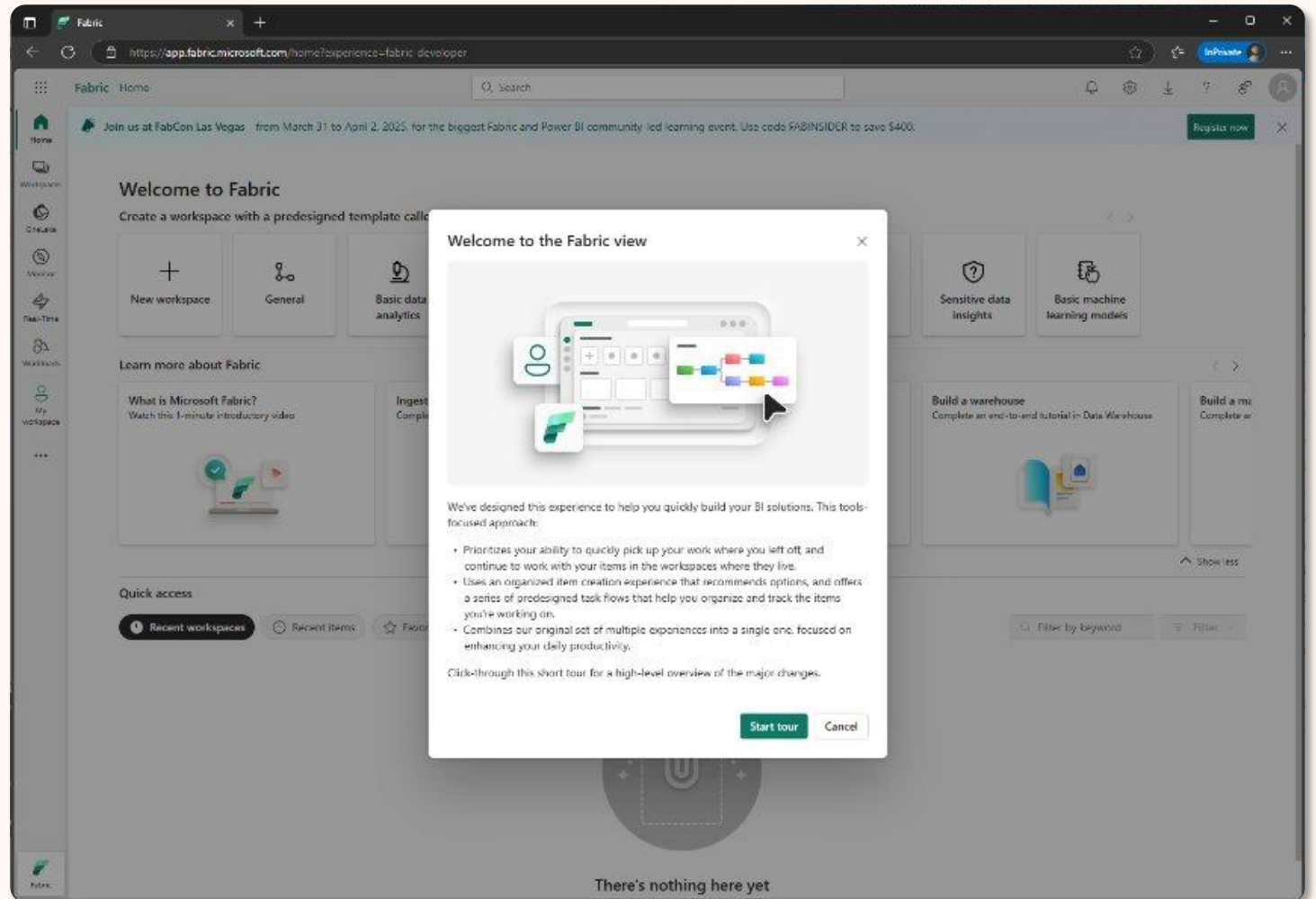
You can select **either option** you prefer for staying signed options



Logging in with your free Fabric Demo User Profile

You are now ready to work with Fabric!

*make sure to **hold onto your handout with your Fabric email (username)**



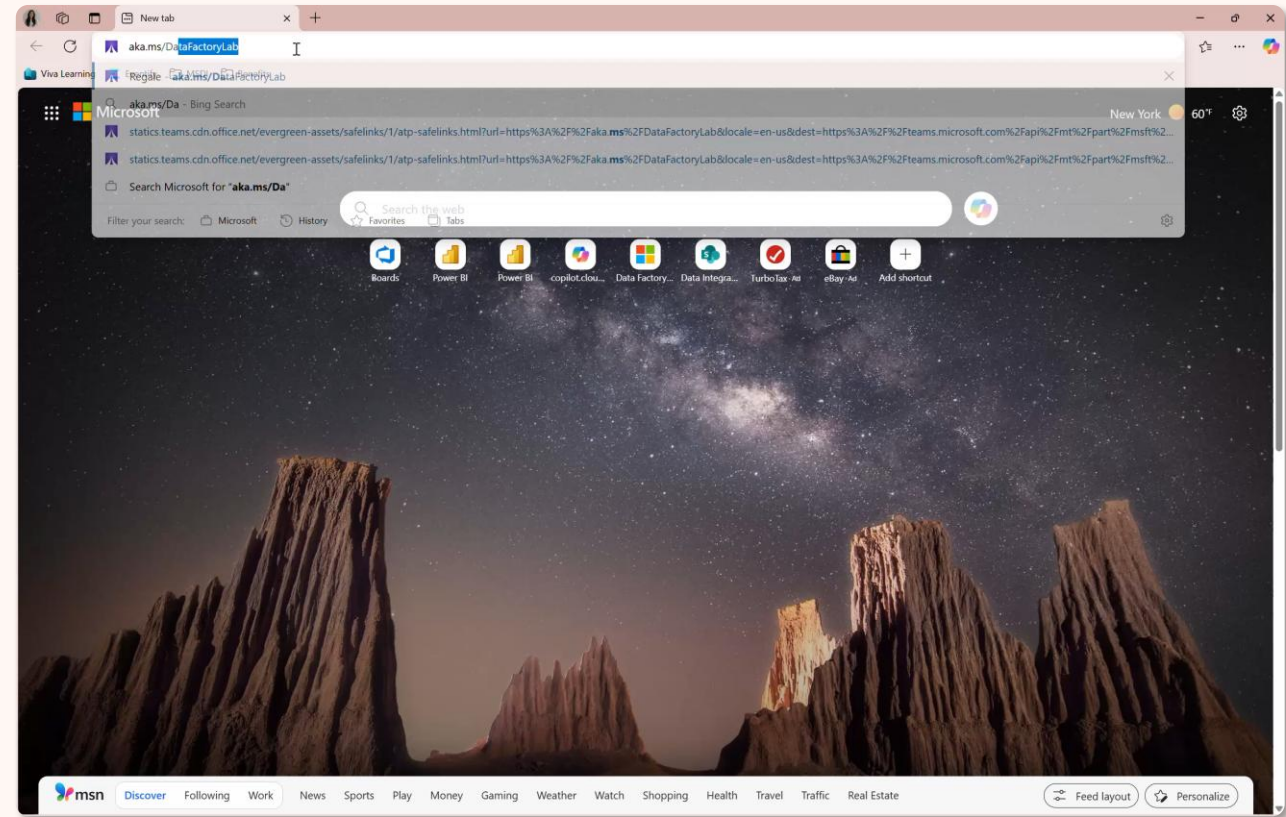
Troubleshooting with the lab

Something not working on Fabric?

Raise your hand, and we'll have one **our proctors help to troubleshoot** your issues!

Click-by-click lab experience:

<https://aka.ms/DataFactoryLab>



Make sure the following links are bookmarked

Lab instructions:

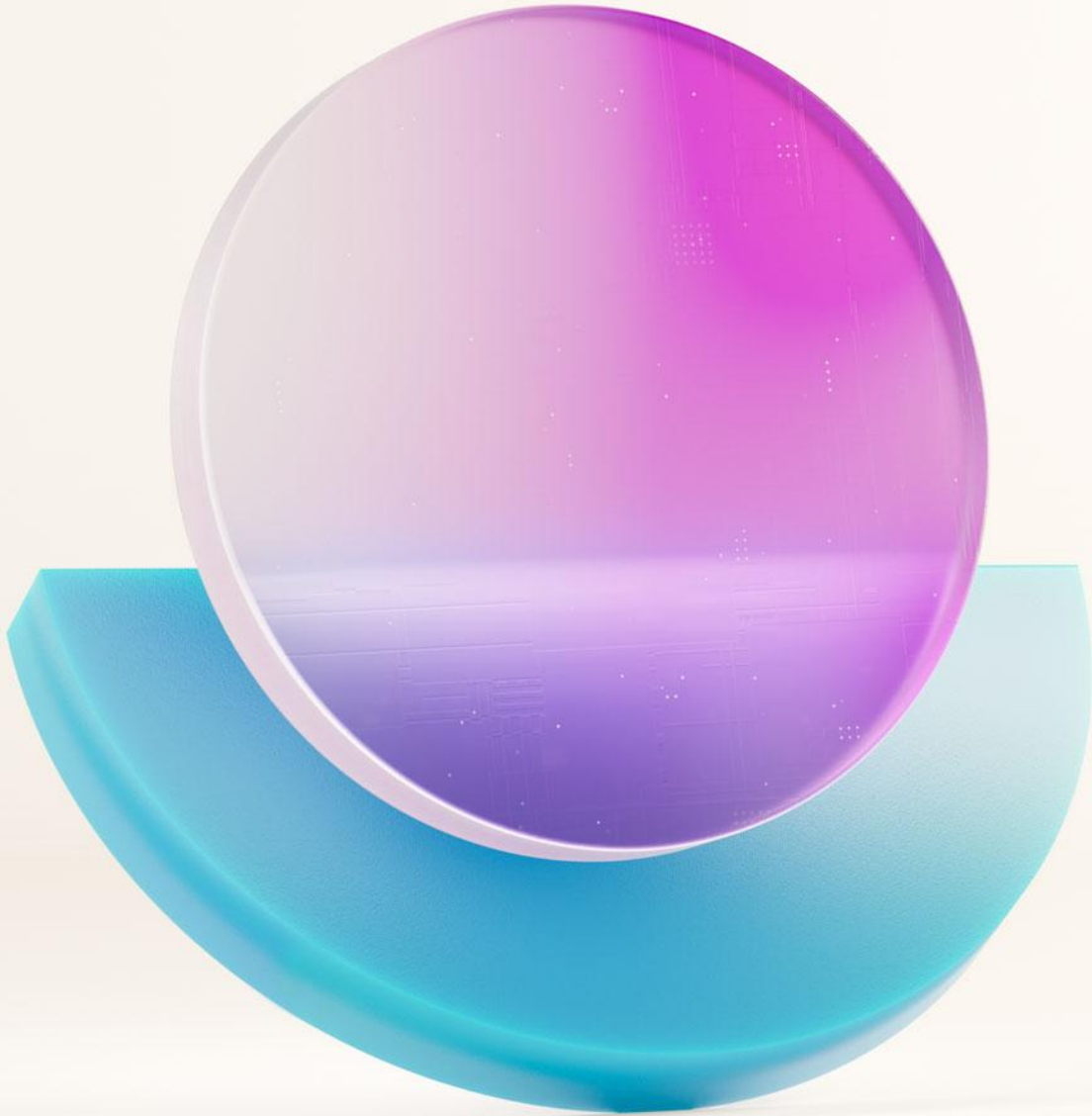
<https://aka.ms/dfiad>

Click-by-click lab:

<https://aka.ms/DataFactoryLab>

Microsoft Fabric
Community Conference

An Introduction to Microsoft Fabric



Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI

Fabric Platform



AI



OneLake



Governance

Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Pipelines



Connectors



Dataflows

Fabric Platform



AI



OneLake

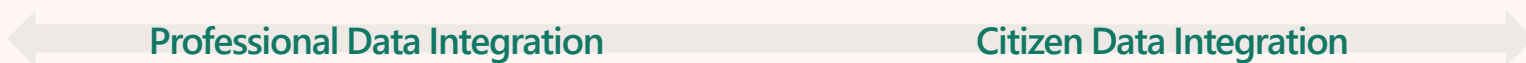


Governance



Microsoft Fabric

The unified data platform for AI transformation



Enterprise data integration in Azure

- Azure Data Factory
- Synapse Link
- Purview Integrated



Self-service data prep with Power Query

- Excel
- Power BI
- Office 365
- Dynamics
- Power Platform



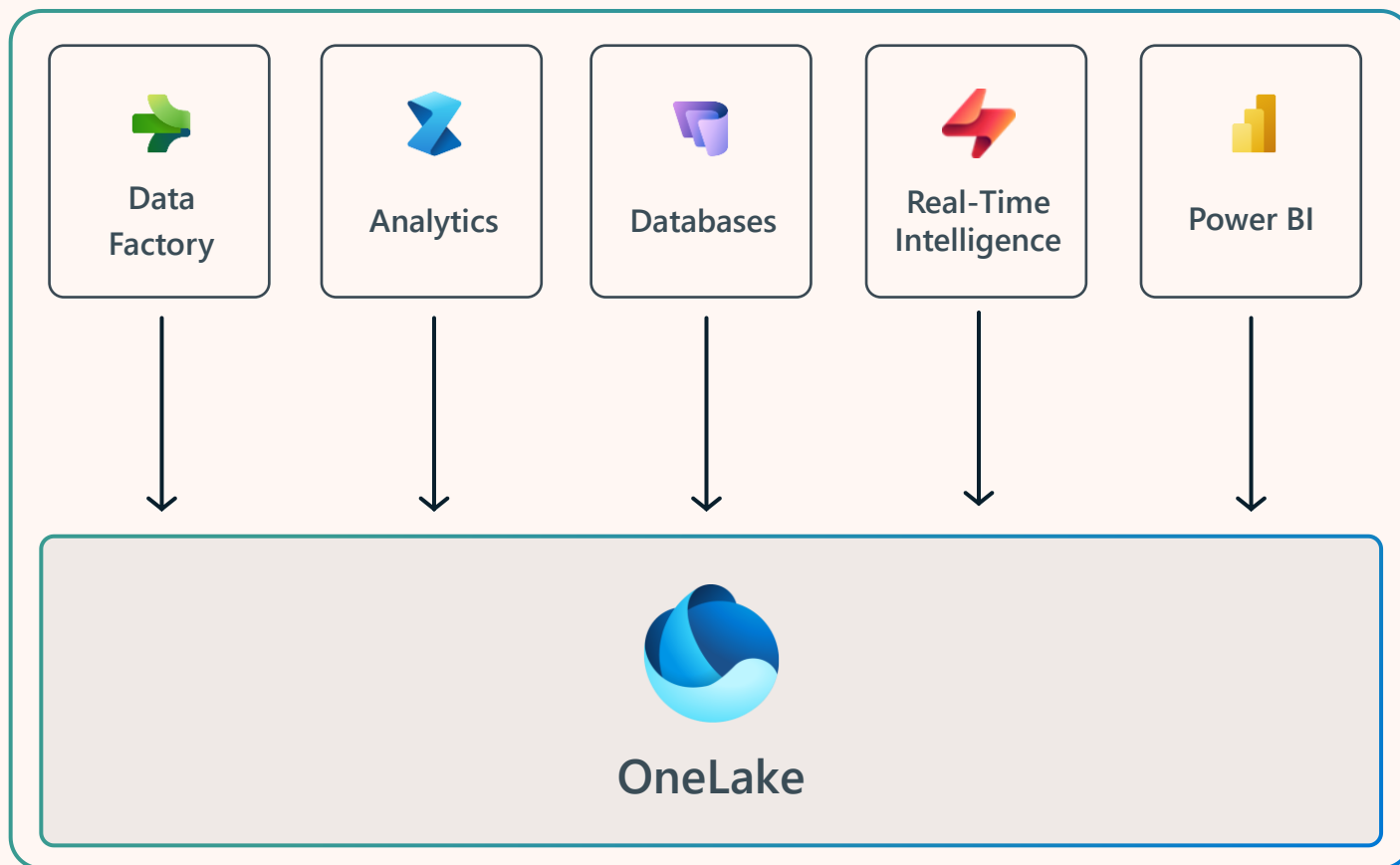
Data Factory in Microsoft Fabric

- Integrated SaaS platform
- No-code & code tools
- Copilot & AI-powered
- Open Source



OneLake for All Data

The OneDrive for Data



A **single SaaS lake** for the whole organization

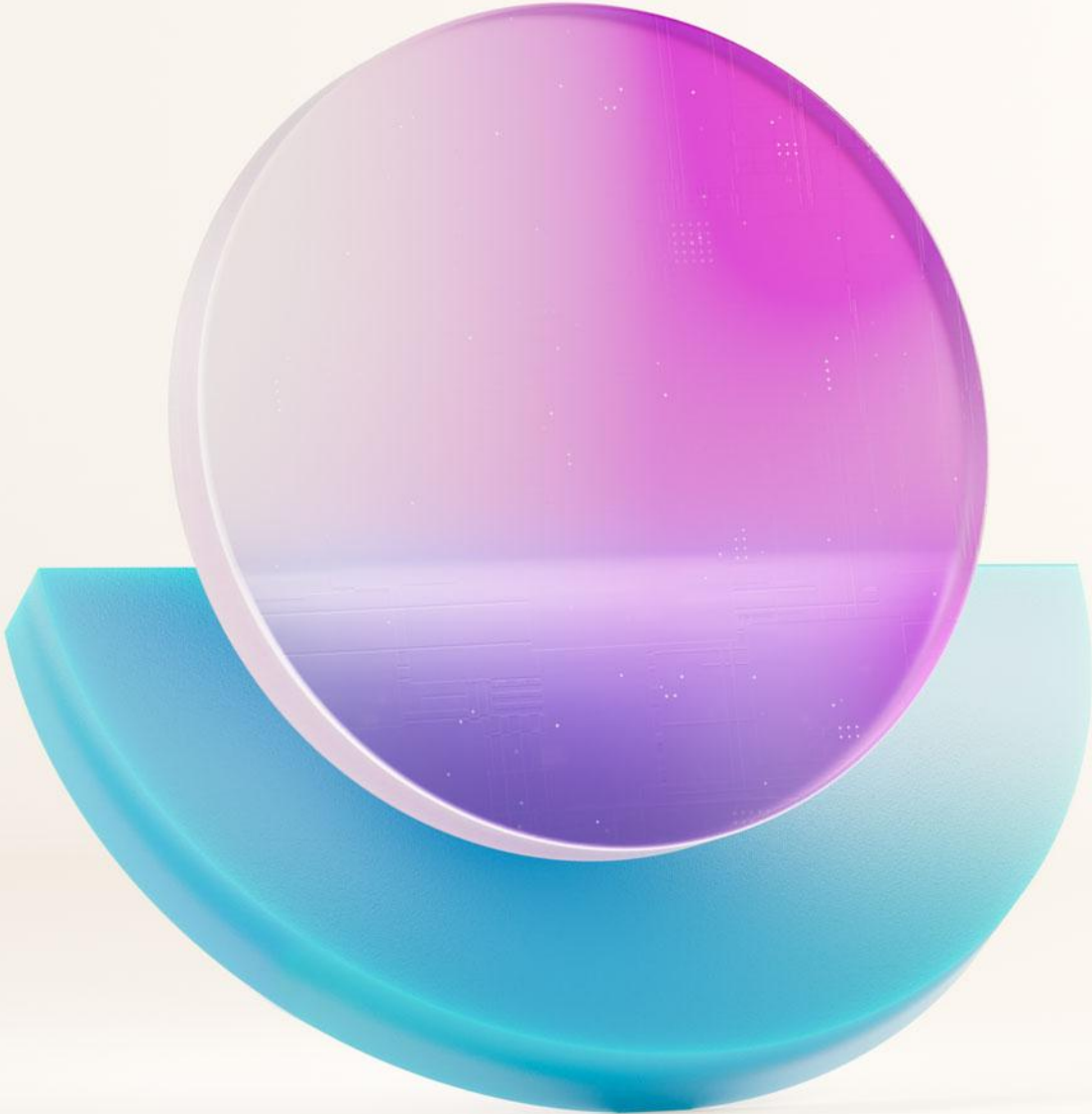
Provisioned **automatically** with the tenant

All workloads automatically store their data in the OneLake workspace folders

All the data is organized in an intuitive **hierarchical namespace**

The data in OneLake is automatically indexed for **discovery, Sensitivity labels, lineage, PII scans, sharing, governance and compliance**

Microsoft Fabric
Community Conference



An Introduction to Fabric Data Factory



Data Factory in Microsoft Fabric



Best-in-class
Connectivity

Orchestration

Transformation

Data Movement



Deployment &
Observability



OneLake

AI-powered Intelligence



Data Factory in Microsoft Fabric

Items in Microsoft Fabric



Data pipeline

Ingest data at scale and schedule data workflows.



Copy Job

Copy data easily with settings for full, incremental, or event-based.



Dataflow *Gen2*

Prep, clean, and transform data.



Apache Airflow Job

Simplify creation and management of Apache Airflow environments.



Mirrored Database

Easily replicate data from an existing source into OneLake.



Azure Data Factory

Mount ADF into Fabric to monitor all pipelines in one platform.



Data pipelines

Ingest and orchestrate activities at scale

No code Data pipelines

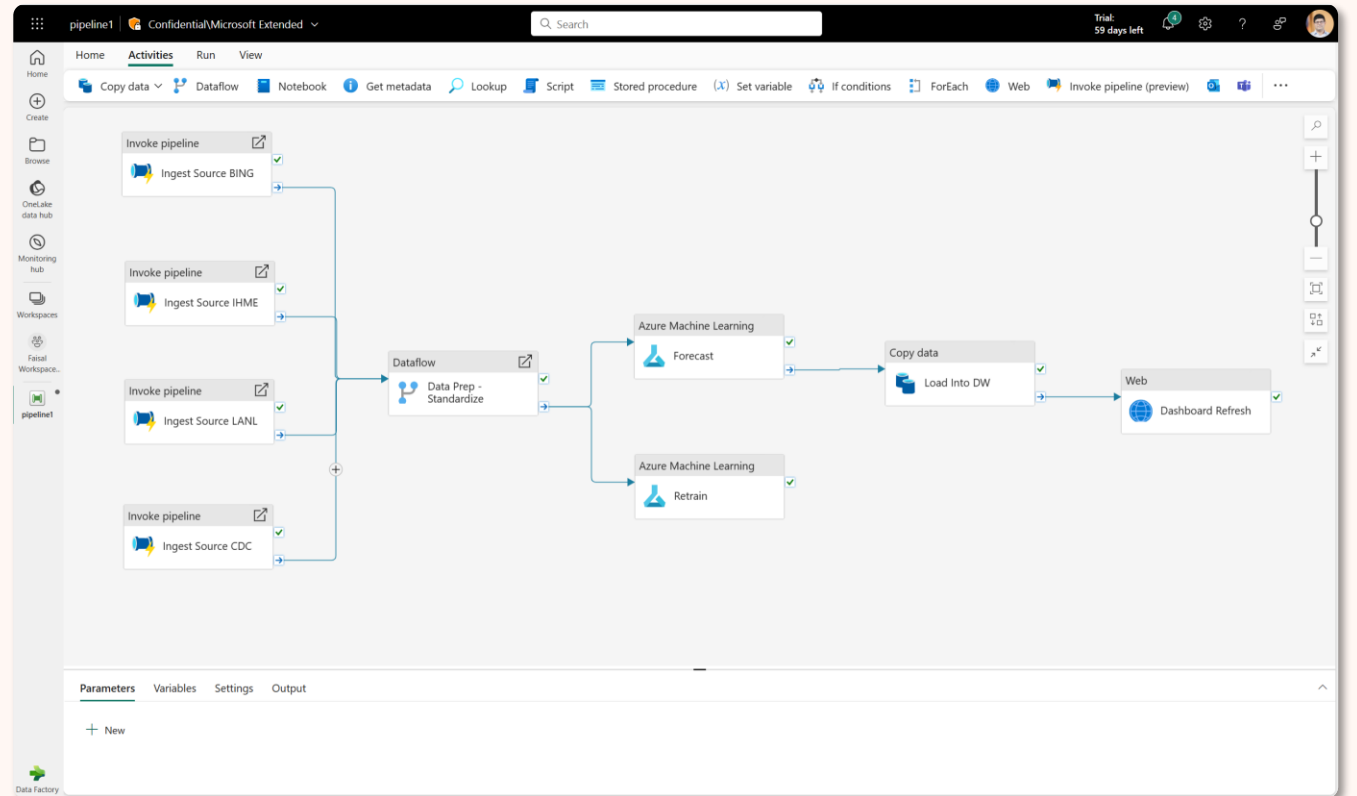
Intuitive, **cloud-based** user experience

Orchestrate data movement, transformation, cleansing, control flow

Triggered and **scheduled** execution models

User Data Functions for specialized functions

AI Powered with **Copilots**

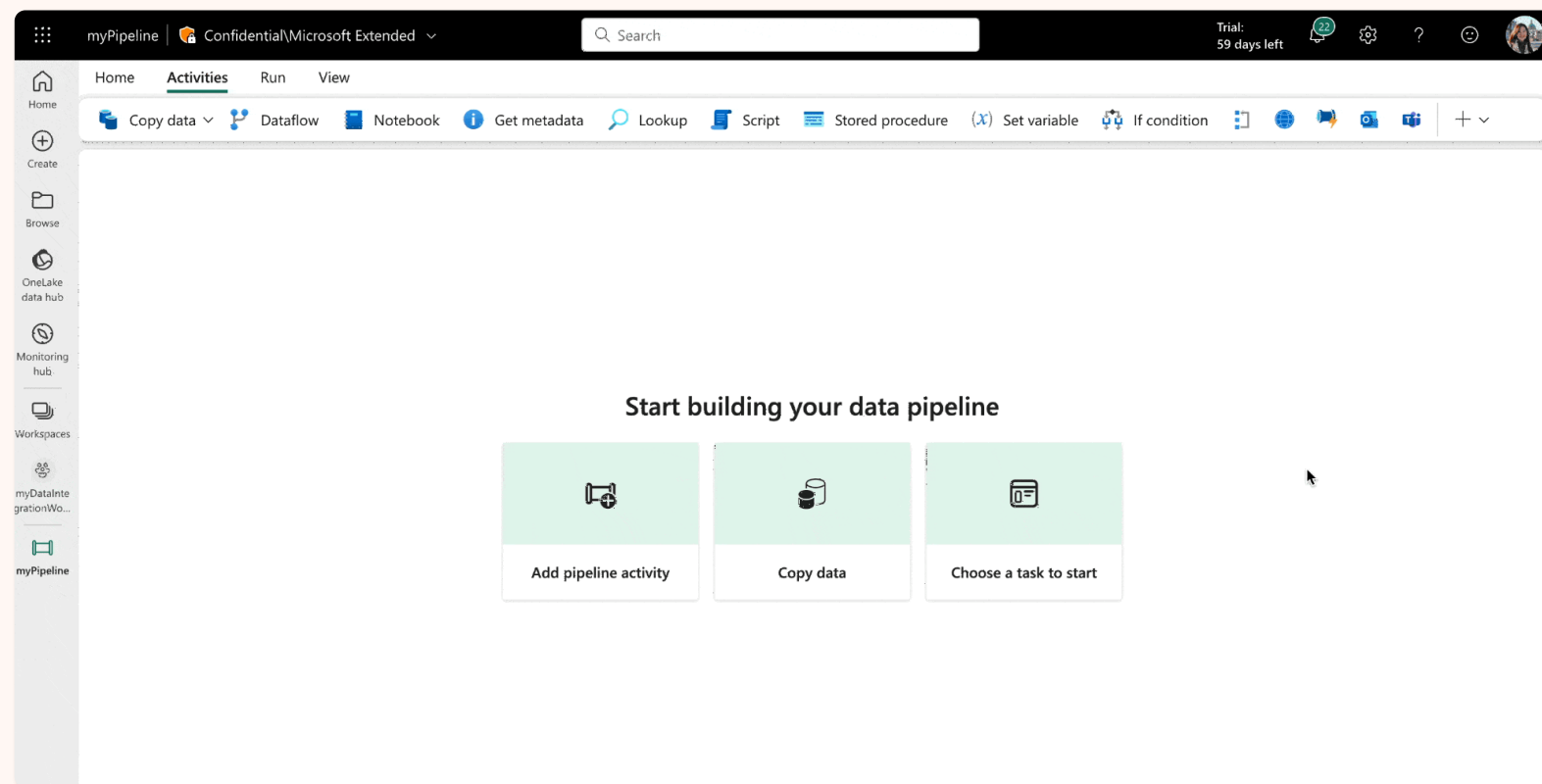




Data pipelines

Ingest and orchestrate activities at scale

Familiar authoring canvas experience



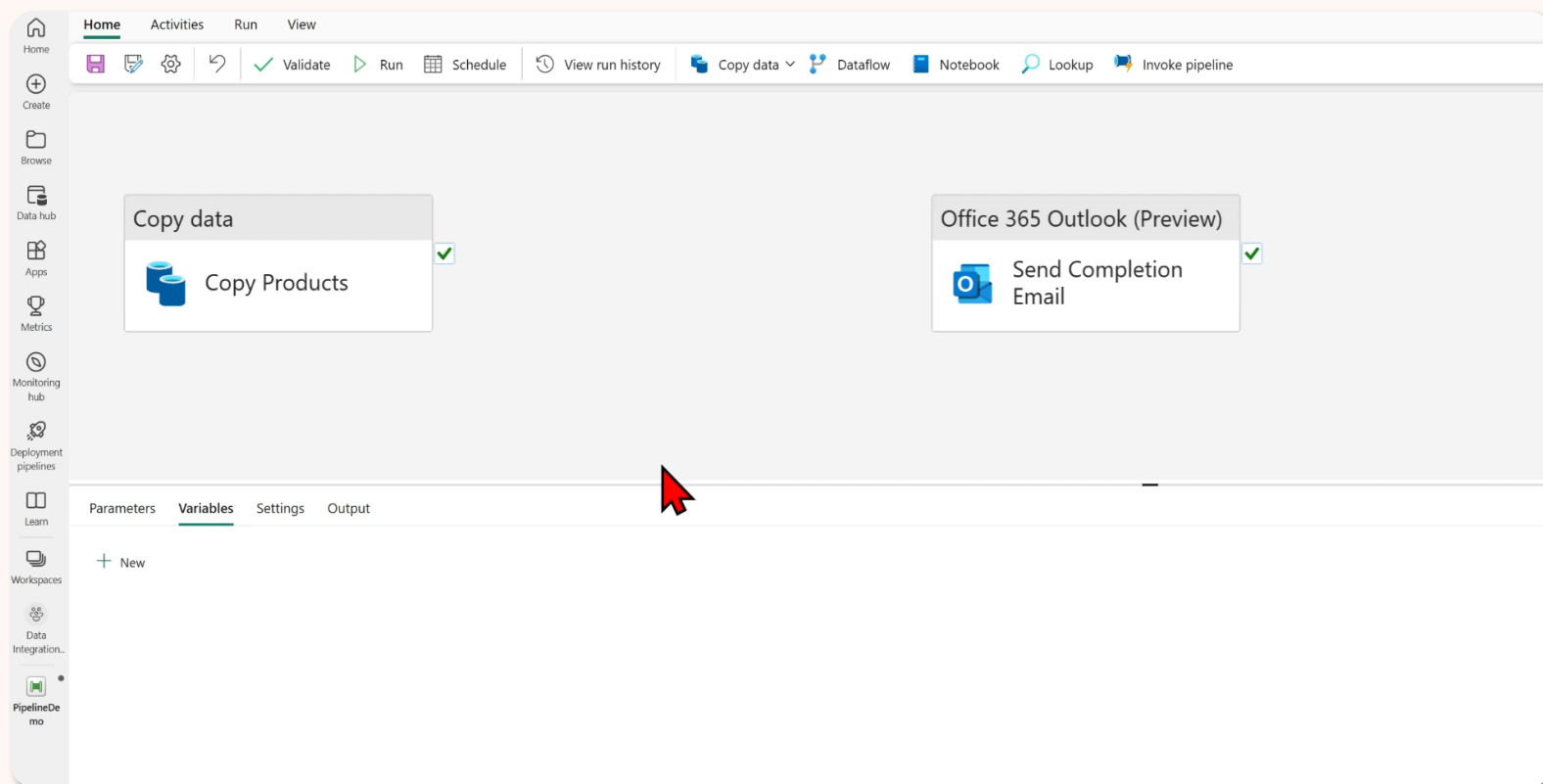
OneLake



Data pipelines

Ingest and orchestrate activities at scale

Empower every person to integrate data



OneLake



Data pipelines

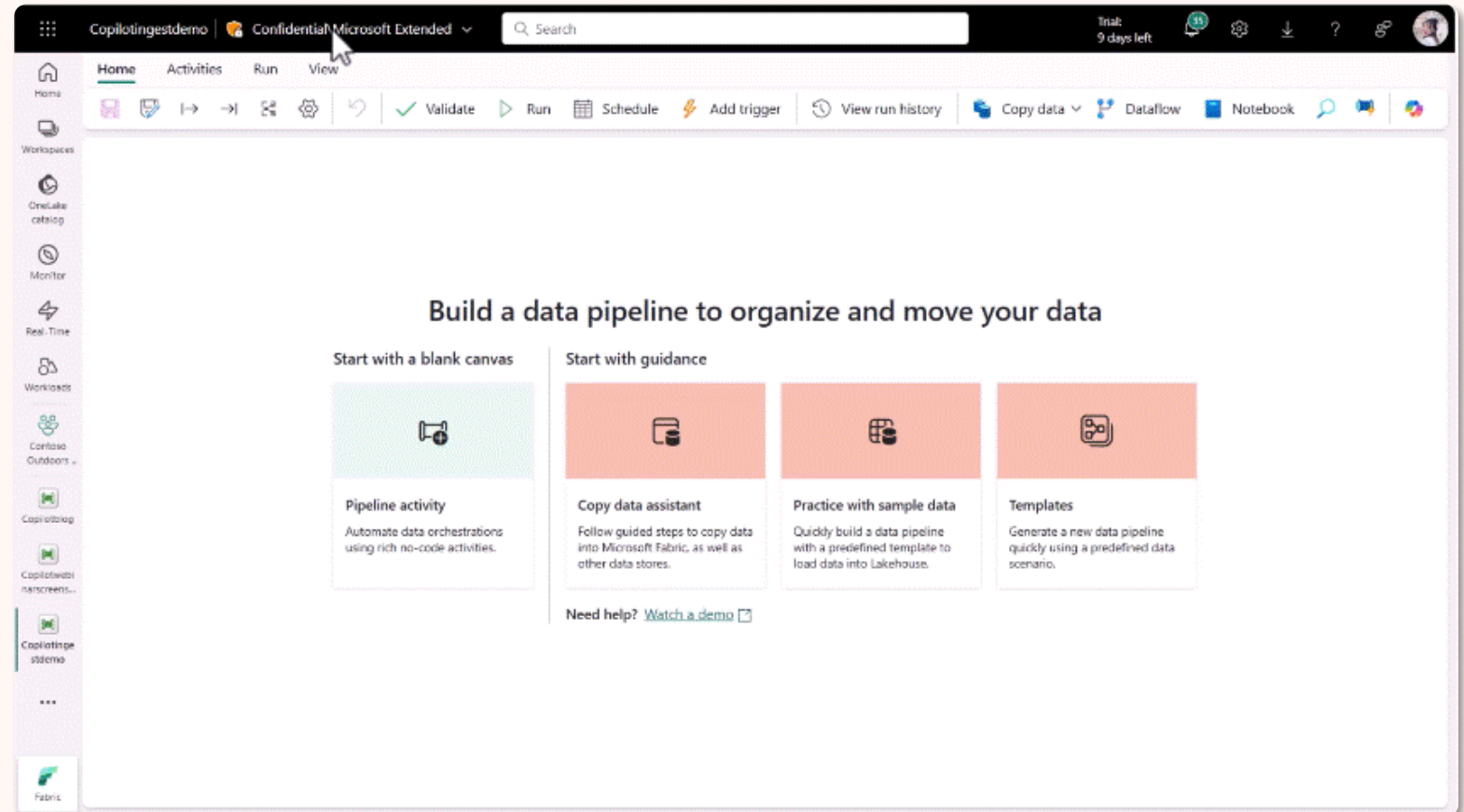
Ingest and orchestrate activities at scale

AI-powered Intelligence

Pipeline generation

Getting from **text to action**

Enabling you to be productive during development of data pipelines

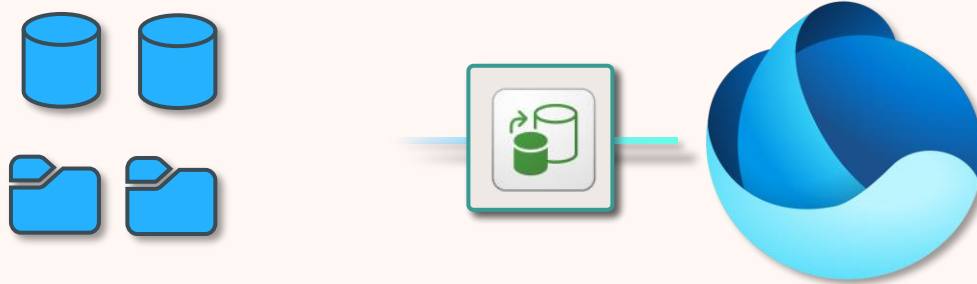




Copy job

Integrate data from built-in connectors

Generally Available



Copy Job item

Streamline copying data into and across OneLake

Setup automatic **incremental refresh**

Built-in support for **copying data between workspaces**



Copy job

Integrate data from built-in connectors

Generally Available

Supports both batch and near real-time incremental copy* (CDC)

(*Incremental copy is still in Public Preview)

New

- Dataflow Gen2 (Preview)
- Data pipeline
- Copy job
- Apache Airflow project
- Data Factory mount

Recommended

- Not sure where to start? Tell Copilot what you want to do and it will do it for you. [Launch Dataflow Copilot](#)
- Not sure where to start? Tell Copilot what you want to do and it will do it for you. [Launch Pipelines Copilot](#)
- Introduction to data integration: Essential concepts to know. [Watch video](#)
- Getting started with dataflow: Learn the basics of dataflow. [Watch the tutorial video](#)
- Getting started with data pipelines: Learn the basics of data pipelines. [Watch the tutorial video](#)

Quick access

Recent Favorites My apps

Name	Type	Opened	Owner	Endorsement	Sensitivity	Workspace
Cool data mart	Datamart	7m ago	Tim Deboar	—	General	Contoso workspace
Data flow for triggers	Data flow	7m ago	Tim Deboar	—	—	Contoso workspace
User data	Datamart	7m ago	Tim Deboar	Certified	—	Contoso workspace
Copy data pipeline	Pipeline	7m ago	Tim Deboar	—	—	Contoso workspace



OneLake



Dataflows Gen2

Ingest and transform data at scale

Dataflows Gen2: Power Query with enterprise scale

Based on **Power Query**, used by millions

File, relational, multi-dimensional, SaaS data experiences

Scalable transformations via Fabric compute

AI Powered with **Copilots**

The screenshot displays the Power Query 'Edit queries' window. The top ribbon includes tabs for Home, Transform, Add column, and View. The main area shows a data flow diagram with queries like 'Orders', 'Order_Details', 'Customers Data', 'Merge', and 'Top Customers'. Below the diagram, a table of data is shown with columns for CustomerID, CompanyName, ContactName, ContactTitle, and Total Sales. The table includes 10 rows of data, such as 'QUICK-Stop', 'SAVEA', 'ERINSH', etc. The bottom status bar indicates 'Completed (11.30 s) Columns: 5 Rows: 10' and 'Column profiling based on top 1,000 rows'.

	CustomerID	CompanyName	ContactName	ContactTitle	1,2 Total Sales
1	QUICK	QUICK-Stop	Horst Kloss	Accounting Manager	117483.39
2	SAVEA	Save-a-lot Markets	Jose Pavarotti	Sales Representative	115673.39
3	ERINSH	Ernst Handel	Roland Mendel	Sales Manager	113236.68
4	HUNGO	Hungry Owl All-Night Grocers	Patricia McKenna	Sales Associate	57317.39
5	RATTC	Rattlesnake Canyon Grocery	Paula Wilson	Assistant Sales Representati...	52245.9
6	HANAR	Hanari Carnes	Mario Pontes	Accounting Manager	34101.15
7	FOLKO	Folk och få HB	Maria Larsson	Owner	32555.55
8	MEREP	Mère Paillard	Jean Fresnière	Marketing Assistant	32203.9
9	KOENE	Königlich Essen	Philip Cramer	Sales Associate	31745.75
10	QUEEN	Queen Cozinha	Lúcia Carvalho	Marketing Assistant	30226.1



OnlineSalesDataflow

No label

Search

Home

Transform

Add column

View

Help

Get data

Enter data

Manage data connections

Options

Manage parameters

Refresh

Advanced editor

Add data destination

Choose columns

Remove columns

Keep rows

Remove rows

Filter rows

Sort

Split column

Group by

Replace values

Transform

Replace values

Merge queries

Append queries

Combine files

Map to entity

Export template

Queries [13]

Data staging [5]

DimCustomer_raw

DimGeography_raw

DimProductCategory_raw

DimProductSubcategory_raw

DimProduct_raw

Data load [3]

DimDate

DimEmployee

DimStore

Data transformation [3]

DimCustomer

DimProduct

FactOnlineSales

fx toGetFact

fx toGetFile

Table: ExpandTableColumn#Merged queries, "DimGeography_raw", ("GeographyType", "ContinentName", "CityName", "StateProvinceName", "RegionCountryName", "Geometry"), ("GeographyType", "CustomerKey", "GeographyKey", "FirstName", "MiddleName", "LastName", "BirthDate", "MaritalStatus", "Suffix", "Title", "EmailAddress", "YearlyIncome", "Education")

Columns: 24 Rows: 9+

Column profiling based on top 1,000 rows

Query settings

Properties

Name

DimCustomer

Entity type

Custom

Applied steps

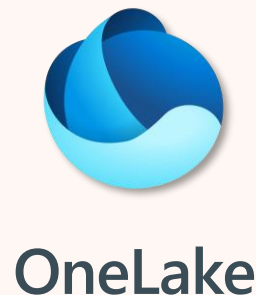
Source

Merged queries

Expanded DimGeography_raw

Data destination

No data destination



OneLake



Dataflows Gen2

Ingest and transform data at scale

AI-powered Intelligence

Transformation using Dataflows Gen2

Transform your data quickly using **Copilot** in Dataflows Gen2

Intelligently create new queries as per your requirements

The screenshot displays the Dataflows Gen2 interface, which is a Power Query editor. The top bar shows 'Copilot for Data Factory' and 'Confidential/Microsoft Extended'. The main workspace is titled 'Power Query' and contains a table of customer data. The table has columns for CustomerID, CompanyName, ContactName, ContactTitle, Address, City, Region, PostalCode, Country, Phone, and Fax. The data is filtered by Country = 'USA'. The interface includes a ribbon with various transformation options like 'Get', 'Enter data', 'Manage connections', 'Default data destination', 'Options', 'Manage parameters', 'Refresh', 'Advanced editor', 'Add data destination', 'Choose columns', 'Remove columns', 'Keep rows', 'Remove rows', 'Filter rows', 'Sort', 'Suggested transforms', 'Split', 'Group by', 'Use first row as headers', 'Replace values', 'Merge queries', 'Append queries', 'Combine files', 'Map to entity', 'Copilot', and 'Export template'. The bottom status bar indicates 'Completed (17:09 s) Columns: 13 Rows: 13'.

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	PostalCode	Country	Phone	Fax
1	GREAL	Howard Snyder	Marketing Manager	2732 Baker Blvd.	Eugene	OR	97403	USA	(503) 555-7555	null
2	HUNGG	Hungry Coyote Import Store	Yoshi Latimer	Sales Representative	City Center Plaza 516 Main St.	Egin	OR	97827	USA	(503) 555-6874
3	LAZYK	Lazy K Kountry Store	John Steel	Marketing Manager	12 Orchestra Terrace	Walla Walla	WA	99362	USA	(509) 555-7969
4	LETS5	Let's Stop N Shop	Jaime Yones	Owner	87 Polk St. Suite 5	San Francisco	CA	94117	USA	(415) 555-5938
5	LONIEP	Lonesome Pine Restaurant	Fran Wilson	Sales Manager	89 Chiaroscuro Rd.	Portland	OR	97219	USA	(503) 555-9573
6	OLDWO	Old World Delicatessen	Rene Phillips	Sales Representative	2743 Bering St.	Anchorage	AK	99508	USA	(907) 555-7504
7	RATTC	Rattlesnake Canyon Grocery	Paula Wilson	Assistant Sales Representative	2817 Milton Dr.	Albuquerque	NM	87110	USA	(505) 555-5939
8	SAVEA	Save-a-lot Markets	Jose Pavarotti	Sales Representative	187 Suffolk Ln.	Boise	ID	83720	USA	(208) 555-8097
9	SPLUR	Spill Rail Beer & Ale	Art Braunschweiger	Sales Manager	P.O. Box 555	Lander	WY	82520	USA	(307) 555-4680
10	THEBI	The Big Cheese	Liz Nixon	Marketing Manager	89 Jefferson Way Suite 2	Portland	OR	97201	USA	(503) 555-9612
11	THECR	The Crack Box	Liu Wong	Marketing Assistant	55 Grizzly Peak Rd.	Butte	MT	59801	USA	(406) 555-5834
12	TRAIH	Trail's Head Gourmet Provisioners	Hervé Hery	Sales Associate	722 Delaware Blvd.	Kirkland	WA	98034	USA	(206) 555-8237
13	WHITC	White Clover Markets	Karl Jablonski	Owner	305 - 14th Ave. S. Suite 3B	Seattle	WA	98128	USA	(206) 555-4112



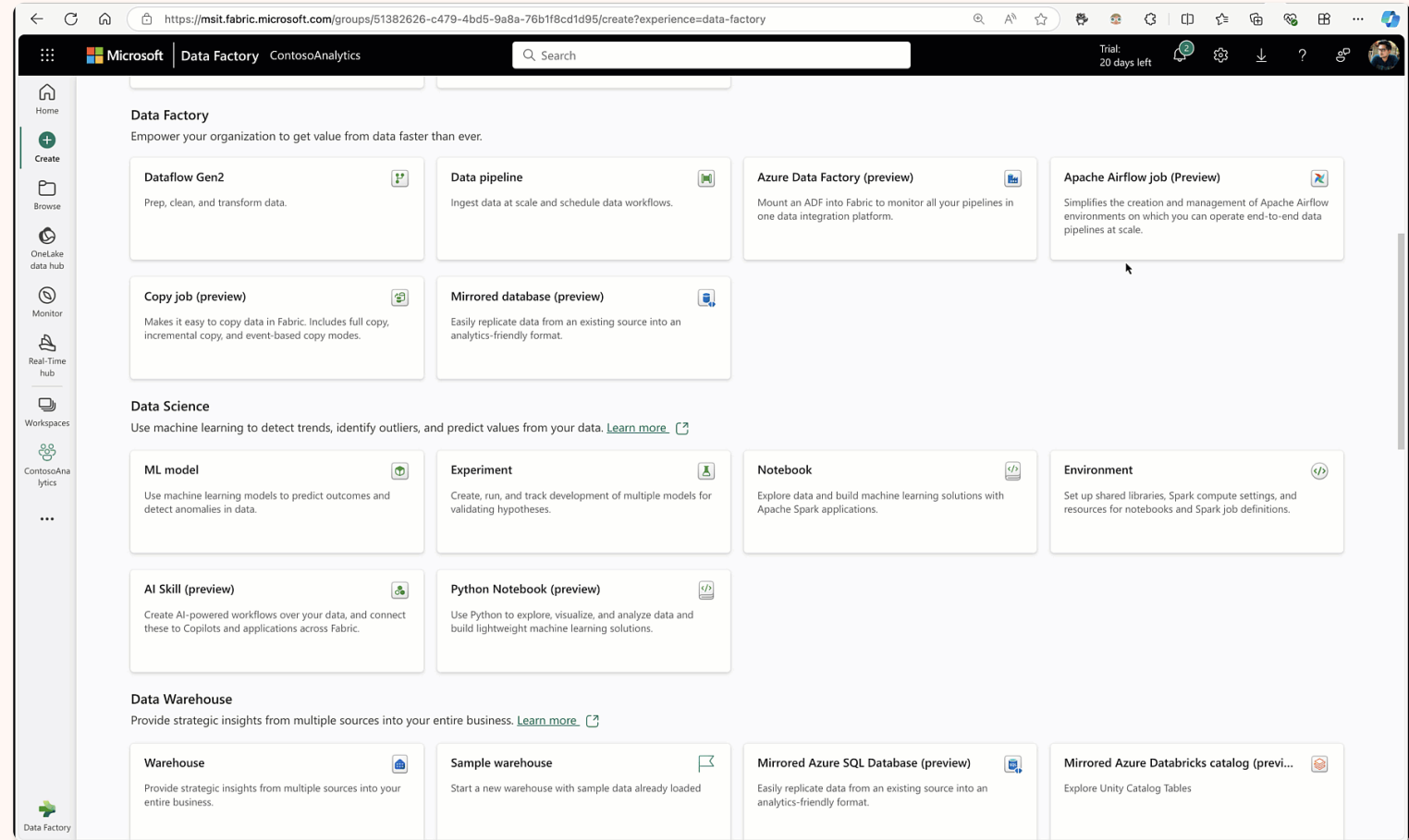
Apache Airflow Job

Data Factory's workflow orchestration manager

Native support for Apache Airflow

Apache Airflow job - Pro code, Python based data orchestration based on **Open-Source Apache Airflow project**

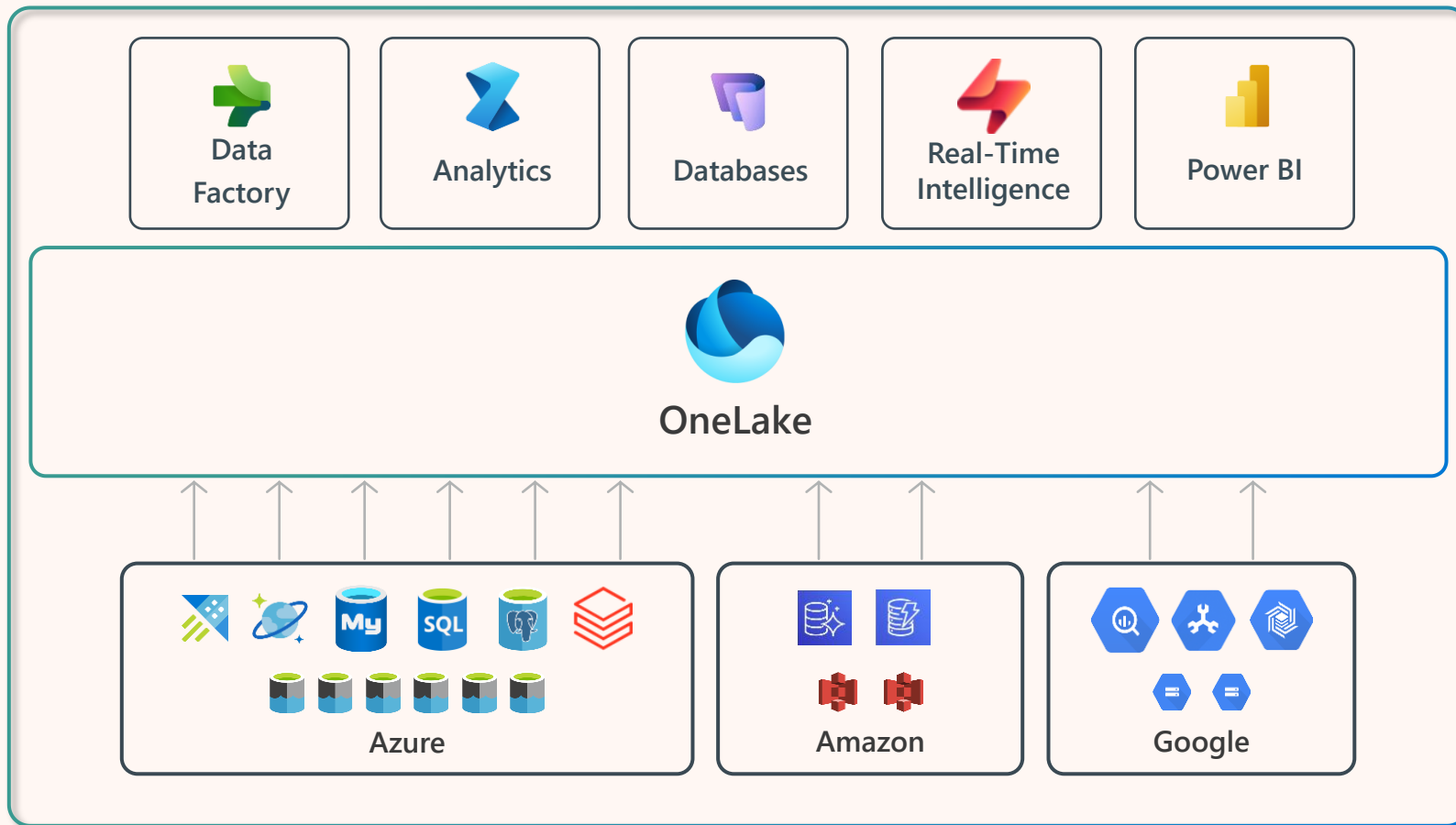
Serverless model with instant provisioning and **automatic scaling**





Mirrored Database

Bring data from various systems together



Frictionless linking of **external databases, with full replicas** created with a couple of clicks

Available for both **multi-cloud and on-premises** databases

Real time updates of the replicas using the CDC feeds of the database

Data is stored in **Delta Parquet** tables, with all Fabric services instantly available



Azure Data Factory

Mount your ADF items to Microsoft Fabric

Upgrade Pathways for ADF & Synapse customers to Fabric



Data Factory in Fabric

Data pipelines in Fabric feature parity with ADF

1

Available today



ADF Connectors in Fabric

Connectors to Lakehouse and Data Warehouse (DW)

2

Available today



Mounting ADF in Fabric

Azure Data Factory works as-is experience in Fabric

3

Available today



Upgrade to Data Factory in Fabric

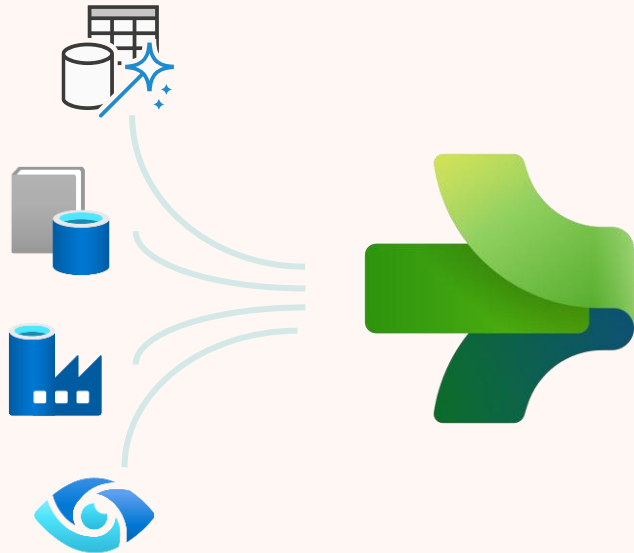
Upgrade your mounted ADF pipelines to native Fabric pipelines

4



Unifying data in OneLake

Seamlessly connect to more than 170+ data sources



 Azure Database for PostgreSQL	 Azure Databricks Delta Lake	 Amazon RDS for Oracle	 Amazon RDS for SQL Server	 Amazon Redshift	 Phoenix	 PostgreSQL	 Presto	 Magento (Preview)
 Azure SQL Database	 Azure SQL Database Managed Instance	 Apache Impala	 Azure SQL Database Managed Instance	 DB2	 SAP BW Open Hub	 SAP BW via MDX	 SAP HANA	 Oracle Eloqua (Preview)
 Azure Table Storage	 MongoDB Atlas	 Drill	 Google AdWords	 Google BigQuery	 SAP Table	 SQL server	 Spark	 PayPal (Preview)
 Azure Cosmos DB (MongoDB API)	 Azure Cosmos DB (SQL API)	 Greenplum	 HBase	 Hive	 Amazon S3	 Amazon S3 Compatible	 FTP	 SAP Cloud For Customer
 Azure Data Lake Storage Gen1	 Azure Data Lake Storage Gen1 for Cosmos Structured Stream	 Informix	 MariaDB	 Microsoft Access	 File system	 Google Cloud Storage (S3 API)	 HDFS	 Salesforce Marketing Cloud
 Azure Data Lake Storage Gen2 for Cosmos Structured Stream	 Azure Database for MariaDB	 MySQL	 Netezza	 Oracle	 HTTP	 Oracle Cloud Storage (S3 API)	 SFTP	 Shopify (Preview)
 Teradata	 Vertica	 ODBC	 OData	 REST	 Amazon Marketplace Web Service	 Concur (Preview)	 Dataverse (Common Data Service for Apps)	 Web Table
 Jira	 Kusto	 SharePoint Online List	 Dynamics 365	 Dynamics AX	 Dynamics CRM	 Cassandra	 Couchbase (Preview)	 More



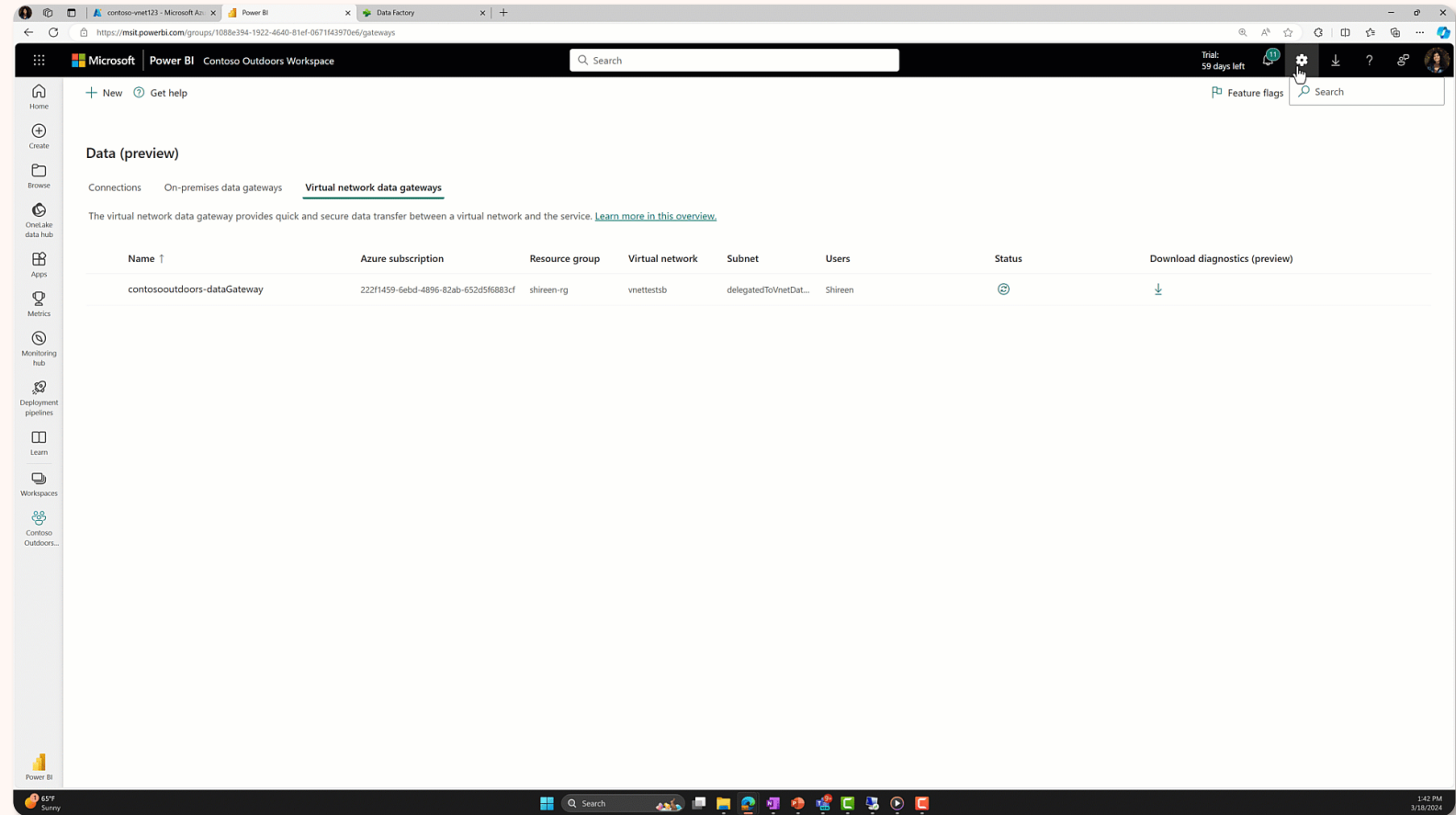
Virtual Network & On Premises Data Gateways

Secure connectivity to your Azure and private data services

VNet & On Prem Gateways

VNet Gateways enable network security compliant access to secured resources

On Prem Gateways enable line of sight to on-prem data sources





CI/CD & Monitoring Hub in Microsoft Fabric

Rich monitoring capabilities and DataOps

Connect to Git with **Azure DevOps** or **GitHub** (cloud) repositories effortlessly

Utilize a code-free UI and API-based deployment for streamlined processes.

Diff and merge Fabric items with ease

View and track all your activities across all workspaces in one place

The screenshot displays the Microsoft Fabric Monitoring hub interface. The main view shows a table of activities with columns for Activity name, Status, Item type, Start time, Submitter, and Location. The table lists various pipelines and notebooks, some succeeded and some failed. A 'Workspace settings' dialog is open, showing options for Git integration, including connecting to Azure DevOps or GitHub, and setting up a repository and branch.

Monitoring hub
Monitoring hub is a station to view and track active activities across different products.

Refresh Filter by keyword Filter (4)

Clear all ContosoAnalytics X Dataflow Gen2 X Data pipeline X Notebook X

Activity name	Status	Item type	Start time ↑	Submitter	Location
demo-pipeline	Succeeded	Data pipeline	11:46 PM, 4/26/23	Shireen Bahadur	ContosoAnalytics
demo-pipeline	Succeeded	Data pipeline	11:48 PM, 4/26/23	Shireen Bahadur	ContosoAnalytics
Contoso_Data_Pipeline	Failed	Data pipeline	5:28 PM, 5/9/23	Abhishek Narain	ContosoAnalytics
Contoso_Data_Pipeline	Failed	Data pipeline	5:31 PM, 5/9/23	Abhishek Narain	ContosoAnalytics
pipeline6	Succeeded	Data pipeline	10:31 AM, 5/10/23	Abhishek Narain	ContosoAnalytics
Contoso_Data_Pipeline	Failed	Data pipeline	1:36 PM, 5/16/23	Abhishek Narain	ContosoAnalytics
pipeline8	Failed	Data pipeline			
Notebook 2_990e4f3d-8fb0-468e-9b4c-844462...	Succeeded	Notebook			
pipeline8	Failed	Data pipeline			
pipeline10	Succeeded	Data pipeline			
pipeline10	Succeeded	Data pipeline			
pipeline10	Succeeded	Data pipeline			

Workspace settings

Search

Git integration
Connect to Git with Azure DevOps to manage your code and back up your work. [Learn more](#)

View Azure DevOps account

Connect Git repository and branch

Organization *
sbahadur0592

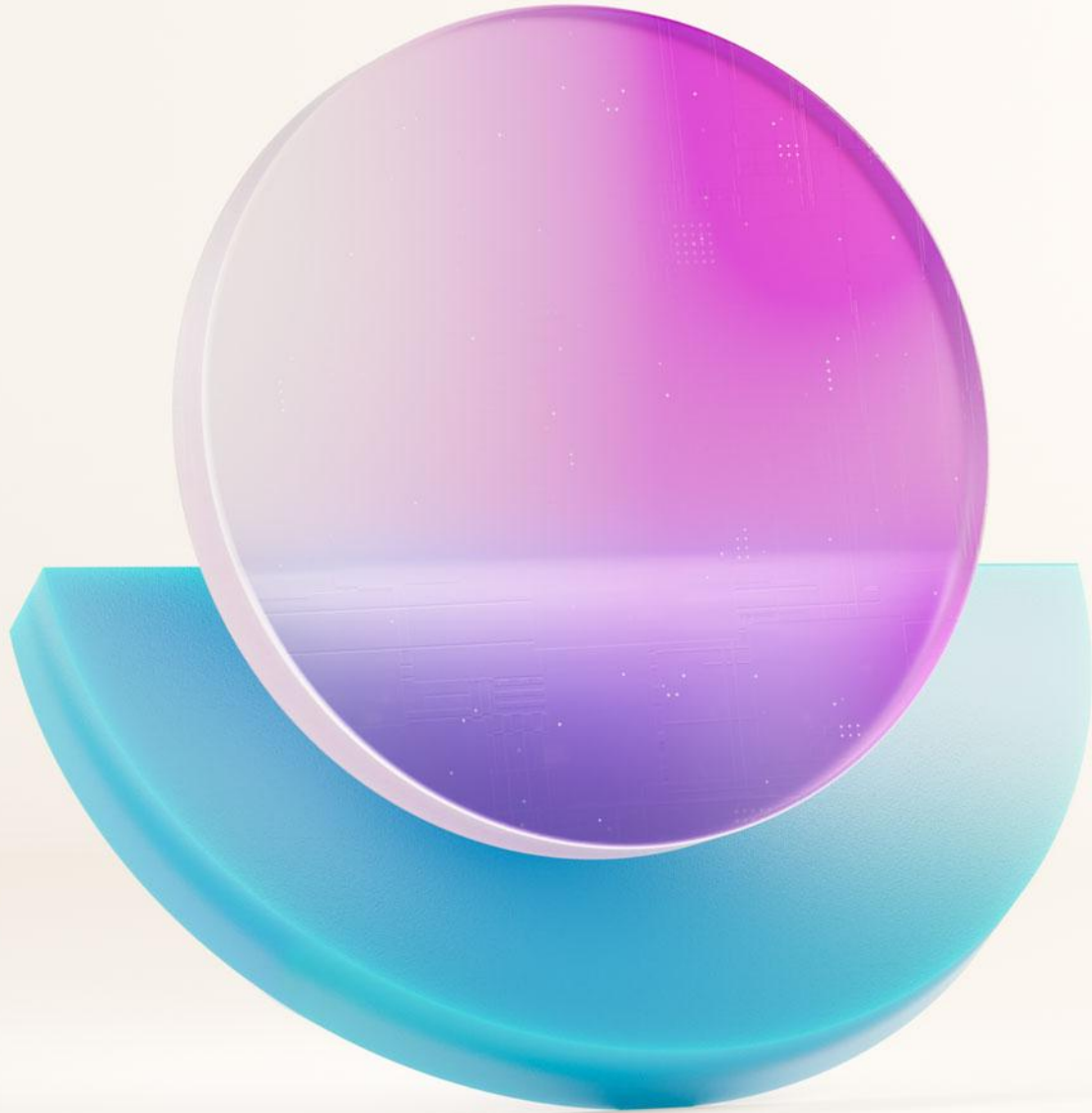
Project *
MS-Sales

Git repository * ⓘ
saledev

Branch * ⓘ
contoso-dev

Git folder ⓘ
Enter name of folder

Connect and sync Cancel



Questions?

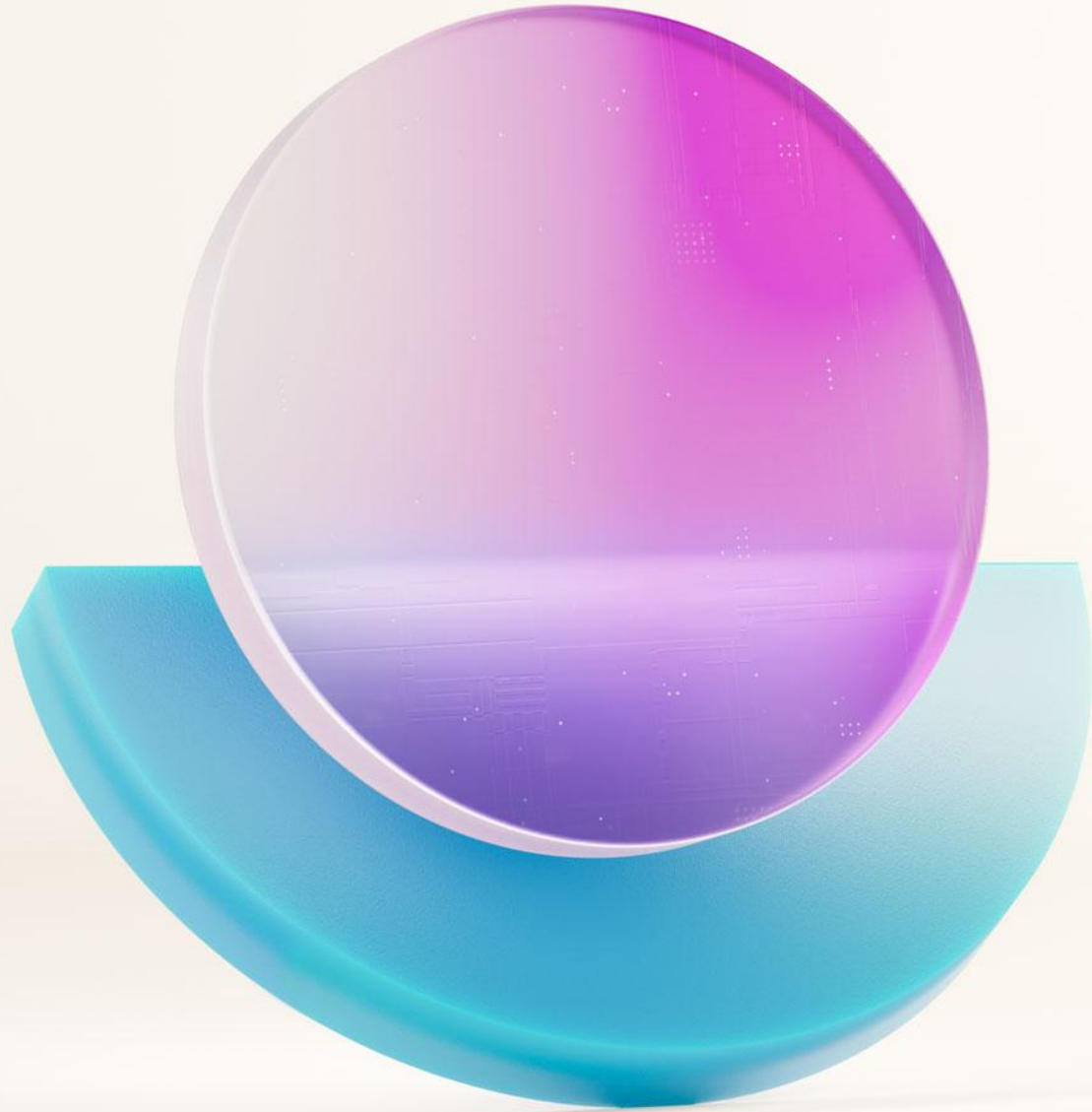


Let's take a short break...

Be back @ 11:00

Influence our product roadmap
and ensure Fabric meets your real-life needs
<https://aka.ms/fabricideas>

Microsoft Fabric
Community Conference



**Unify your data
with Lakehouses**



Unified data management with a Lakehouse in Microsoft Fabric

Store, manage and **analyze** all your data in a single location and easily share across the entire enterprise

Flexible and scalable solution that enables organizations to handle large data volumes of all types and sizes

Built-in SQL endpoint unlocks Data Warehouse capabilities on top of your Lakehouse with no data movement

Address the challenges of traditional Data Lakes by adding a **Delta Lake storage** layer directly on top of the cloud Data Lake

The screenshot displays the Microsoft Fabric Lakehouse interface. The top navigation bar includes the Lakehouse1 name, a search bar, and a 'Trial: 12 days left' indicator. The left sidebar contains navigation icons for Home, Create, Browse, OneLake, Apps, Metrics, Monitor, Learn, Real-Time, Workloads, Workspaces, and Lakehouse1. The main content area shows the 'Explorer' view with a search bar and a tree structure under 'Lakehouse1' containing 'Tables' and 'Files'. The 'Tables' folder is expanded, showing a 'sales' table. The 'sales' table is displayed in a grid view with columns: SalesOrderNumber, SalesOrderLine..., OrderDate, CustomerName, EmailAddress, Item, Quantity, UnitPrice, and TaxAmount. The table contains 28 rows of data. A status bar at the bottom indicates 'Succeeded (21 sec 721 ms)' and 'Columns 9 Rows 1,000'.

	ABC SalesOrderNumber	123 SalesOrderLine...	OrderDate	ABC CustomerName	ABC EmailAddress	ABC Item	123 Quantity	12 UnitPrice	12 TaxAmount
1	SO51555	7	6/23/2021 12:00:00 AM	Chloe Garcia	chloe27@adventure...	Patch Kit/8 Patches	1	2.29	0.1832
2	SO54042	7	8/9/2021 12:00:00 AM	Logan Collins	logan29@adventure...	Half-Finger Gloves, L	1	24.49	1.9592
3	SO54784	7	8/22/2021 12:00:00 AM	Autumn Li	autumn3@adventure...	All-Purpose Bike Stand	1	159	12.72
4	SO58572	7	10/25/2021 12:00:00 ...	Cesar Sara	cesar9@adventure-w...	Short-Sleeve Classic J...	1	53.99	4.3192
5	SO58845	7	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Sport-100 Helmet, Bla...	1	34.99	2.7992
6	SO58845	8	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Long-Sleeve Logo Jer...	1	49.99	3.9992
7	SO60233	7	11/16/2021 12:00:00 ...	Jason Mitchell	jason40@adventure...	Sport-100 Helmet, Bla...	1	34.99	2.7992
8	SO61412	7	12/3/2021 12:00:00 AM	Nathaniel Cooper	nathaniel9@adventur...	Short-Sleeve Classic J...	1	53.99	4.3192
9	SO62984	7	12/29/2021 12:00:00 ...	Miguel Sanchez	miguel72@adventure...	Racing Socks, M	1	8.99	0.7192
10	SO51555	6	6/23/2021 12:00:00 AM	Chloe Garcia	chloe27@adventure...	Mountain Bottle Cage	1	9.99	0.7992
11	SO52058	6	7/4/2021 12:00:00 AM	Elijah Ross	elijah7@adventure-w...	Short-Sleeve Classic J...	1	53.99	4.3192
12	SO53255	6	7/28/2021 12:00:00 AM	Edward Taylor	edward31@adventur...	Short-Sleeve Classic J...	1	53.99	4.3192
13	SO53852	6	8/5/2021 12:00:00 AM	Maria Reed	maria4@adventure-w...	Hydration Pack - 70 oz.	1	54.99	4.3992
14	SO54042	6	8/9/2021 12:00:00 AM	Logan Collins	logan29@adventure...	Short-Sleeve Classic J...	1	53.99	4.3192
15	SO54377	6	8/15/2021 12:00:00 AM	Ashlee Xu	ashlee12@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
16	SO54784	6	8/22/2021 12:00:00 AM	Autumn Li	autumn3@adventure...	Patch Kit/8 Patches	1	2.29	0.1832
17	SO55957	6	9/10/2021 12:00:00 AM	Melissa Richardson	melissa31@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
18	SO56533	6	9/21/2021 12:00:00 AM	Max Alvarez	max5@adventure-wo...	Bike Wash - Dissolver	1	7.95	0.636
19	SO58845	6	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Water Bottle - 30 oz.	1	4.99	0.3992
20	SO59161	6	11/1/2021 12:00:00 AM	Isabella Long	isabella20@adventur...	Hydration Pack - 70 oz.	1	54.99	4.3992
21	SO59384	6	11/4/2021 12:00:00 AM	Miranda Ross	miranda4@adventure...	Hydration Pack - 70 oz.	1	54.99	4.3992
22	SO60232	6	11/16/2021 12:00:00 ...	Kristi Malhotra	kristi21@adventure-w...	Hitch Rack - 4-Bike	1	120	9.6
23	SO60233	6	11/16/2021 12:00:00 ...	Jason Mitchell	jason40@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
24	SO62210	6	12/16/2021 12:00:00 ...	Billy Ortega	billy22@adventure-w...	Bike Wash - Dissolver	1	7.95	0.636
25	SO62208	6	12/16/2021 12:00:00 ...	Emily Flores	emily37@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
26	SO62934	6	12/28/2021 12:00:00 ...	Brenda Arun	brenda10@adventure...	Patch Kit/8 Patches	1	2.29	0.1832
27	SO63095	6	12/31/2021 12:00:00 ...	Austin Johnson	austin40@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
28	SO51370	6	6/11/2021 12:00:00 AM	Christopher Harris	christopher14@adven...	Sport-100 Helmet, Blue	1	34.99	2.7992



Automatic discovery of Delta Lake tables in a Lakehouse

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

DETeamTests

NYCCabData

Data Engineering

Home

Get data

New Power BI dataset

Open notebook

Explorer

NYCCabData

- Tables
- Files
- TaxiData

Files > TaxiData

Name	Date modified	Type	Size
taxi_zone_lookup.csv	5/3/2023 11:30:02 AM	CSV	12 KB



Automatic discovery of Delta Lake tables in a Lakehouse

The Lakehouse explorer provides a **tree-like view** of the objects in the Microsoft Fabric Lakehouse item.

It has a key capability of discovering and displaying tables that are described in the metadata repository and in OneLake storage.

Delta Lake - Achieve seamless data access across all compute engines in Microsoft Fabric

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

My workspace

TFLakehouse

A SQL endpoint for SQL querying and a default dataset for reporting were created and will be updated with any tables added to the lakehouse.

Explorer

TFLakehouse

Tables

taxi_zone_lookup

123 LocationID

ABC Borough

ABC Zone

ABC service_zone

Files

TaxiData

taxi_zone_lookup

	LocationID	Borough	Zone	service_zone
1	1	EWB	Newark Air...	EWB
2	132	Queens	JFK Airport	Airports
3	138	Queens	LaGuardia ...	Airports
4	264	Unknown	NV	N/A
5	265	Unknown	NA	N/A
6	4	Manhattan	Alphabet City	Yellow Zone
7	12	Manhattan	Battery Park	Yellow Zone
8	13	Manhattan	Battery Par...	Yellow Zone
9	24	Manhattan	Bloomingd...	Yellow Zone



Navigating the Fabric Lakehouse explorer in Microsoft Fabric

Home

Create

Browse

Data hub

Monitoring hub

Workspaces

LHRedesign_AviWS

ContosoDailySales

ContosoDailySales

More...

Power BI

Home

Get data

New Power BI dataset

Open notebook

A SQL endpoint for SQL querying and a default dataset for reporting were created and will be updated with any tables added to the lakehouse. You can access the SQL endpoint using the dropdown.

Lakehouse explorer

ContosoDailySales

Tables

Customer

inventory

product

sales

Transactions

Unidentified

CustomerFeedbackAudio

Files

Employee

Product

Sales

Transaction

Customer

Showing <1000> rows

	Index	UserId	FirstName	LastName	Sex	Email	Phone	DateOfBirth	JobTitle
1	2	3d5AD30A...	Jo	Rivers	Female	fergusonkat...	-10395	7/26/1931	Dancer
2	3	810Ce0F27...	Sheryl	Lowery	Female	fhoward@e...	(599)782-0...	11/25/2013	Copy
3	5	9afFEafAe1...	Lindsey	Rice	Female	elin@exam...	(390)417-1...	4/15/1923	Biomedical ...
4	13	CDA21B6e8...	Eddie	Barnes	Female	brandy23@...	801.809.91...	2/27/1975	Dramathera...
5	14	1CC30c5F2...	Ralph	Lowe	Female	dleon@exa...	+1-511-127...	4/10/1938	Presenter, b...
6	16	bFCFDdE54...	Carly	Abbott	Female	stricklando...	(416)979-0...	10/27/2007	Therapeutic...
7	18	aCeffF56E59...	Natasha	Macias	Female	dorothyme...	(929)366-8...	10/31/1971	Recruitmen...
8	19	CF091D6b9...	Courtney	Jenkins	Female	estesana@...	(973)243-9...	1/20/1948	Accounting...
9	20	462EF46dca...	Perry	Mcmahon	Female	allison66@...	060-611-93...	11/24/2006	Education o...
10	24	3Cb9Fe3aB...	Norman	Walton	Female	samanthas...	(590)187-8...	6/19/1973	Personnel o...
11	25	be68Ba9EB...	Roger	Sweenev	Female	leblanciohn...	-8153	9/9/2008	Race relatio...



Navigating the Fabric Lakehouse explorer in Microsoft Fabric

Navigation

1. Refresh Lakehouse metadata
2. Get data
3. Lakehouse explorer for tables and files

The screenshot shows the Microsoft Fabric Lakehouse explorer interface. The top navigation bar includes a 'Home' button (1), a 'Get data' button (2), and a 'New Power BI dataset' button. The 'Lakehouse explorer' pane on the left shows a tree view of the lakehouse structure. The 'Tables' folder is highlighted (3), and the 'Files' folder is also highlighted. The main area displays a table of customer data.

Index	UserId	FirstName	LastName	Sex	Email	Phone	DateOfBirth	JobTitle	
1	2	3d5AD30A...	Jo	Rivers	Female	fergusonkat...	-10395	7/26/1931	Dancer
2	3	810Ce0F27...	Sheryl	Lowery	Female	fhoward@e...	(599)782-0...	11/25/2013	Copy
3	5	9affEafAe1...	Lindsey	Rice	Female	elin@exam...	(390)417-1...	4/15/1923	Biomedical ...
4	13	CDA21B6e8...	Eddie	Barnes	Female	brandy23@...	801.809.91...	2/27/1975	Dramathera...
5	14	1CC30c5F2...	Ralph	Lowe	Female	dleon@exa...	+1-511-127...	4/10/1938	Presenter, b...
6	16	bFCFDdE54...	Carly	Abbott	Female	stricklando...	(416)979-0...	10/27/2007	Therapeutic...
7	18	aCeFF56E59...	Natasha	Macias	Female	dorothyme...	(929)366-8...	10/31/1971	Recruitmen...
8	19	CF091D6b9...	Courtney	Jenkins	Female	estesana@...	(973)243-9...	1/20/1948	Accounting...
9	20	462EF46dca...	Perry	McMahon	Female	allison66@...	060-611-93...	11/24/2006	Education o...
10	24	3Cb9Fe3aB...	Norman	Walton	Female	samanthas...	(590)187-8...	6/19/1973	Personnel o...
11	25	be688a9EB...	Roder	Sweeney	Female	leblanciohn...	-8153	9/9/2008	Race relatio...

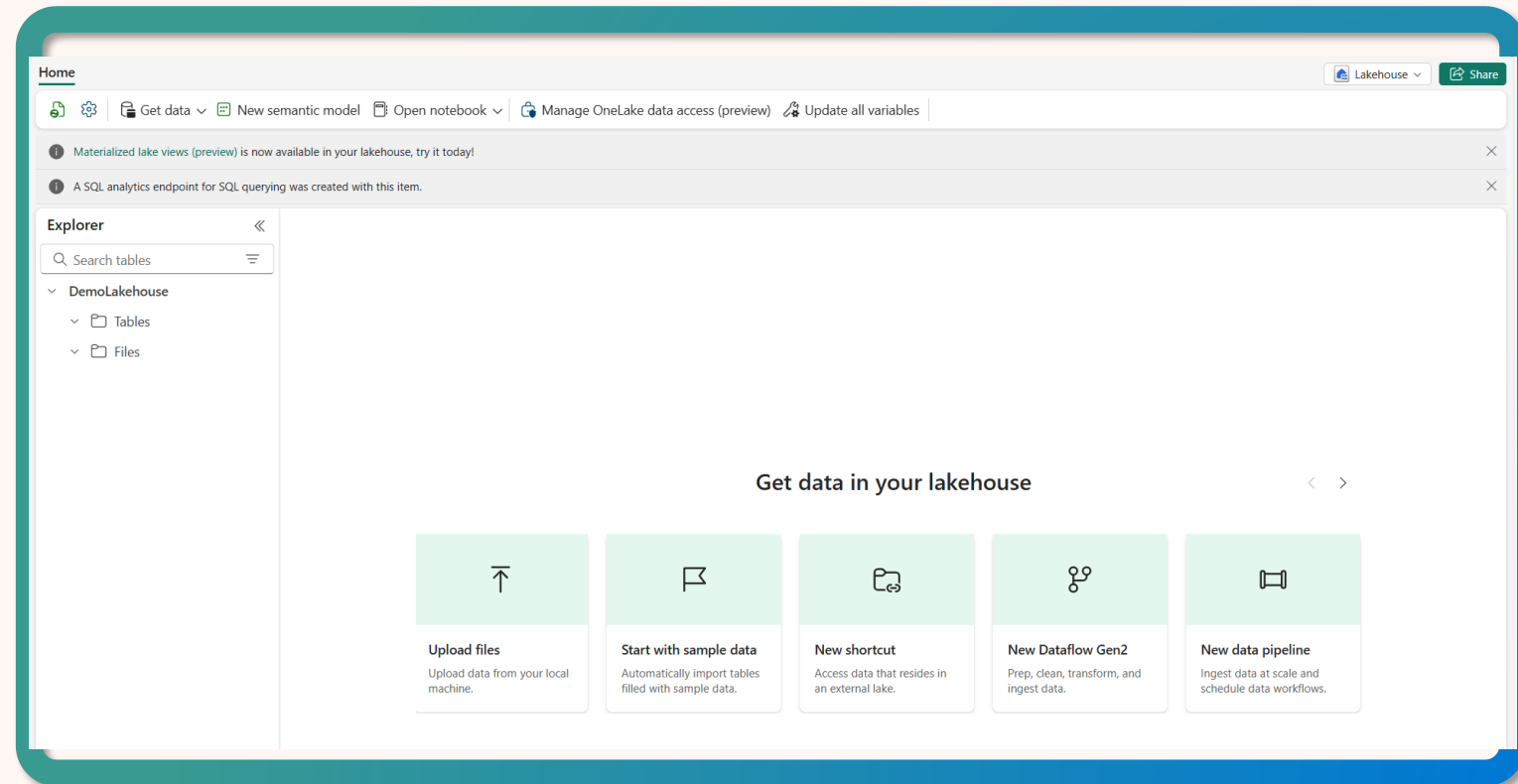


Lakehouse SQL endpoint in Microsoft Fabric

The Lakehouse creates a serving layer by automatically generating a SQL endpoint

You can **only read data** from delta tables using SQL endpoint

Save functions, views, and set SQL object-level security





Share your Lakehouse with consumers in Microsoft Fabric

Grant access **without adding users** to your workspace

Use **SQL Security** or custom permissions for SQL Endpoint access

Discover accessible Lakehouses in the OneLake catalog

Access Lakehouse data via Spark, SQL Endpoint, and Power BI semantic models

The screenshot displays the Microsoft Power BI OneLake data hub interface. The top navigation bar includes the Microsoft logo, 'Power BI', and 'OneLake data hub'. A search bar is located on the right. The left sidebar contains navigation icons for Home, Create, Browse, Data Lake, Monitoring hub, Metrics, Apps, Deployment pipelines, Learn, Workspaces, UxRedesign_AviWS, ContosoDailySales, and Power BI.

The main content area is titled 'OneLake data hub' and includes a subtitle 'Discover, manage, and use data from across your org.' with a link to 'Learn more about OneLake data hub'. Below this, a 'Recommended' section displays six lakehouse cards: ContosoDailySales, Test, Test2, Customer360, LakehouseTest1 (marked as 'Promoted'), and ContosoTransactionsDetails. Each card shows the owner 'Avinanda Chattapadday' and a 'Details' link.

Below the recommended cards is a table titled 'Explorer' showing accessible data. The table has columns for Name, Type, Owner, Location, Refreshed, Endorsement, and Sensitivity. The data is filtered by 'All' and 'My data'.

Name	Type	Owner	Location	Refreshed	Endorsement	Sensitivity
ContosoDailySales	Lakehouse	Avinanda Chattapad...	LHRedesign_AviWS	-	-	Confidential/Microsoft Ext...
Test	Lakehouse	Avinanda Chattapad...	Customer360WS	-	-	Confidential/Microsoft Ext...
Test2	Lakehouse	Avinanda Chattapad...	Customer360WS	-	-	Confidential/Microsoft Ext...
Customer360	Lakehouse	Avinanda Chattapad...	Customer360WS	-	-	Confidential/Microsoft Ext...
Test2	Warehouse (default)	Avinanda Chattapad...	Customer360WS	12/31/52, 4:07:02 PM	-	Confidential/Microsoft Ext...
Test2	Dataset (default)	Avinanda Chattapad...	Customer360WS	4/10/23, 1:52:44 PM	-	Confidential/Microsoft Ext...



Open and accessible endpoints in Microsoft Fabric



Lakehouse

SQL analytics
endpoint / Azure
Blob File System
(ABFS) / URL
endpoint



Warehouse

SQL endpoint /
OneLake
availability



Eventhouse

Cluster address
(Kusto) / OneLake
availability



Semantic model

Extensible Markup
Language for
Analysis (XMLA)
endpoint address
/ OneLake
availability



GraphQL

URL endpoint



Delta table and column name validation and rules in Microsoft Fabric

Tables

Table names can only contain **alphanumeric** characters and **underscores**



Text files

Text files without column headers are replaced with **standard col# notation** as the table column names



Columns

Column names are **validated during the load action** and the action fails if column names are invalid



Interacting with a Lakehouse in Microsoft Fabric

OneLake file explorer

Explore data in the Lakehouse using Windows File Explorer

Notebooks

Use the Notebook to write code to read, transform and write directly to Lakehouse as tables and/or folders

Data pipelines

Use pipeline copy tool to pull data from other sources and land into the Lakehouse

Copy job

Copy data easily with settings for full, incremental, or event-based.

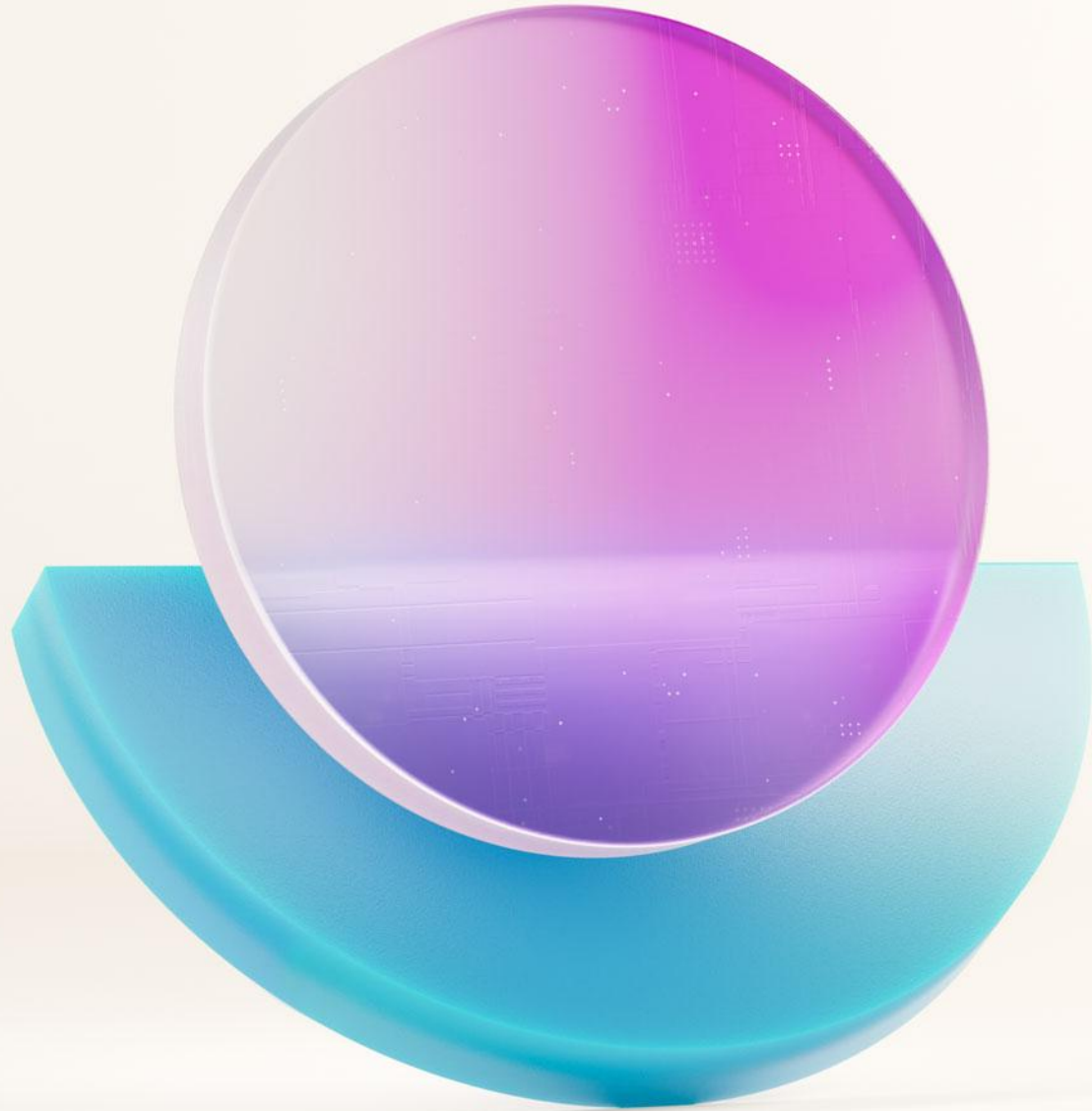
Dataflows Gen2

Use Dataflows Gen2 to ingest and prepare the data

Apache Spark job definitions

Develop robust applications and orchestrate the execution of compiled Spark jobs in Java, Scala, and Python

And more...



Create Medallion Architecture in a Lakehouse



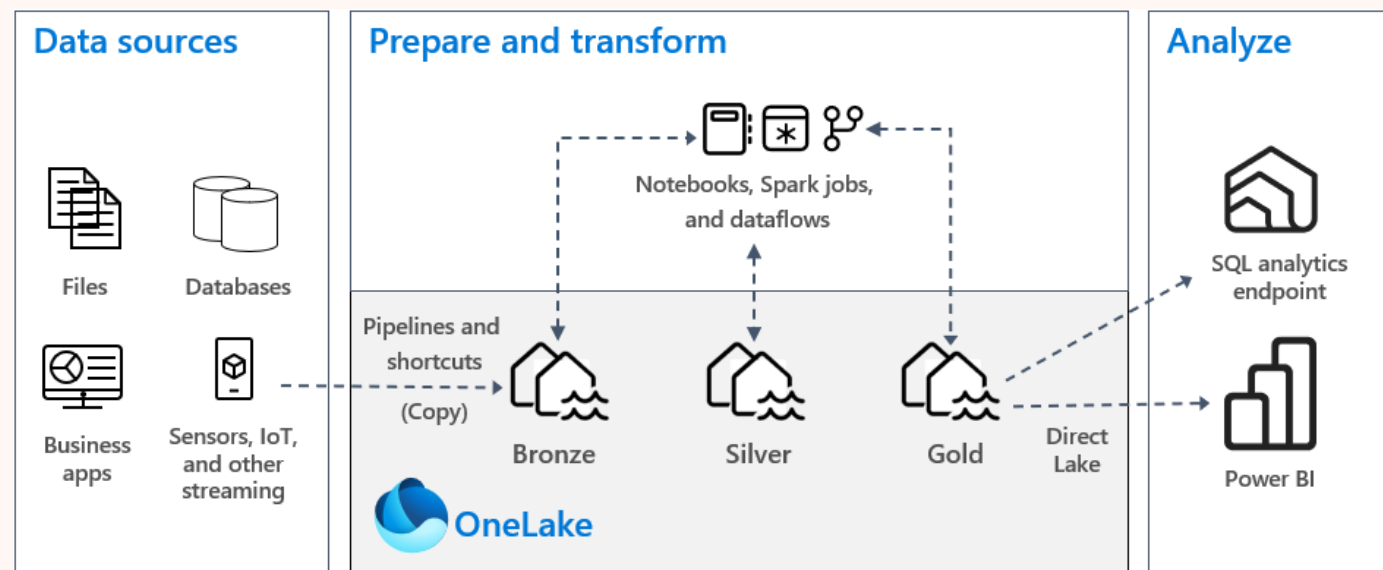
Medallion architecture

Build a single source of truth for enterprise data products with a multi-layered approach

Medallion Lakehouse architecture

Medallion architecture has three layers that represent different data quality levels: **bronze (raw data)**, **silver (validated data)**, and **gold (enriched data)**

As data moves through these layers, it undergoes validations and transformations to ensure it meets the **ACID properties** (Atomicity, Consistency, Isolation, and Durability) and is optimized for analytics





Medallion architecture

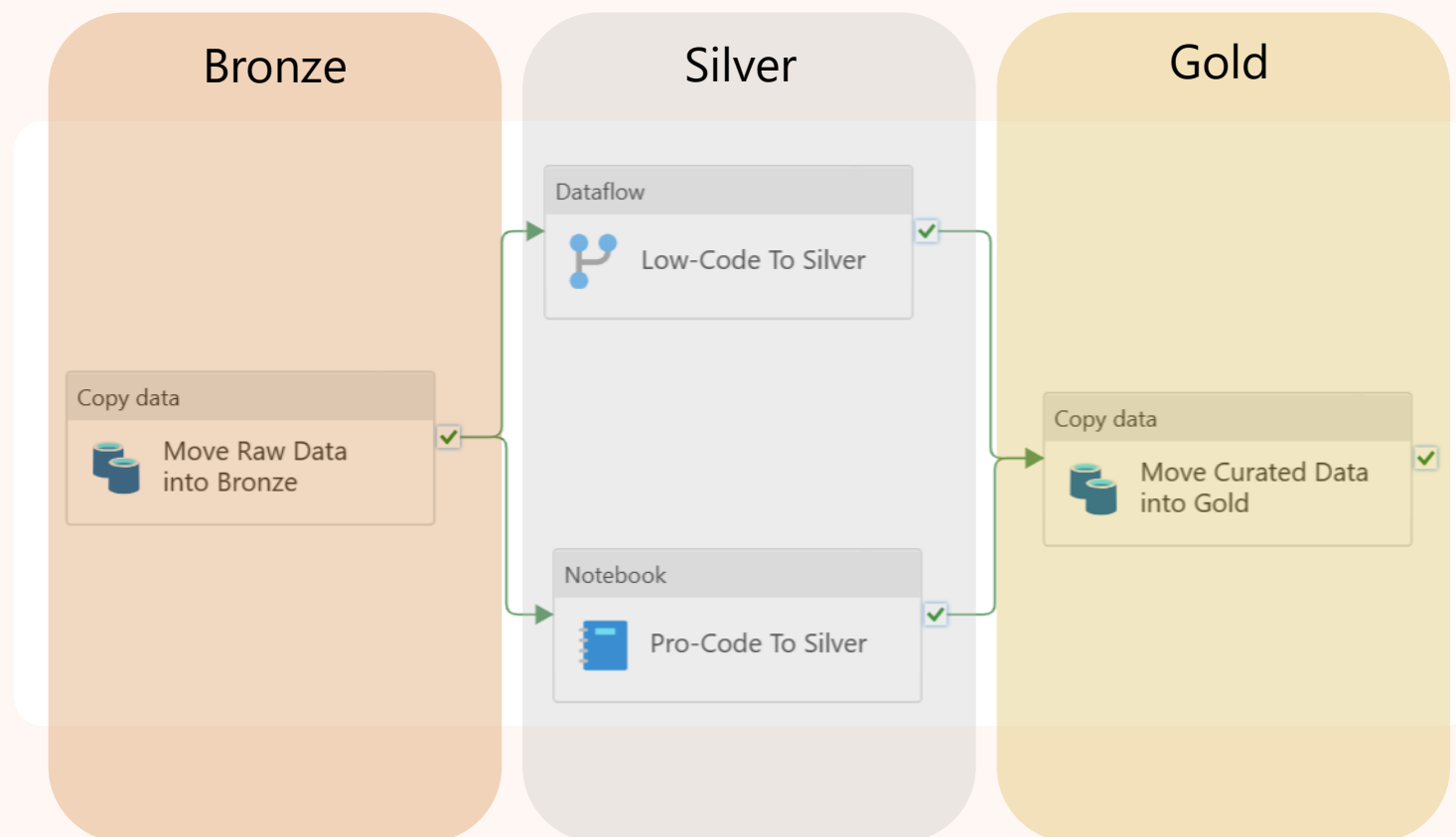
Build a single source of truth for enterprise data products with a multi-layered approach

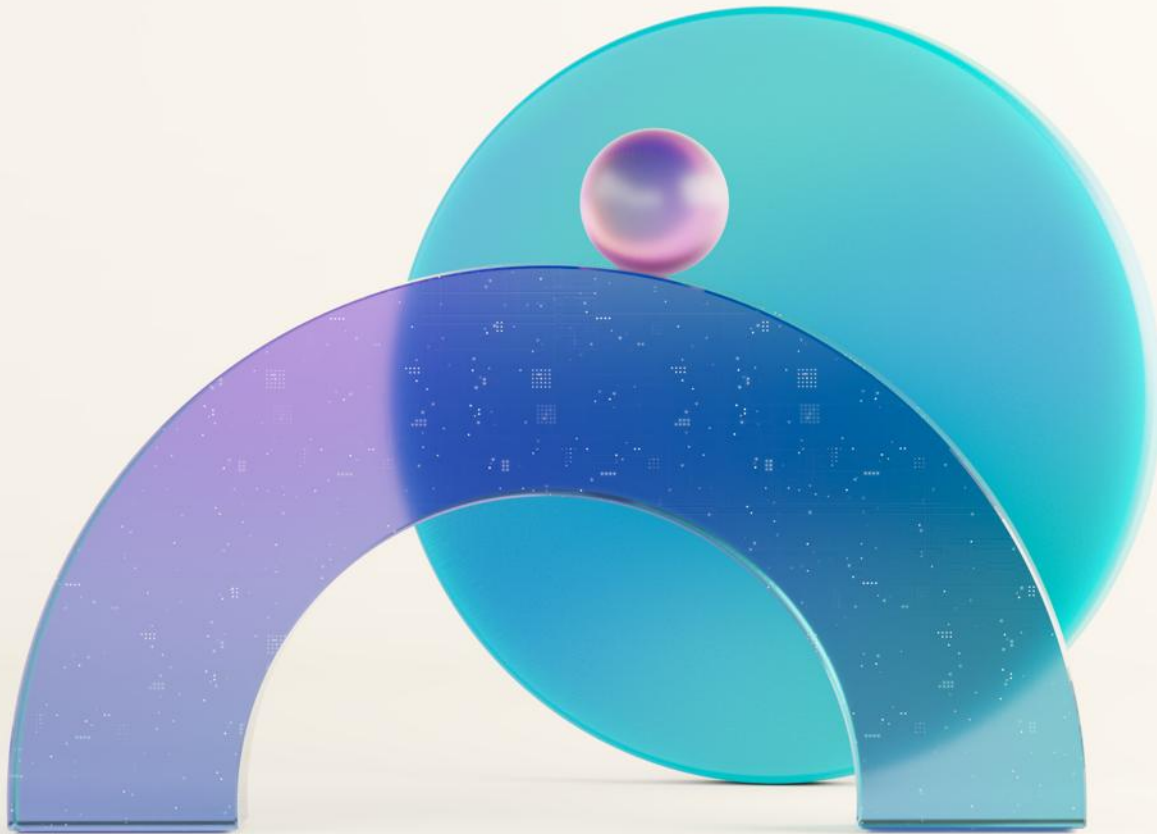
Distinct layers (or zones)

Bronze (raw zone): This layer stores source data in its original format, which is typically append-only and immutable.

Silver (enriched zone): Data from the bronze layer is cleansed, standardized, and structured as tables. It may also be integrated with other data to provide a comprehensive view of business entities like customers and products.

Gold (curated zone): Data from the silver layer is refined to meet specific business and analytics requirements. Tables in this layer usually follow a star schema design, optimizing them for performance and usability.





Lab Demo: Let's Get Started

- Create a **Medallion task flow** with a Lakehouse + Data pipeline
- Use a **Copy activity** in Data pipeline
- Explore **Data pipeline's authoring canvas**

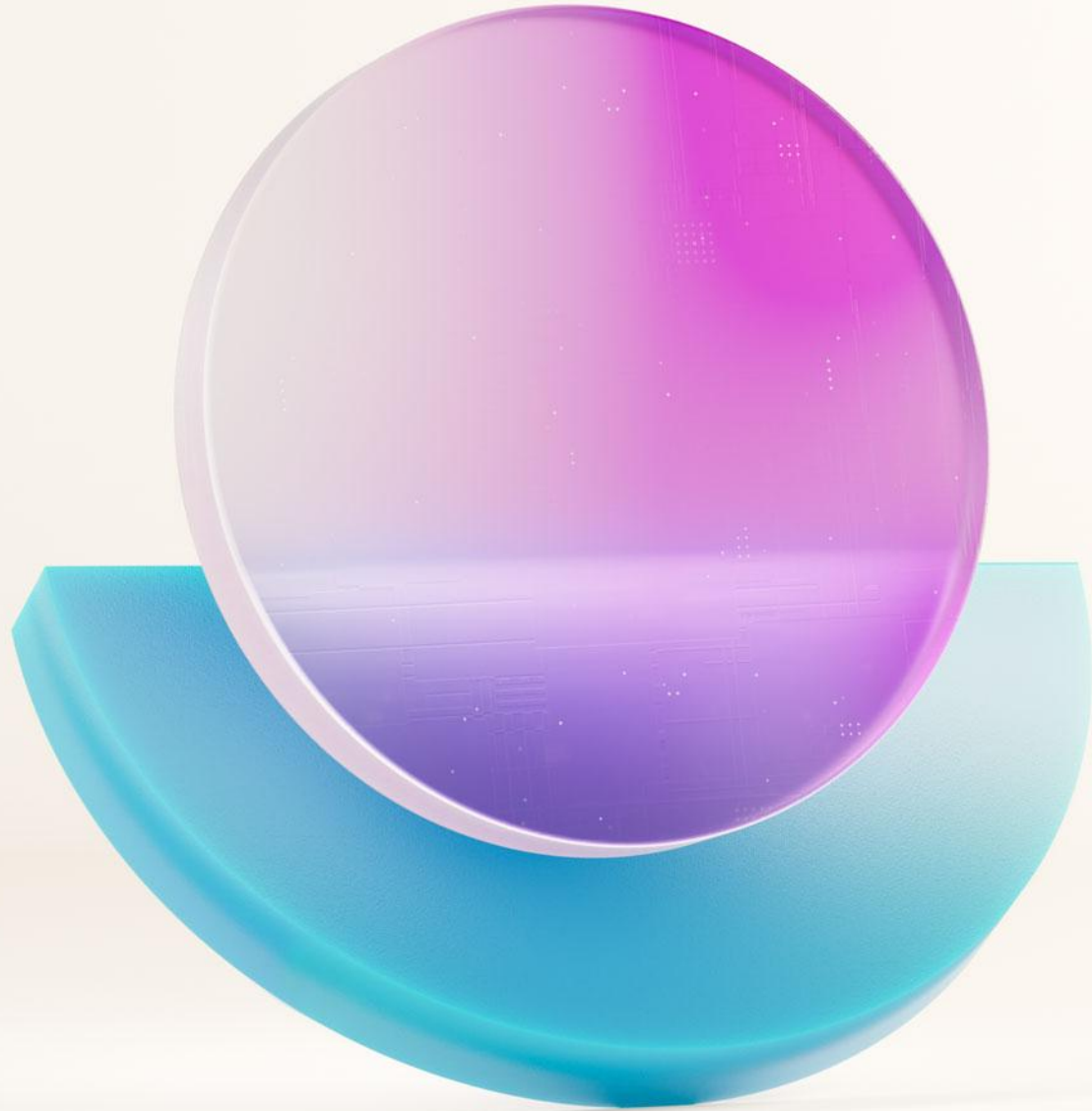
Reminder of how to access lab instructions

All materials are available at:

<https://aka.ms/dfiad>

Lab Instructions:

Getting started



Getting data into a Lakehouse



Getting data into a Fabric Lakehouse

File upload from local computer or OneLake file explorer

Run a Copy job, Data pipeline, Dataflow Gen2 or Mirrored database to **ingest data**

Ingest and transform data with a Dataflow Gen2 or Eventstream

Apache Spark libraries in **Notebook code**

The screenshot shows the Microsoft Fabric interface. At the top, there's a 'Home' tab and a navigation bar with icons for 'Get data', 'New semantic model', 'Open notebook', and 'Manage OneLake data'. Below this, a dropdown menu for 'Get data' is open, showing options: 'Upload files', 'New data pipeline', 'New Dataflow Gen2', 'New shortcut', and 'New Eventstream'. The 'Upload files' option is highlighted with a blue border. In the background, the 'Explorer' pane shows a tree view with 'testLake' expanded, containing 'Date' and 'DimCustomer'. The 'DimCustomer' table is selected, and its data is displayed in a table on the right.

	12L CustomerKey	12L GeographyK...	ABC FirstName
1	3704	586	Clifford
2	4260	443	Franklin
3	4279	546	Brandon
4	4668	674	Hector
5	7900	628	Glenn
6	8284	626	Adrian
7	9832	515	Drew
8	10097	480	Jon

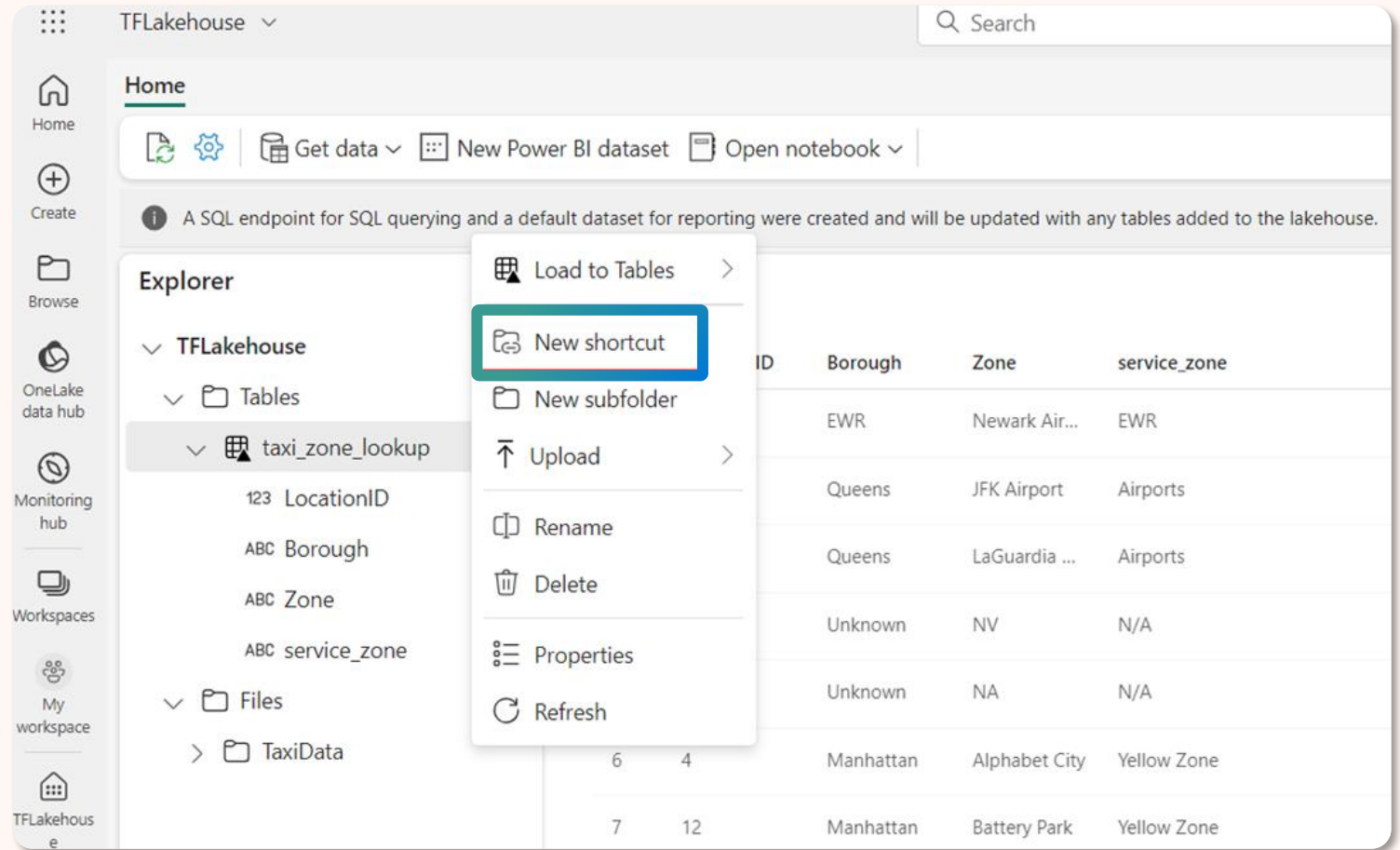


Shortcuts in a Fabric Lakehouse

Shortcuts in a Lakehouse allow users to reference data without copying it from **Tables** or **Files**

Shortcuts unify data from different Lakehouses, workspaces, or external storage, such as:

- ADLS Gen2
- AWS S3
- Dataverse

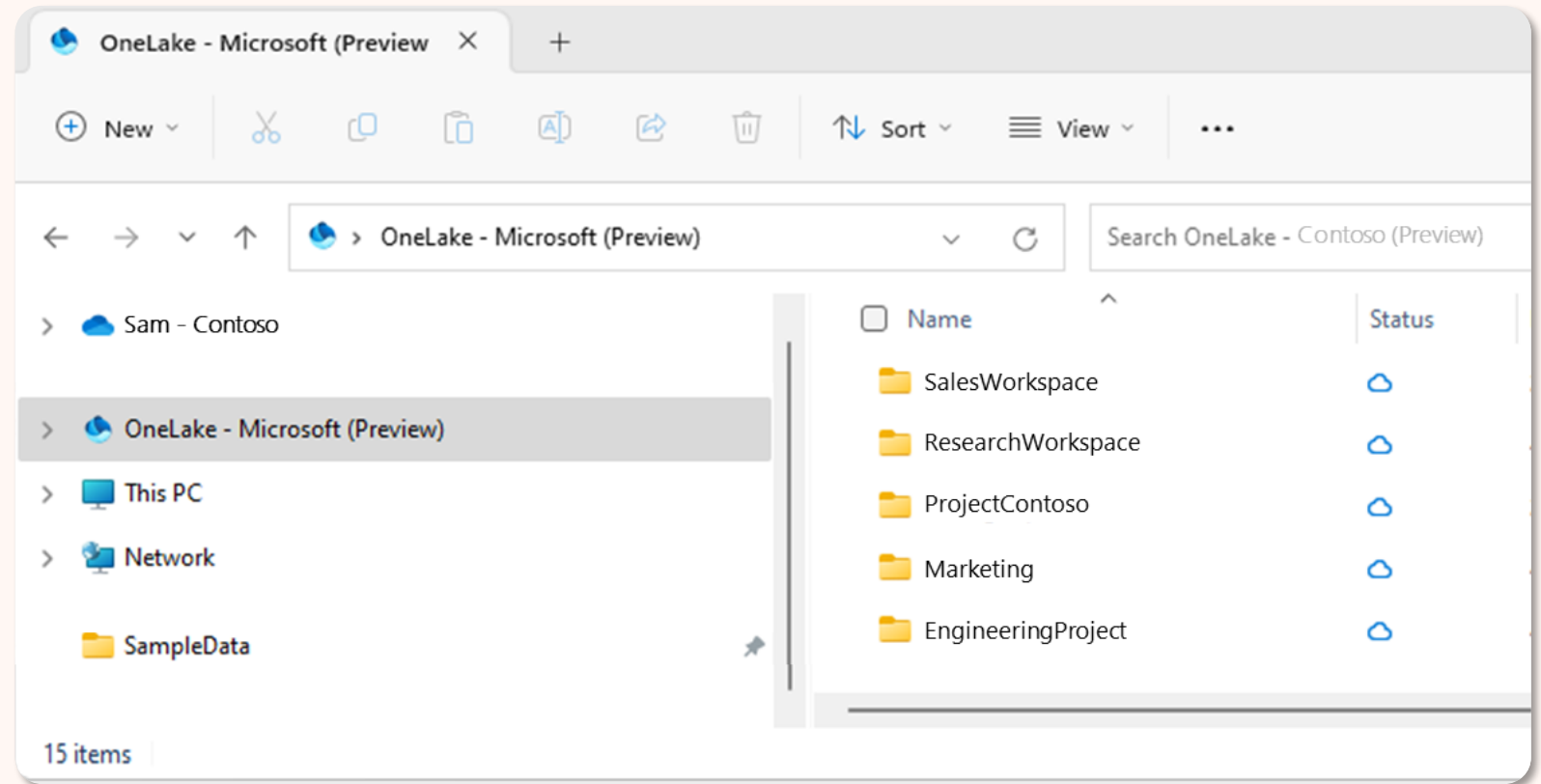




OneLake file explorer

Seamlessly integrates OneLake with Windows File Explorer, **automatically syncing** all OneLake items that you have access to in file explorer

Syncing doesn't download the data, it creates placeholders. You must double-click on a file to download the data locally





Check your knowledge

Data ingestion scenarios

Connecting to existing SQL Server and **copying** data into Delta table on the Lakehouse

Uploading files from your computer

Copying and **merging** multiple tables from other Lakehouses into a new Delta table

Referencing data without copying it from other internal Lakehouses or external sources



Considerations when loading data

Use case

Recommendation

Small file upload from local machine



Use OneLake file explorer / Upload file

Data ingestion only



Use Copy job or Data pipeline

Orchestrated data ingestion and transformation



Use a Data pipeline and Dataflow Gen2

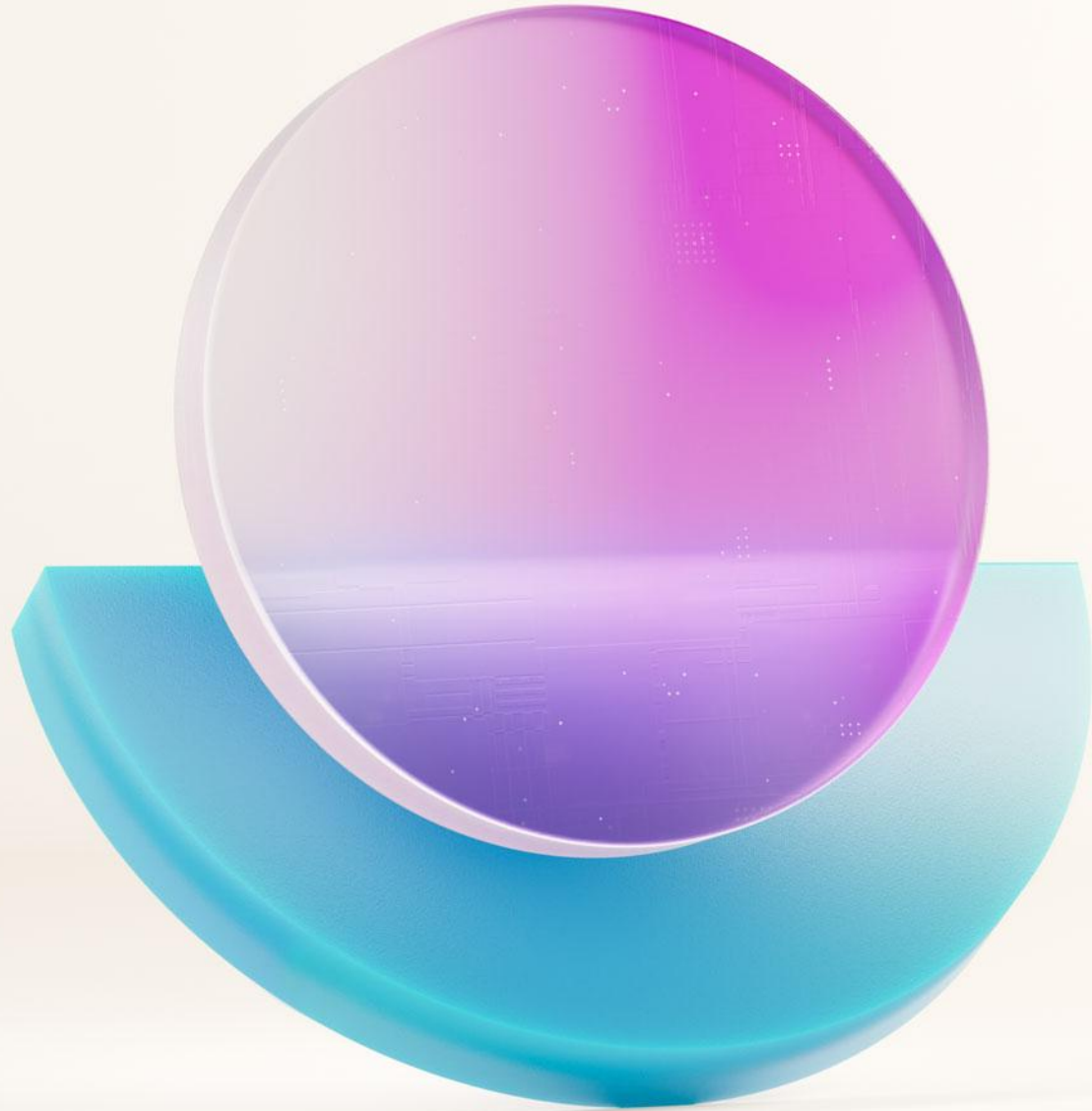
Complex data transformations and orchestration



Use Data pipeline and Notebook

Data ingestion decision guide

	Copy job	Data pipeline	Dataflow Gen2	Spark
Use case	Data ingestion	Data ingestion, lightweight transformation	Data ingestion, data transformation, data wrangling, data profiling	Data ingestion, data transformation, data processing, data profiling
Primary developer persona	Data integrator	Data engineer, data integrator	Data engineer, data integrator, business analyst	Data engineer, data scientist, data developer
Primary developer skill set	ETL	ETL, SQL, JSON	ETL, M, SQL	Spark (Scala, Python, Spark SQL, R)
Code written	No code	No code, low code	No code, low code	Code
Data volume	Low to high	Low to high	Low to high	Low to high
Development interface	Assistant	Assistant, canvas	Power Query	Notebook, Spark job definition
Sources	8+ connectors	30+ connectors	150+ connectors	Hundreds of Spark libraries
Destinations	10+ connectors	18+ connectors	Lakehouse, Warehouse, Azure SQL database, Azure Data explorer	Hundreds of Spark libraries
Transformation complexity	Low: Lightweight – column mapping, table mapping	Low: lightweight - type conversion, column mapping, merge/split files, flatten hierarchy	Low to high: 300+ transformation functions	Low to high: support for native Spark and open-source libraries



Overview: Data engineering in Fabric



Data engineering in Fabric

Empowers data engineers to transform data at scale and build a Lakehouse architecture



Build a Lakehouse
for all your
organizational data



Spark runtime with great
out of the box
performance and robust
admin controls



Delightful authoring
experience in your
tools of choice



Completely
integrated into the
Fabric foundation



Data engineering in Fabric

Empowers data engineers to transform data at scale and build a Lakehouse architecture



Lakehouse

Store and manage structured and unstructured data in a single location



Apache Spark job definition

Submit batch/streaming job to Spark cluster, apply different transformation logic to the data



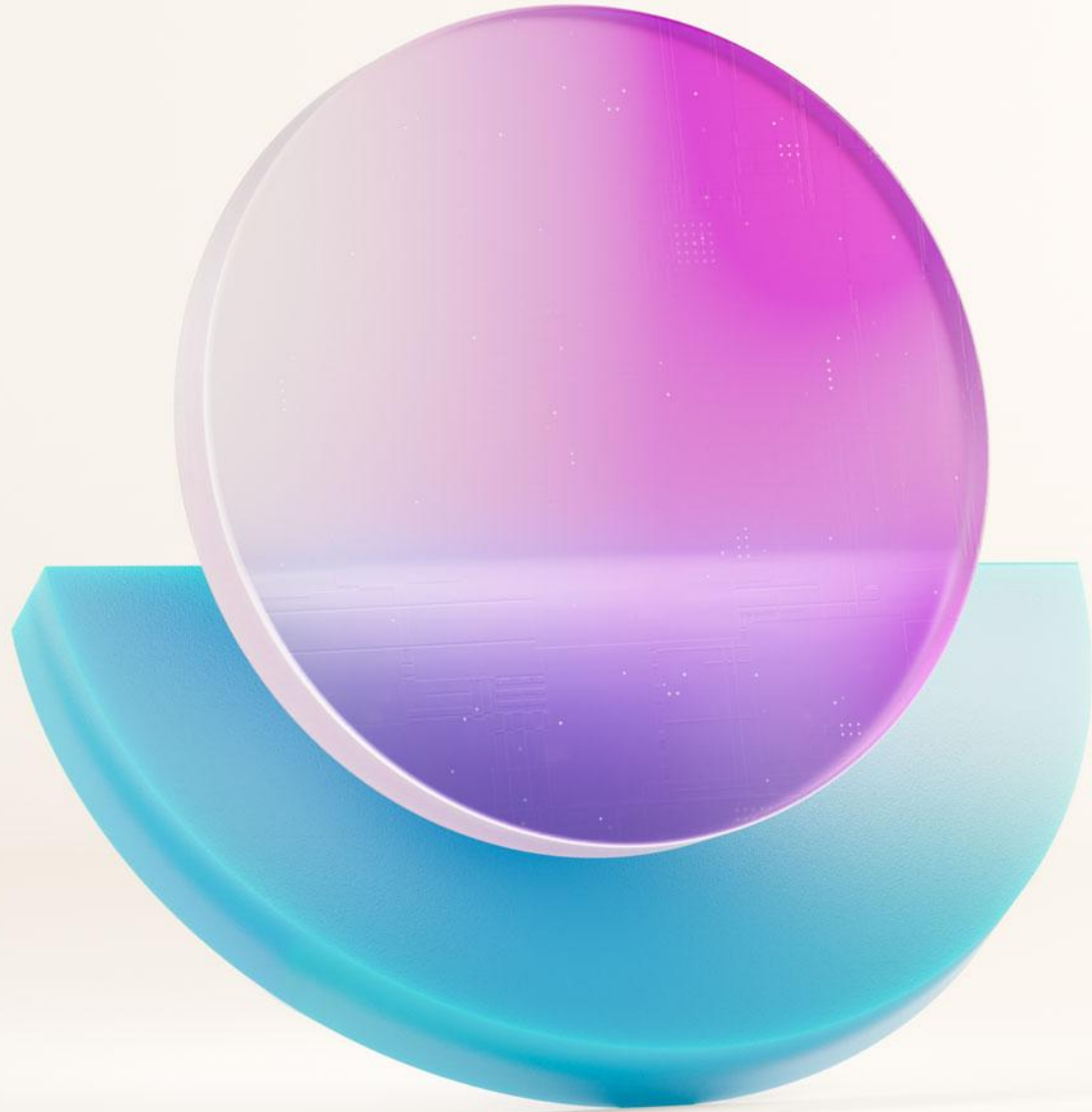
Notebook

Create and share documents that contain live code, equations, visualizations, and narrative text



Data pipeline

Collect, process, and transform data from its raw form to a format that you can use for analysis and decision-making



Ingesting and orchestrating data with Data pipelines



Data pipelines in Fabric Data Factory

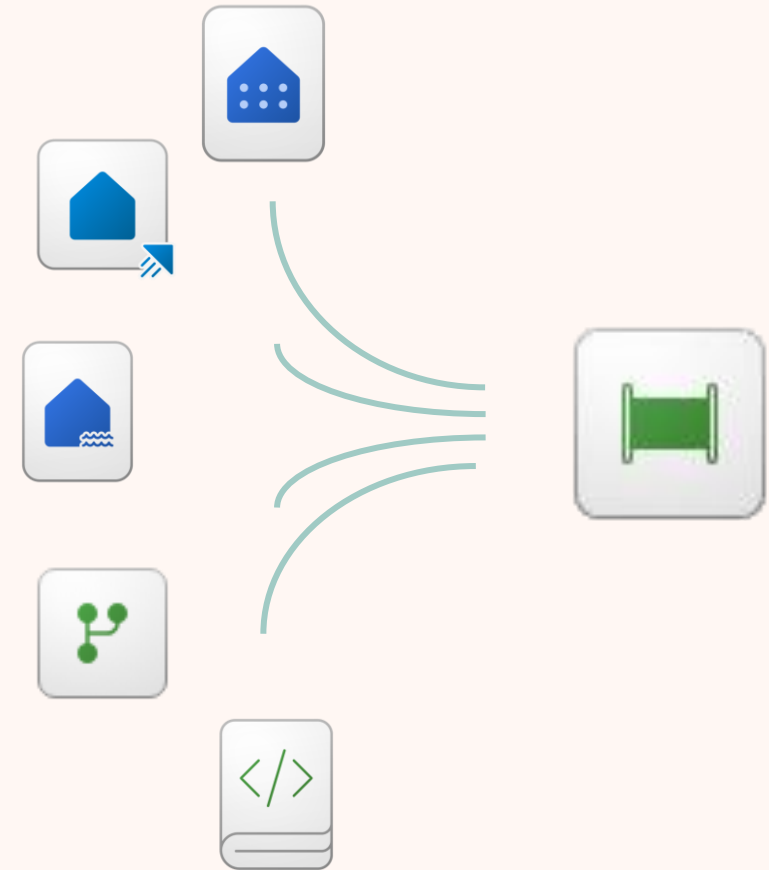
Seamlessly connect and ingest data into Fabric using a no-code interface

Evolution of **Azure Data Factory** pipelines

Rich library of **activities**

Fast copy of binary files using Az-Copy

Jumpstart with **Copy Assistant**





User Data Functions in the Functions activity in Data pipelines

Public Preview

Run specific functions within the pipeline

Fabric User Data Functions allows for custom functions designed for **event-driven scenarios**

Azure functions are serverless functions that can be triggered by various events

New item

☆ Favorites

☑ All items

Develop data

Create and build your software, applications, and data solutions.

User data functions (preview)



Author, host, and manage serverless user data functions optimized for Fabric.



function3 | Confidential\Microsoft Extended

Search

Home Edit



Language

Python



Generate invocation code



Manage connections



Library management



Functions explorer ⓘ

Functions explorer shows only published functions.

fx hello_fabric

```
1 import datetime
2 import fabric.functions as fn
3 import logging
4
5 udf = fn.UserDataFunctions()
6
7 @udf.function()
8 def hello_fabric(name: str) -> str:
9     logging.info('Python UDF trigger function processed a request.')
10
11     return f'Welcome to Fabric Functions, {name}, at {datetime.datetime.now()}'
```



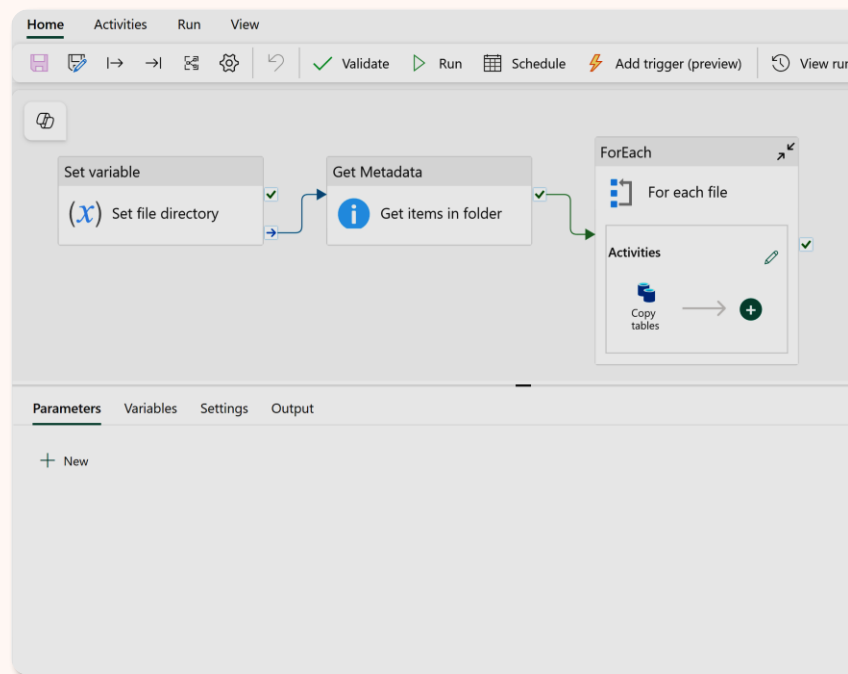
Copilot for Data Factory in Data pipelines

AI-powered Intelligence

Ingest data into Fabric via pipeline's Copilot

Transform data by invoking a Dataflow Gen2 into your pipeline with Copilot

Receive a **summary** of your pipeline in natural language



Copilot

12:52

What would you like to do?

Choose an option below or type a request to get started.

Ingest data
Bring data into Microsoft Fabric

Build the pipeline using a screenshot
Upload a screenshot of what you want in this pipeline, and Copilot will do the rest

Summarize this pipeline
Understand this pipeline's purpose and functions with summarized explanations

Describe the summary you want or ask a question.
Type '/' to list all existing connection references that you'd like to provide.



AI-generated content can have mistakes. Make sure it's accurate and appropriate before using it. [Review Terms](#)



Copy activity in Data pipelines

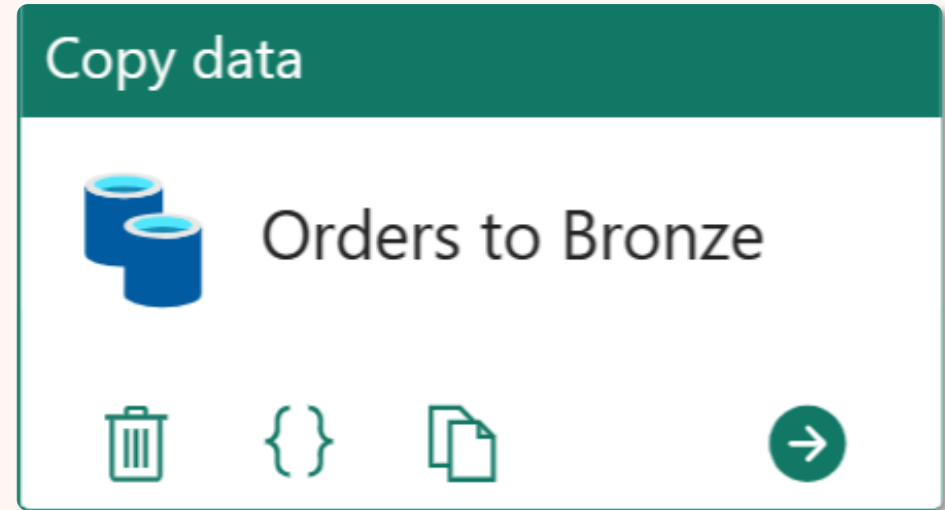
Copy activity is the core of data pipelines

Jumpstart with **Copy job** or **Copy assistant**

Fine grain control of file type conversion, column mapping, merging/splitting of files, and more..

Workspace source and destinations include Lakehouse (tables & files), Warehouse, SQL Database in Fabric, and KQL Database

Supports external sources and destinations



And so much more...

Capability

Description

Control flow	→	Conditional paths Loops
Nesting	→	Invoking data pipelines from other data pipelines Bi-directional communication via output variables
Parameterization	→	Parameters of varying data types
Expression language	→	Almost every field can be made dynamic Extensive built-in functions allow for massive customizations (data driven designs)
Repeatable	→	Schedulable



Platform scheduler

Schedule events at desired intervals

Supports **Data pipelines**, **Copy job**, **Dataflow Gen2**, and more.

Deeply integrated with **monitoring hub**

Scheduled:

- By the minute
- Hourly
- Daily
- Weekly
- Monthly

 **scheduling**
Data pipeline

[About](#)
[Sensitivity label](#)
[Endorsement](#)
[Schedule](#)

No previous history
The scheduled refresh is turned off
Run

 Schedule

Scheduled run
☐ On ☒ Off

Repeat

By the minute ▾

Every

minute(s)

Start date and time

End date and time

Time zone

(UTC-06:00) Central Time (US and Canada) ▾

Apply

Discard

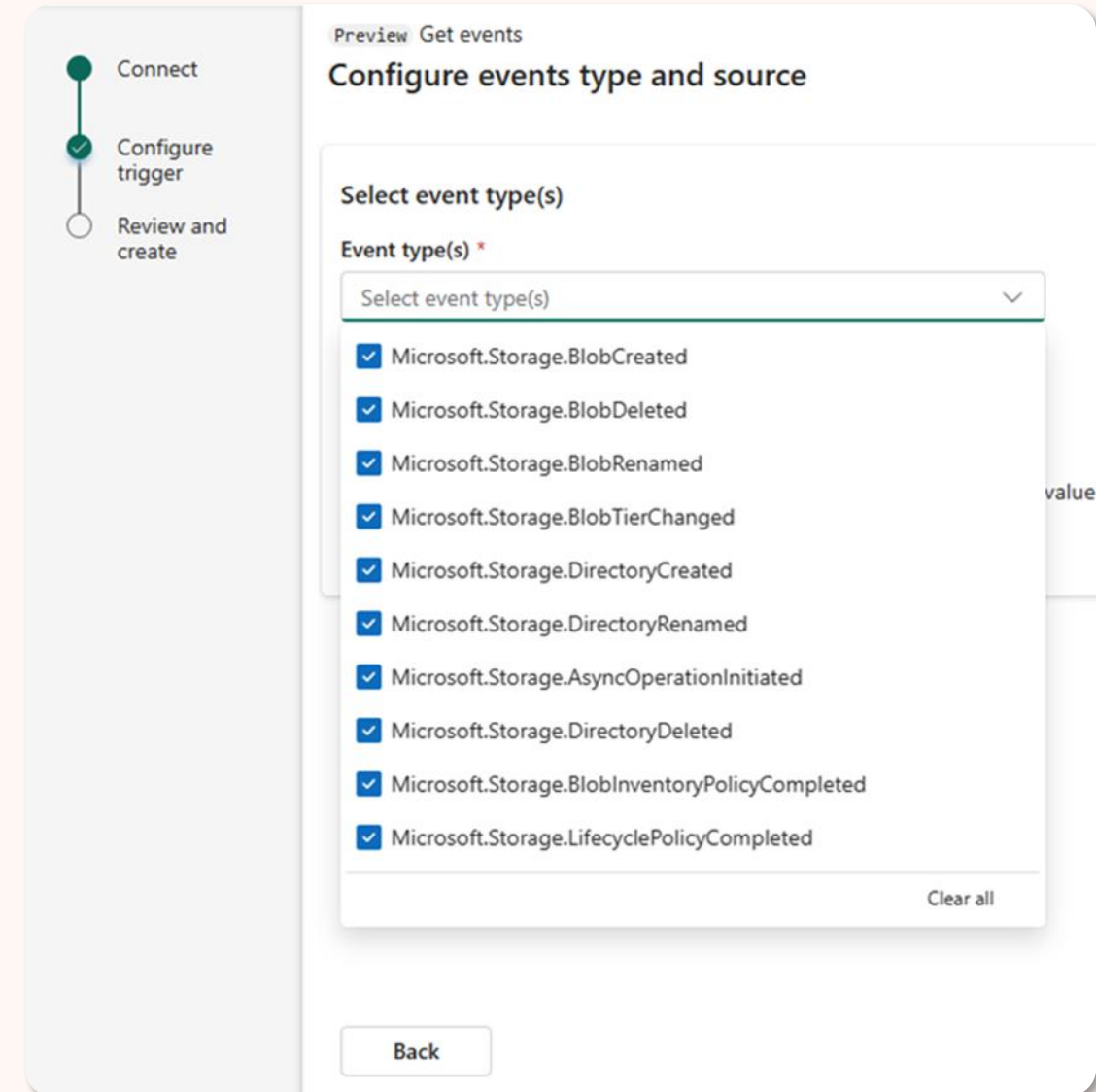
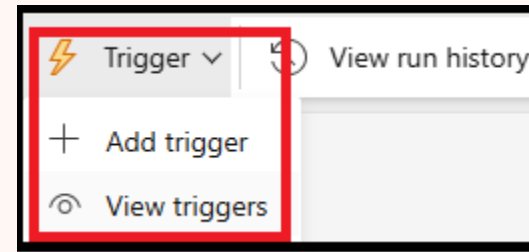


Event-driven triggers in Data pipelines

Invoke a Data pipeline upon a file event using
Eventstreams and **Reflex triggers**

Extensive event types and filtering capabilities

Ability to link multiple pipelines to a single event





Pipeline expression builder in Fabric Data Factory

Create and manage dynamic data-driven content

Wide range of functions and operators that can be used to manipulate and transform data

Greater **flexibility** and **reusability**, and to perform advanced data transformations without the need for custom code

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@string(add(mul(int(if(
    startswith(
        split(split(pipeline().parameters.interval, ':')[0],
            '.')[0]
        , '0')
    , substring(
        split(split(pipeline().parameters.interval, ':')[0],
            '.')[0]
        , 1
```

[Clear contents](#)

Parameters

System variables

Functions

Variables

☐ Expand all

☐ Collection Functions

☐ Conversion Functions

☐ Date Functions

☐ Logical Functions

☐ Math Functions

☐ String Functions

Add dynamic content [Alt+Shift+D]



Conditional paths in Data pipelines

Enable you to build robust pipelines with **error handling and branching logic**

There are four types:

- **On Skip**
- **On success**
- **On fail**
- **On completion**

The screenshot shows the configuration for a 'Set variable' task. The task name is 'Set variable1' with a blue 'x' icon. Below the task name are icons for deleting, editing code, copying, and a green arrow for next steps. To the right of the task configuration are four conditional path buttons: 'On skip' (grey), 'On success' (green), 'On fail' (red), and 'On completion' (blue). Each button has a corresponding icon in a small box to its left: a curved arrow for 'On skip', a green checkmark for 'On success', a red 'X' for 'On fail', and a blue arrow for 'On completion'.

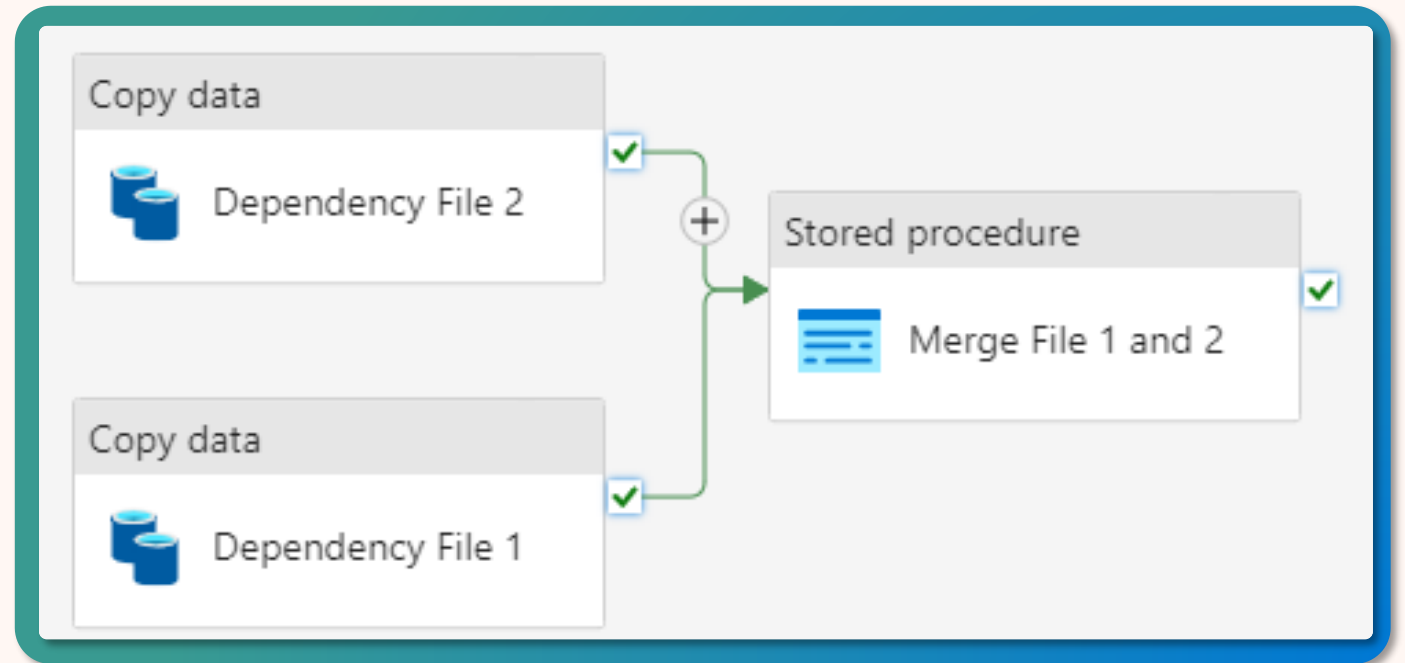
Conditional Path	Icon
On skip	Curved arrow
On success	Green checkmark
On fail	Red X
On completion	Blue arrow



Conditional paths in Data pipelines

Define different execution paths based on the outcome of a previous activity

- **Blocking dependencies**
- Non-blocking dependencies
- Error handling



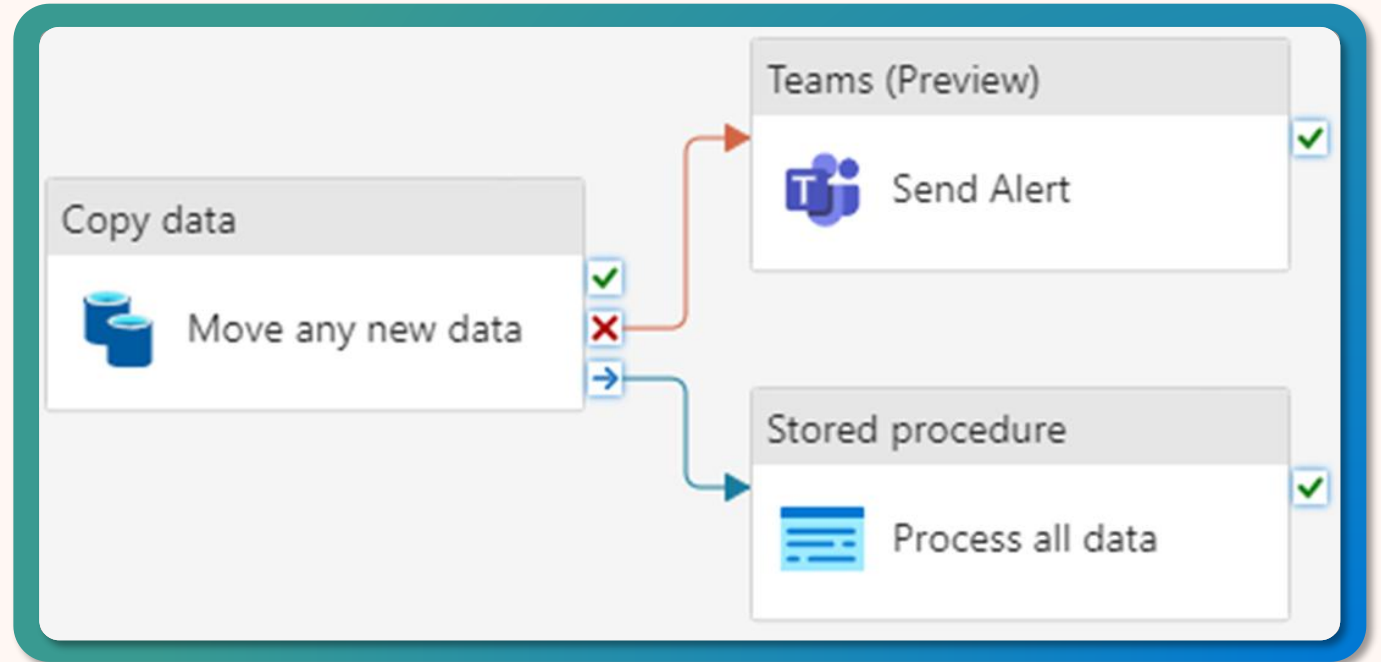
Both activities must be successful before proceeding.



Conditional paths in Data pipelines

Define different execution paths based on the outcome of a previous activity

- Blocking dependencies
- **Non-blocking dependencies**
- Error handling



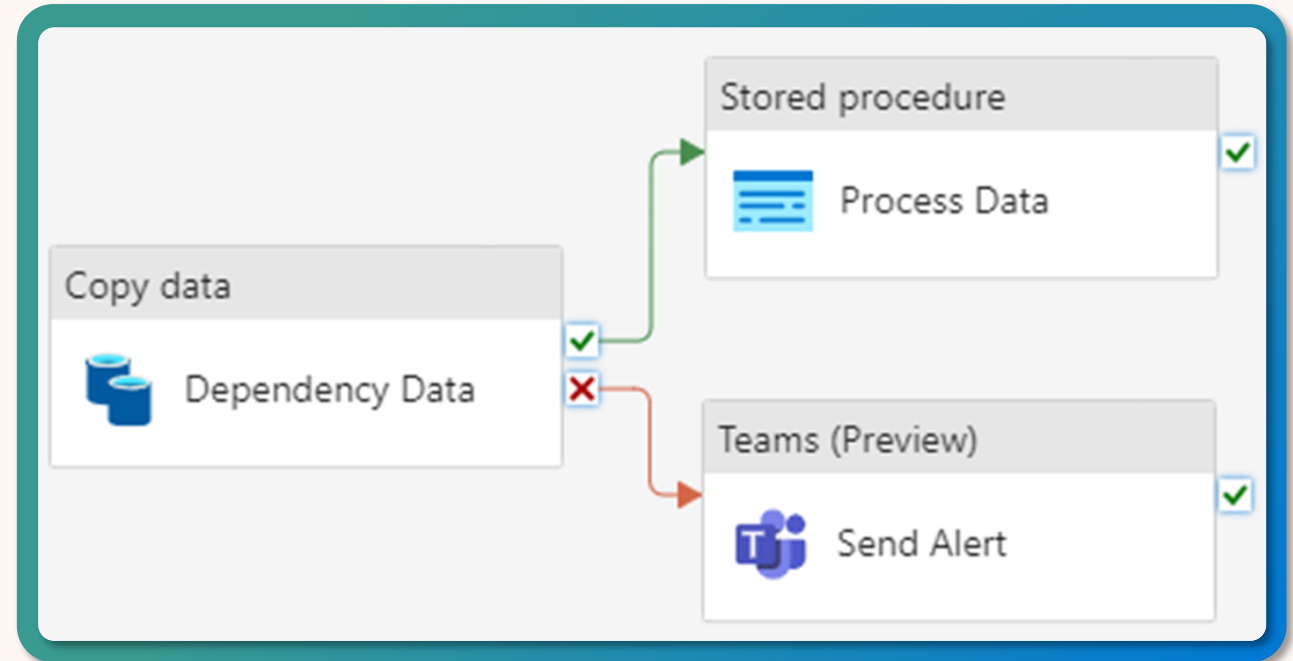
Regardless of the success or failure of an activity, the orchestrator can proceed to the next activity.



Conditional paths in Data pipelines

Define different execution paths based on the outcome of a previous activity

- Blocking dependencies
- Non-blocking dependencies
- **Error handling**



Creating a conditional path for both success and failure events.



Troubleshooting In Data pipelines

Troubleshooting tips

Leverage the **Output window** to find and review errors

Determine if it is an **activity error** or an **error from the source/destination**

Set a **retry attempt** for activities

The screenshot displays the Azure Data Factory console. At the top, there's a navigation bar with tabs: Home, Activities, Run, and View. Below this is a toolbar with icons for Validate, Cancel, Schedule, View run history, Copy data, Dataflow, Notebook, Lookup, and Invoke pipeline. The main workspace shows a 'Copy data' activity named 'orders_bronze' with a green checkmark icon. Below the activity, the 'Output' window is open, showing the pipeline run ID: b83f0502-a8a7-42f1-9923-9dfe548c247b. The output table lists the activity 'orders_bronze' with a status of 'In progress'.

Name	Type	Run start	Duration	Status	Run ID
orders_bronze	Copy data	5/21/2023, 9:48:31 PM	00:00:13	In progress	56829a27-d1ce-41e4-8e78-ba92f94f1



Data pipelines

Exploring data pipelines

Navigation

1. Command ribbon
2. Authoring canvas
3. Properties/output

The screenshot displays the Microsoft Fabric Data Factory interface. The top navigation bar includes a search bar and a 'Fabric Trial: 38 days left' notification. The left sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, SQL Saturday, and Data Factory. The main area is divided into two sections: the Authoring canvas (top) and the Properties/output pane (bottom).

Authoring canvas: The canvas shows a pipeline named 'activities_control_flow'. It contains a 'Set variable' activity (labeled 'Set v_Array') followed by a 'ForEach' loop. The 'ForEach' loop has a 'Filter' activity (labeled 'Filter1') and an 'Activities' sub-canvas. The 'Activities' sub-canvas contains a 'FE Index in v_Array' activity and a 'Wait for some even num of sec' activity. The 'Filter' activity is connected to the 'FE Index in v_Array' activity.

Properties/output pane: The 'Output' tab is selected, showing a table of pipeline run results. The table has columns for Activity name, Activity status, Run start, Duration, Input, and Output. The table shows 7 items, including the 'Set v_Array' activity, 'Filter1', and the 'FE Index in v_Array' loop.

Activity name	Activity status	Run start	Duration	Input	Output
(X) Set v_Array	Succeeded	2/7/2024, 10:49:37 AM	Less than 1s	->	->
Filter1	Succeeded	2/7/2024, 10:49:38 AM	Less than 1s	->	->
FE Index in v_Array	In progress	2/7/2024, 10:49:38 AM	2s	->	->
If index is Even	Succeeded	2/7/2024, 10:49:39 AM	Less than 1s	->	->
Append v_Array_Odds	Succeeded	2/7/2024, 10:49:39 AM	Less than 1s	->	->
If index is Even	In progress	2/7/2024, 10:49:40 AM	Less than 1s	->	->
Wait for some even num of sec	In progress	2/7/2024, 10:49:41 AM	Less than 1s	->	->



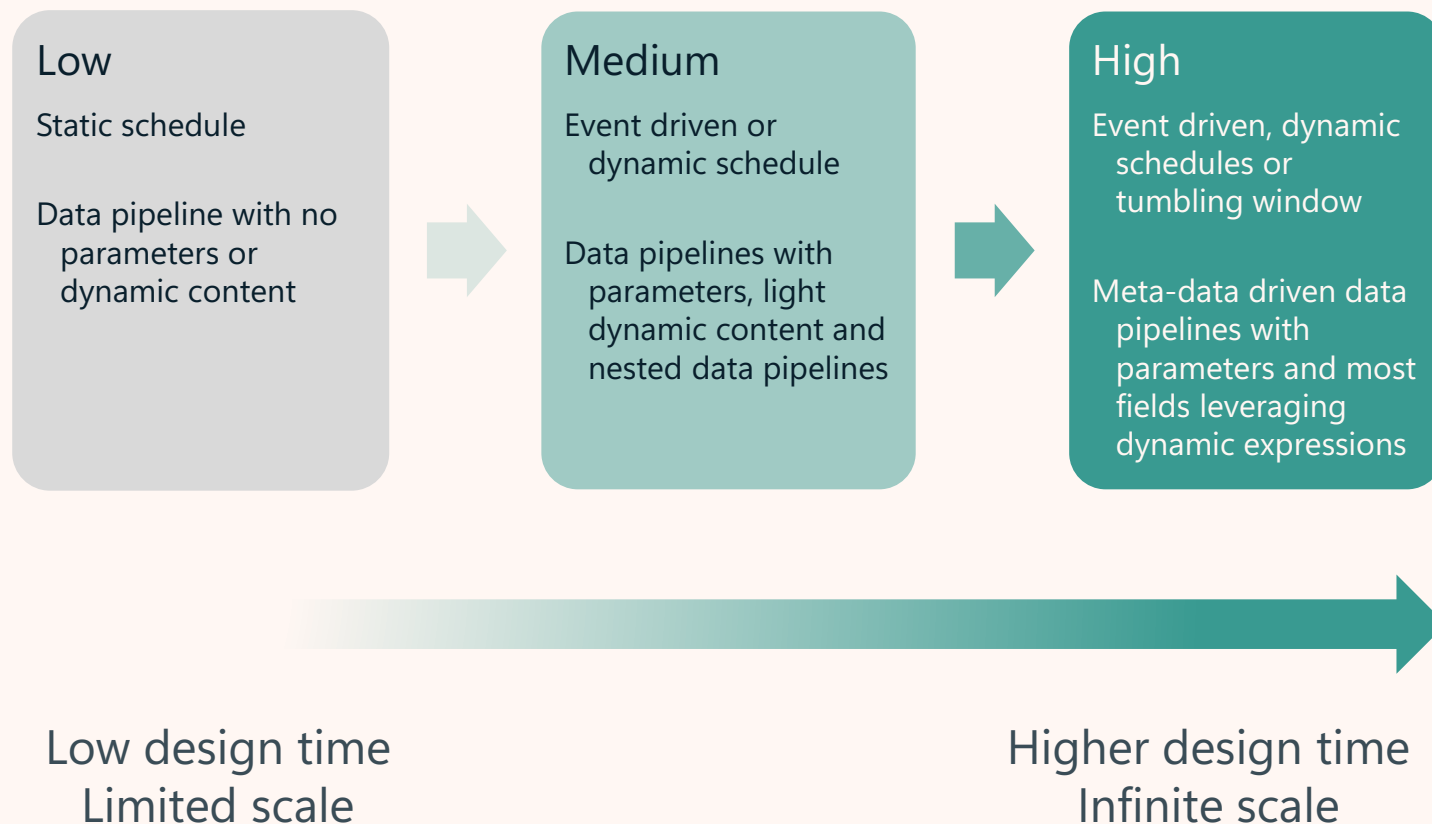
Dynamic expression driven orchestration

Build enterprise data products with a multi-layered approach

Varied Complexity: Architecture designs can range widely in complexity due to the vast array of design options.

Reusability: Design with reusability in mind, incorporating parameters and components that can be reused, such as invoking other pipelines, running dataflows, notebooks or data functions.

Dynamic Designs: Aims for designs that are fully dynamic and driven by expressions, utilizing metadata parameterization.



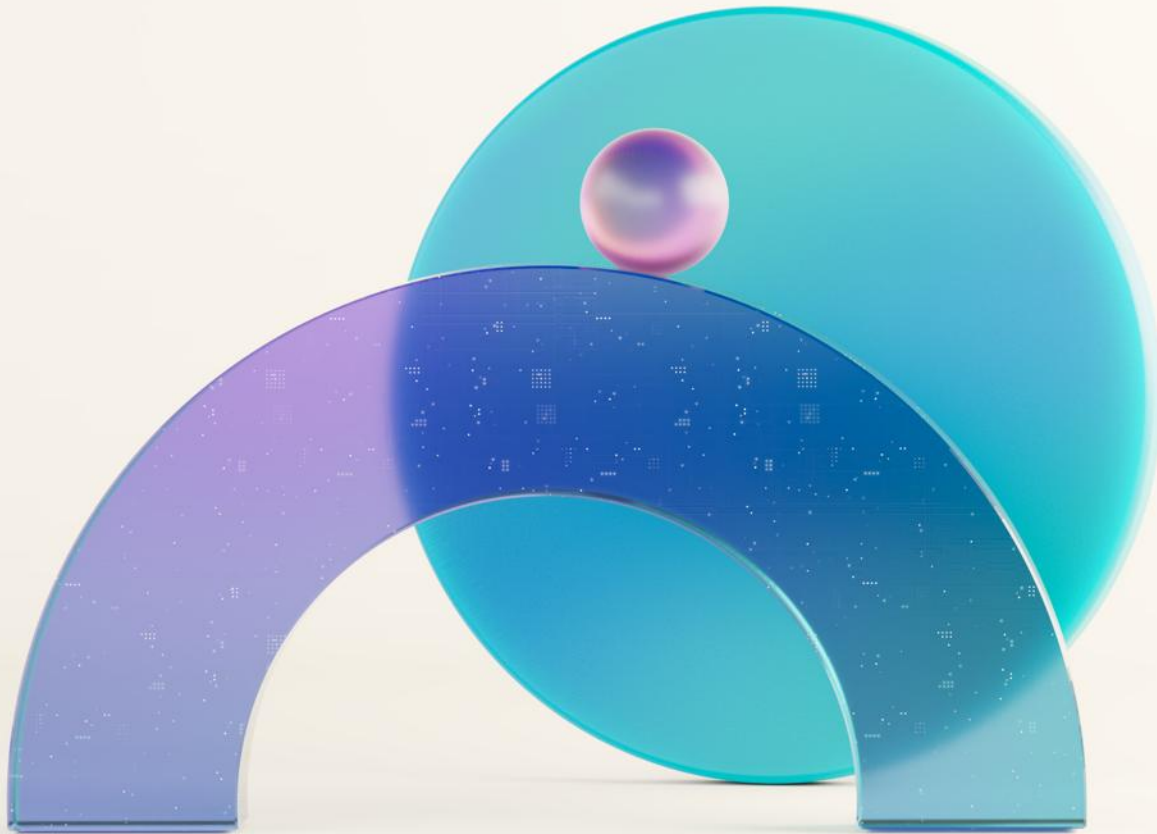
Migrate from ADF/Synapse Pipelines to Fabric Data pipelines

Feature	Fabric Data Factory	Azure Data Factory	Azure Synapse Pipelines
Office 365 Outlook Activity (Email activity)	Yes	*No	*No
MS Teams Activity	Yes	*No	*No
Refresh a Dataflow Gen2	Yes	*No	*No
Refresh a Power BI semantic model	Yes	*No	*No
Change Data Capture	Yes	Yes	No
Disable Activity	Yes	Yes	Yes
Managed Airflow	Yes	Yes (Preview)	No
Validation activity	No	Yes	Yes

* Supported by external services / REST APIs

Lab Demo: Orchestrating Data Movement

- Create a dynamic **Data pipeline** with **pipeline expression builder**
- Navigate **Copy activity** settings
- **Enrich raw data** with **variables**, **loops**, and **conditional paths**
- Utilize **AI-powered Copilot** to summarize Data pipelines
- Move data between **medallion layers** in the **task flow**



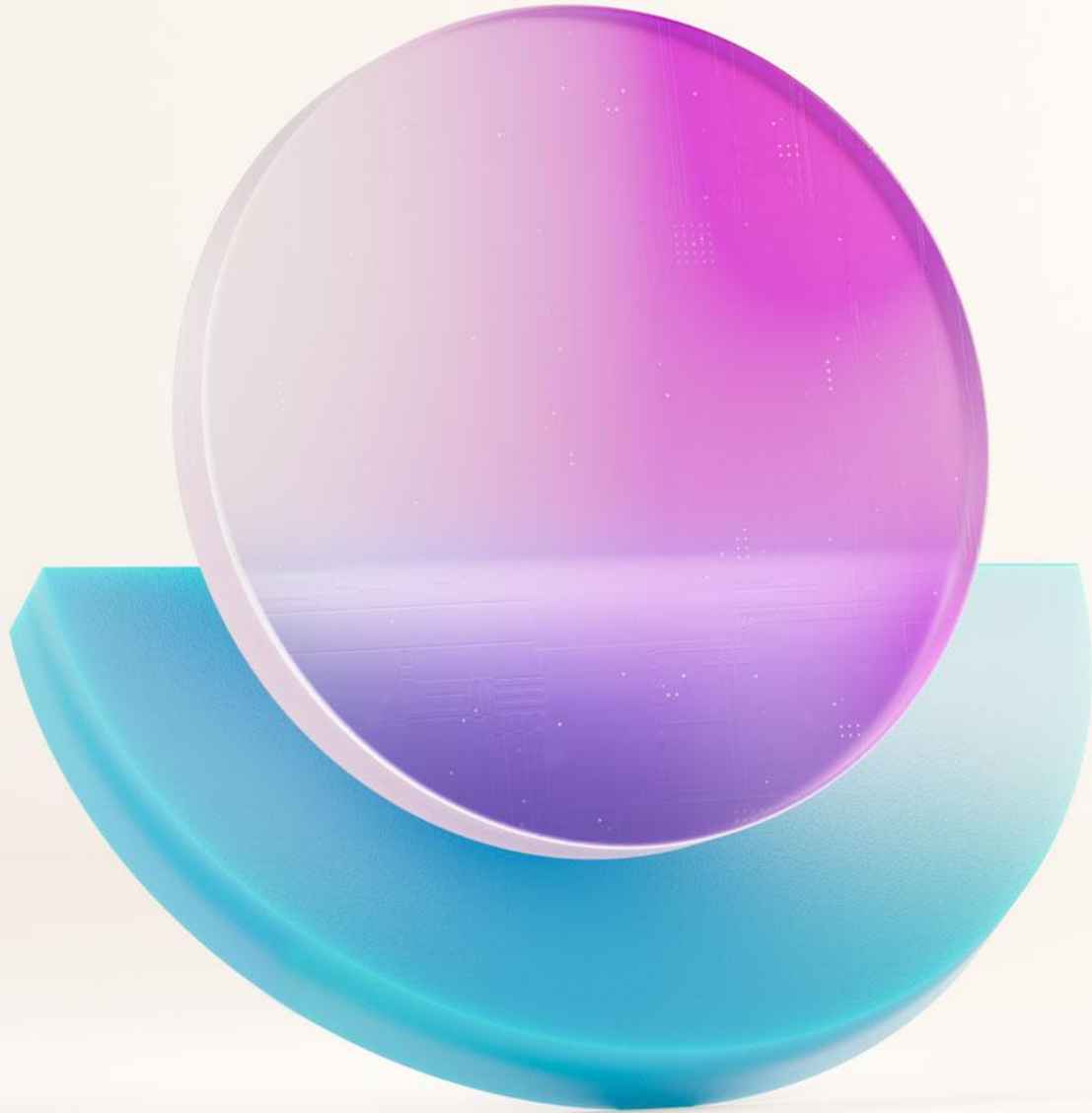
Reminder of how to access lab instructions

All materials are available at:

<https://aka.ms/dfiad>

Lab Instructions:

Orchestrating data movement



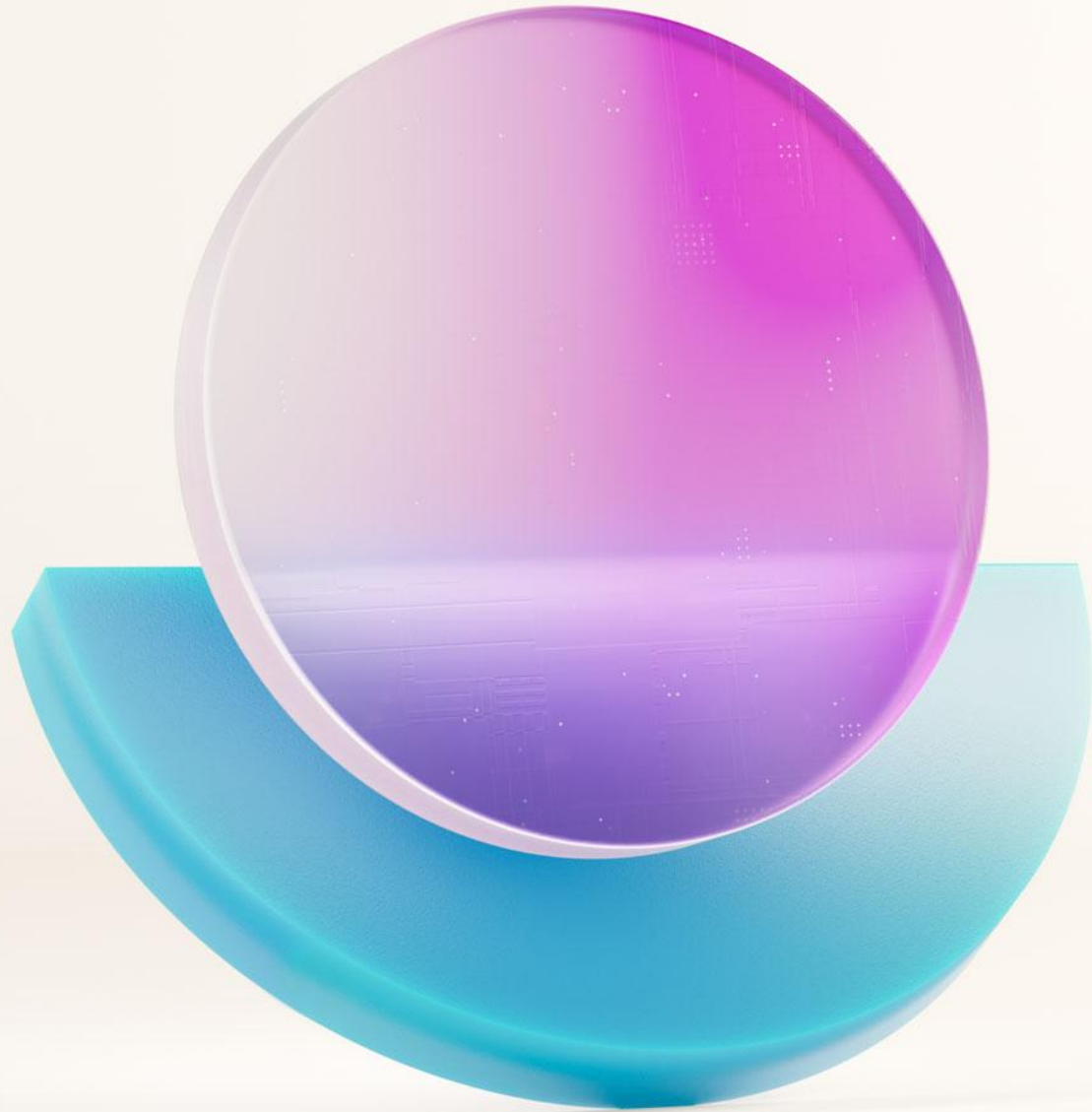
Questions?



Let's take a short break...

Be back @ 2:00 PM

Influence our product roadmap
and ensure Fabric meets your real-life needs
<https://aka.ms/fabricideas>



Ingesting and transforming data with Dataflow Gen2



Next generation of data preparation with Dataflows Gen2 in Fabric Data Factory

Easy to use, no-code ETL & ELT

Includes **smart AI**-based data prep

More than **300+** transformations

Output data destinations

Write output of dataflows to Azure SQL database, Data warehouse, Lakehouse and more

The screenshot displays the Microsoft Fabric Dataflows Gen2 interface. The main workspace shows a query named "Table.ExpandTableColumn('Merged queries', 'DimGeography_raw', {'GeographyType', 'ContinentName', 'CityName', 'StateProvinceName', 'RegionCountryName', 'Geometry'}, {'GeographyType'})". The query is applied to a data source named "DimGeography_raw". The interface includes a sidebar with navigation options like Home, Create, Browse, and a list of data sources and transformations. The main area shows a data table with columns like CustomerKey, GeographyKey, FirstName, MiddleName, LastName, BirthDate, MaritalStatus, Suffix, Title, EmailAddress, YearlyIncome, and Education. The table is populated with data from the "DimGeography_raw" source. The bottom right corner shows the "Query settings" panel with properties like Name, Entity type, and Applied steps.

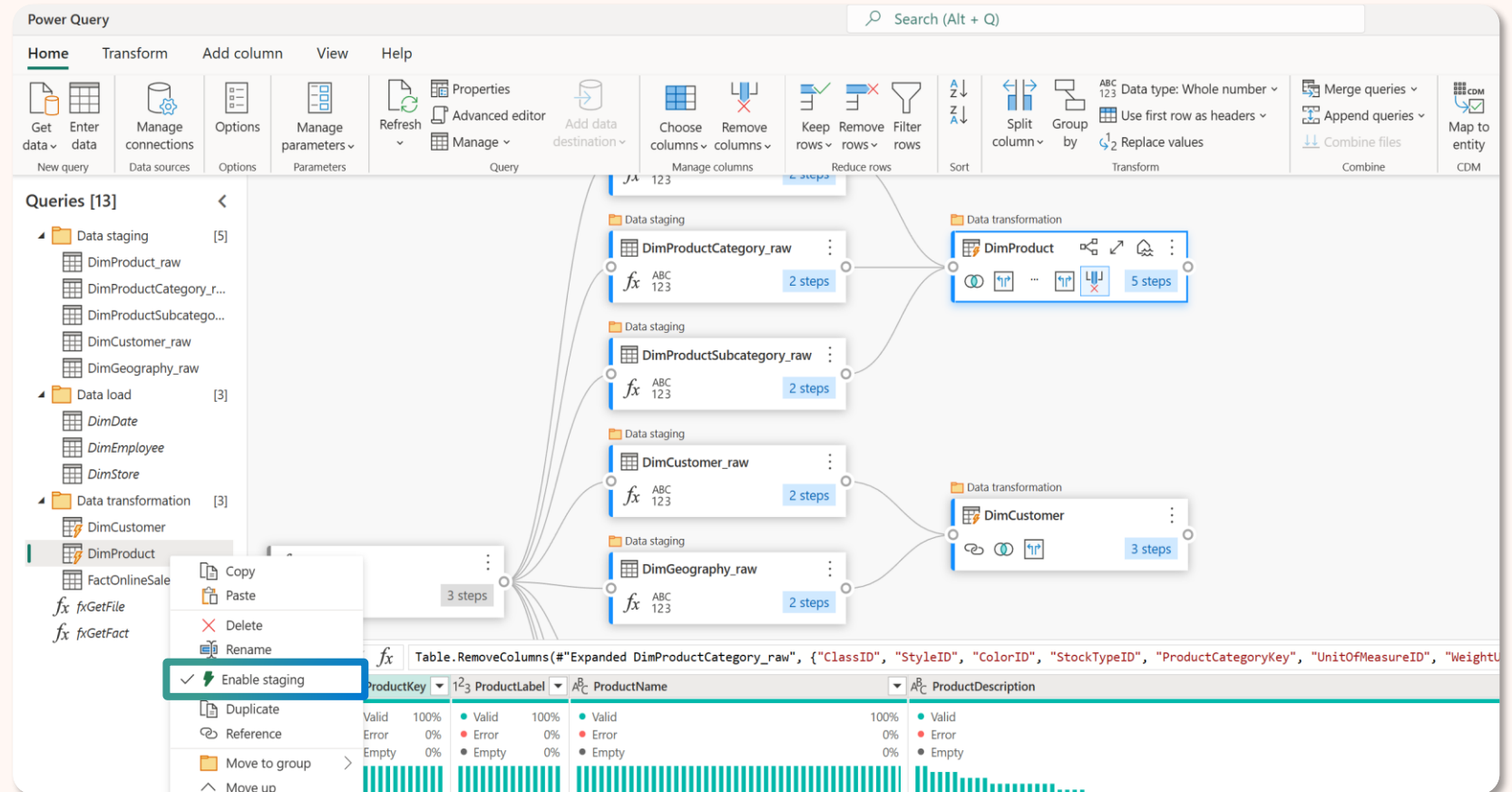


Optimized performance using Fabric compute and staging in Dataflows Gen2

Highly scalable using
Fabric compute

A **seamless experience** -
yielding fast, easy and powerful
results

Abstracts away the
complexities of traditional ETL
and ELT



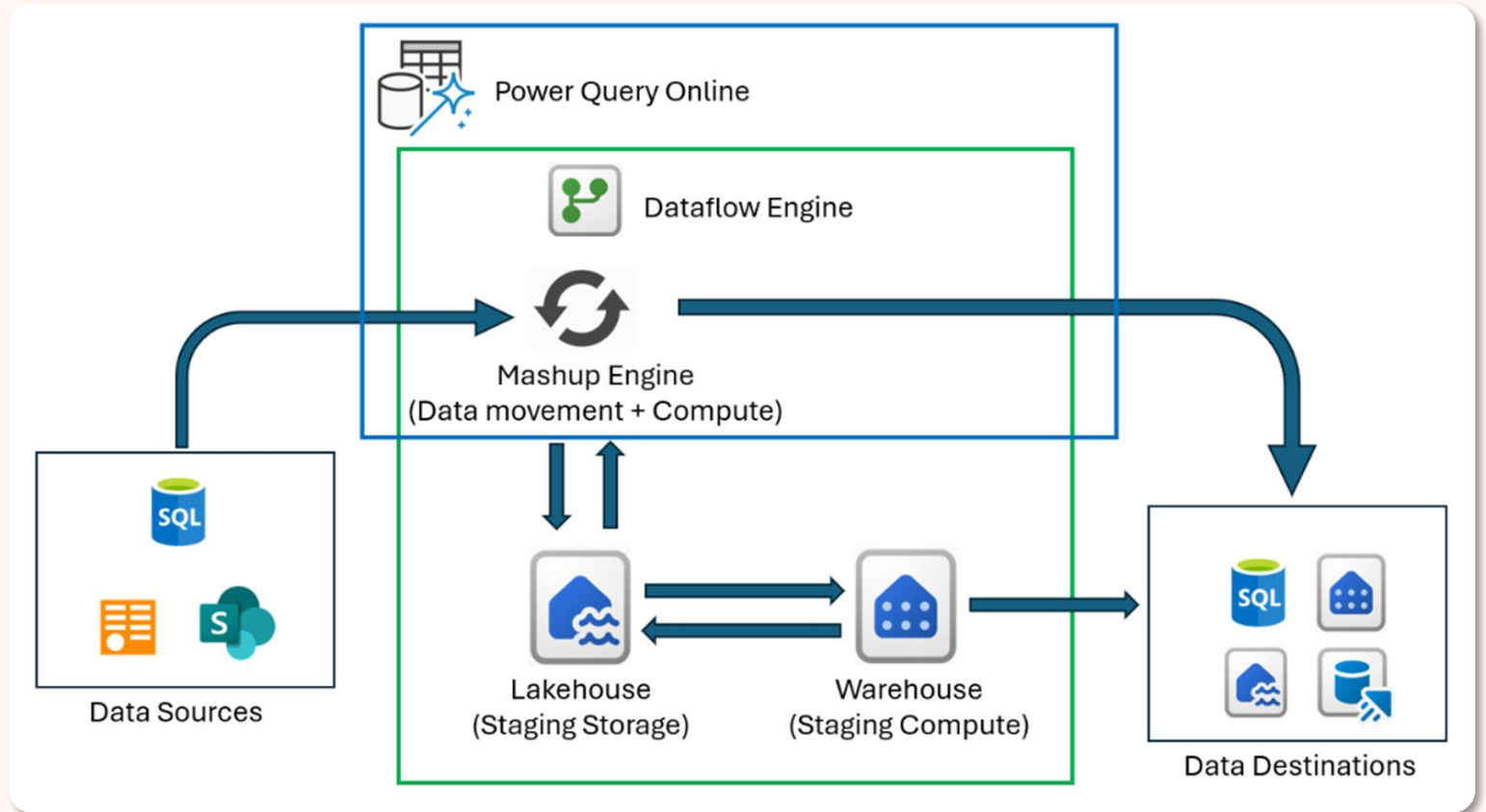


Optimized performance using Fabric compute and staging in Dataflows Gen2

Connect to your data and **copy** it into Fabric using the ***Enable staging** option
(*On by default)

Create a **reference** query in a new query

Apply **transformation steps** to the **computed** table for complex ETL operations such as join, distinct, filter and group by – leveraging Fabric compute





High scale data movement with Fast Copy in Dataflows Gen2

General Availability

Ingest terabytes of data **effortlessly** with dataflows, powered by the scalable backend of the Copy activity

A limited set of transformations are supported:

- Combine files
- Select or remove columns
- Change data types
- Rename a column

The screenshot displays the 'Source' step configuration in the Dataflow Designer. The step is named 'Source' and is configured to use 'Azure Data Lake Storage'. A tooltip indicates that this step will be evaluated by the data source, with a 'Learn more' link. Below the tooltip, a message states: 'This step is going to be evaluated with fast copy.' The 'Applied steps' list on the right includes: 'Source', 'Filtered hid...', 'Invoke cust...', 'Renamed c...', 'Removed o...', 'Expanded t...', and 'Changed c...'. The 'Changed c...' step is currently selected.

1 ² 3 r_1	1 ² 3 r_2	1 ² 3 r_3	1 ² 3 r_4	1 ² 3 r_5
27450000	3240000	26640000	22770000	2016
1080000	28800000	29070000	15210000	2547
16020000	24210000	26460000	15750000	846
2070000	24840000	12150000	3150000	1926
30870000	9720000	18270000	8010000	594
4320000	17640000	21420000	18990000	1458



Optimized performance using incremental refresh In Dataflows Gen2

General Availability

Process only changed data since the last refresh to **save time and resources**

- Evaluate changes by comparing the maximum **DateTime** value with the previous refresh.
- Retrieve and load data for changed buckets in parallel, loaded to staging
- Replaces the destination data with the new data, affecting only updated buckets.

The screenshot shows the 'Incremental refresh' dialog box in the Power Query interface. The dialog has a title bar with a question mark icon. It contains the following options and settings:

- ☒ Enable incremental refresh
- Choose a DateTime column to filter by *: ReportDate (dropdown menu)
- Extract data from the past *: 6 (input field) Years (dropdown menu)
- Bucket size *: Quarter (dropdown menu)
- Only extract new data when the maximum value in this column changes *: ChangeDate (dropdown menu)
- ☐ Only extract data for concluded periods
- > Advanced options

At the bottom right of the dialog are 'OK' and 'Cancel' buttons.

The screenshot shows a context menu for the 'Orders' table. The menu includes the following options:

- Copy
- Paste
- Delete
- Rename
- ✓ Enable staging
- Incremental refresh** (highlighted with a blue border)
- Duplicate
- Reference
- Move to group >
- Move up
- Move down
- fx Create function...
- Convert to parameter
- Advanced editor
- Properties...



Data destinations in Dataflows Gen2

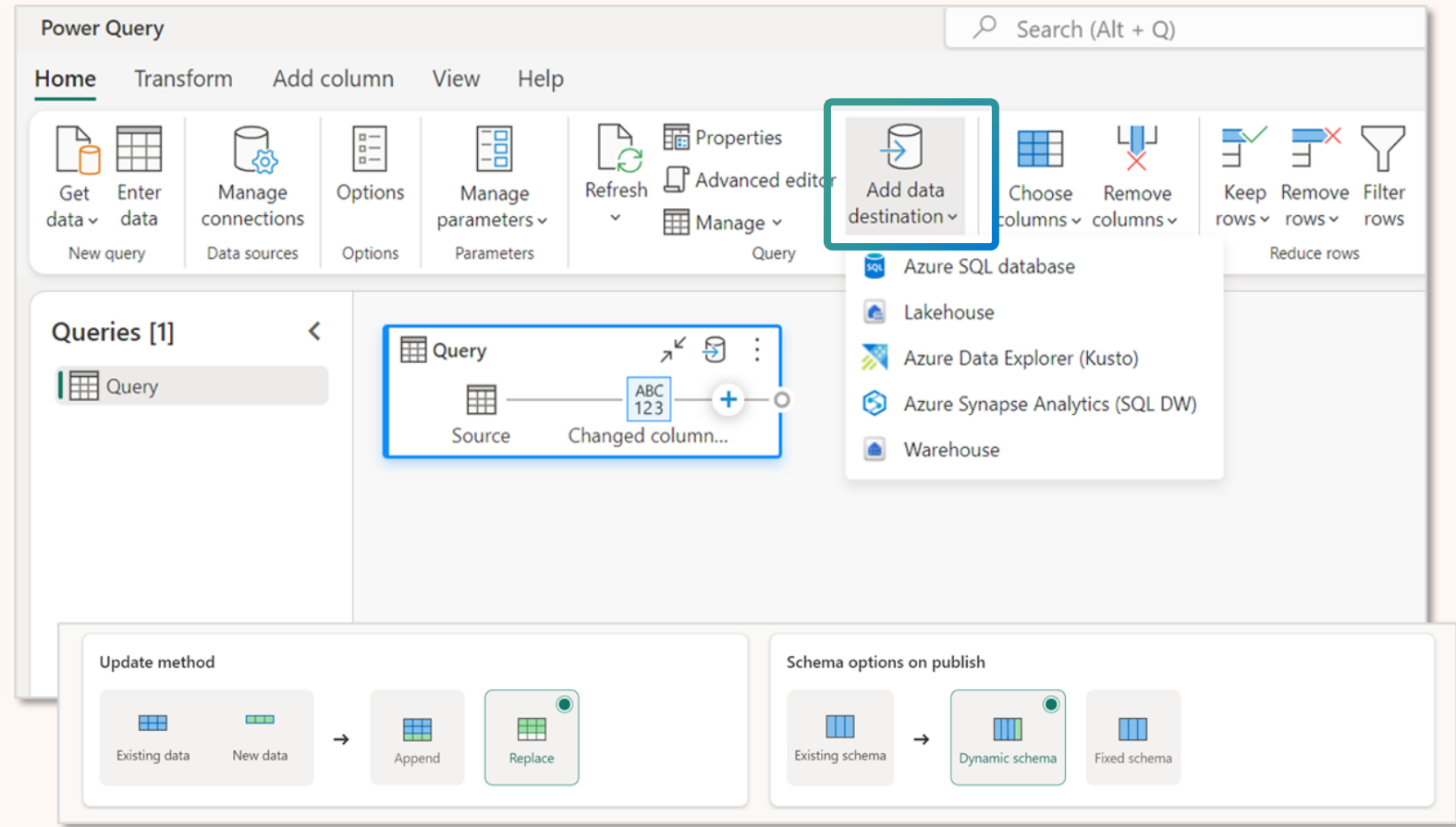
Land your data into a destination for further analysis and reporting

Supported destinations:

- Lakehouse
- Warehouse
- Eventhouse
- SQL database in Fabric
- Azure SQL database
- Azure Synapse Analytics
- Azure Data Explorer

Update methods:

- Replace
- Append





Dataflows Gen2 in Fabric Data Factory

AI-powered Dataflows Gen2 with Copilot

Utilize the **get data** experience to ingest data into your dataflows

Transform data by using natural language to add a step to your query

Receive a **summary** of your queries in your Dataflows Gen2

The screenshot displays the Dataflows Gen2 interface. At the top, a toolbar includes options like 'Group by', 'Merge queries', 'Append queries', 'Map to entity', and 'Export template'. The 'Copilot' icon is highlighted. Below the toolbar, the 'Query settings' panel is visible, showing properties for a query named 'DimProducts'. The 'Applied steps' section lists 'Source', 'Expanded DimProductSubcategory', 'Merged queries', and 'Expanded DimProductCategory'. The 'Data destination' section shows 'No data destination'. On the right, the 'Copilot' sidebar is open, displaying a chat interface with the prompt 'What would you like to do?' and suggestions like 'Get data from...', 'Add a step that...', and 'Describe this query'. A blue box highlights a specific instruction: 'Keep the ProductKey column and remove all other columns containing ID, Key or Label in the name.' At the bottom, there is a disclaimer: 'AI-generated content can have mistakes. Make sure it's accurate and appropriate before using it. [Review terms](#)'.

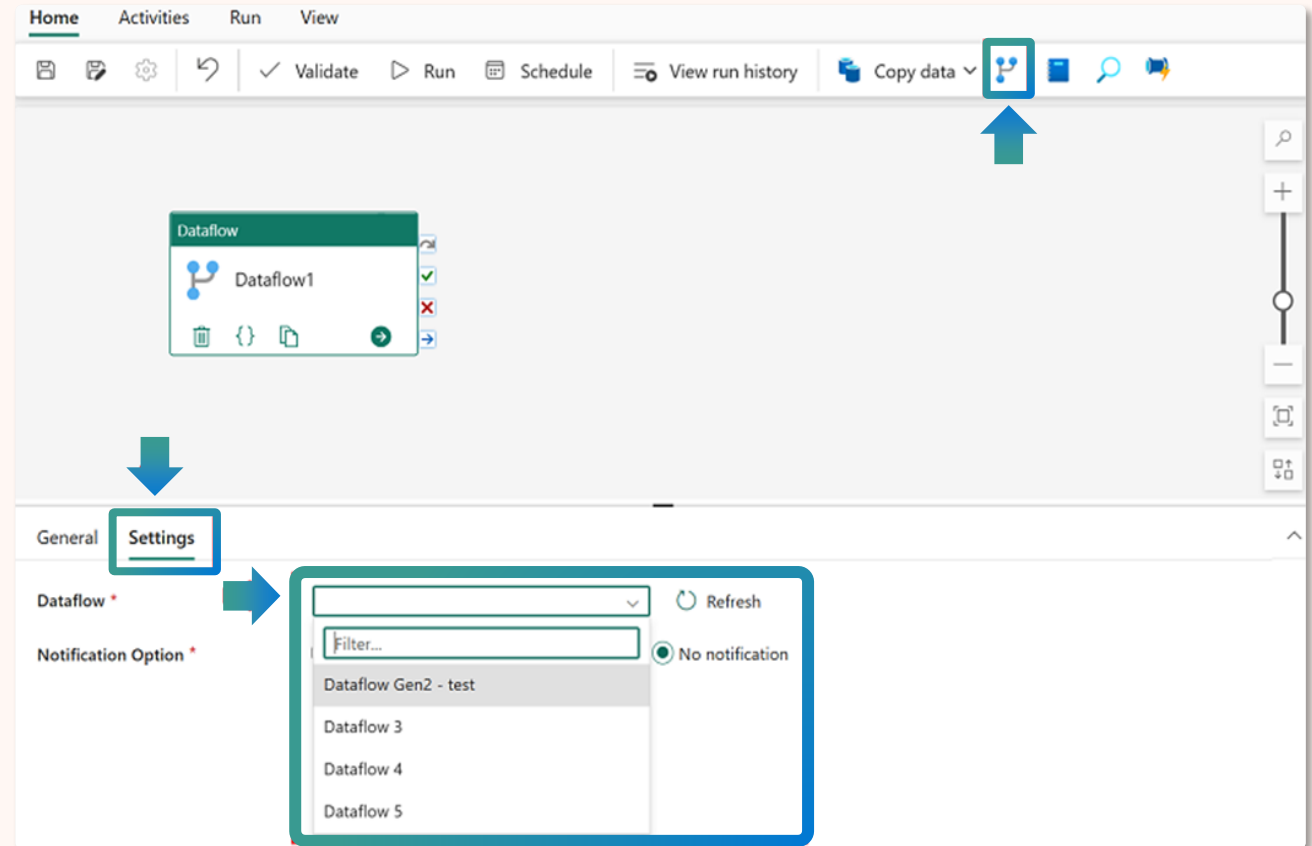


Dataflows Gen2 in Fabric Data Factory

Orchestrating Dataflows Gen2 with Data pipelines

Create a dataflow for **data ingestion and transformation**, and landing into a Lakehouse using dataflows

Then **incorporate the dataflow into a pipeline** to orchestrate additional activities



Check your knowledge | Dataflows Gen2

A single dataflow has a limit of how many queries/tables?

25

50

100



Dataflows Gen2 in Fabric Data Factory

Dataflows Gen2 scale recommendations

Update method separation

Separate **dimension** tables and **fact** tables into separate dataflows based on update method

Staging separation

Separate tables with **staging** and **without staging** into separate dataflows

Fast copy separation

Separate tables with **fast copy** support and **without fast copy** support into separate dataflows

Query folding separation

Separate tables based on the data source and if they **support query folding** or **do not support query folding**

Item distinction

Use copy job, mirroring or data pipelines to **ingest data** and dataflows to **transform**

Migrate from Power BI dataflow to Fabric dataflow gen2

Feature	Power BI Dataflow	Dataflow Gen2
Incremental refresh	Yes	Yes
Accessible file outputs	*No	Yes
Fast copy	No	Yes
Data destination output	No	Yes
Premium capacity required	No	Yes
AI Insights	Yes	*No
AutoML	Deprecated	Deprecated
Attach Common Data Model (CDM) folder	Yes	No
Linked Tables	Yes	No (use Shortcuts)
On-premises data gateway	Yes	Yes
Vnet data gateway	No	Yes

* Supported by external services / REST APIs

Upgrade Pathways for Power BI dataflows customers to Fabric



New Dataflow

Copy-Paste queries from Dataflow Gen1 to Dataflow Gen2

1

General Availability



Power Query Templates

Export template from an existing dataflow, create new dataflow from template

2

General Availability



Save as Dataflow Gen2

Save a copy of an existing dataflow as a Dataflow Gen2

3

General Availability



Dataflows Gen2

Exploring Dataflows Gen2

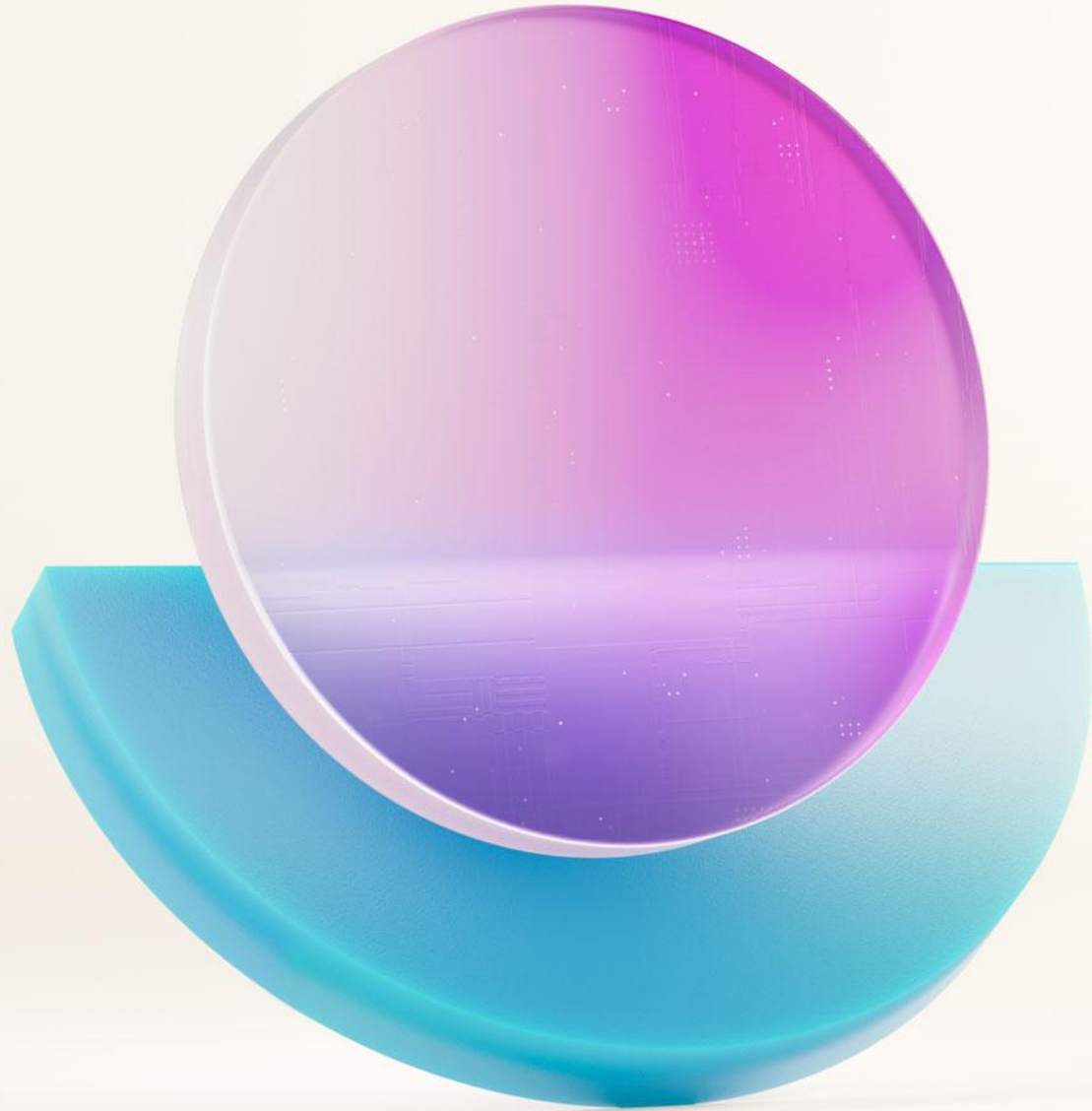
Navigation

1. Command ribbon
2. Queries pane
3. Visual diagram
4. Data preview grid
5. Query settings

The screenshot displays the Power Query interface with the following components:

- Command ribbon:** Located at the top, featuring tabs for Home, Transform, Add column, View, and Help. The Home tab is active, showing various data manipulation options like Get data, Enter data, Options, Manage parameters, Refresh, Advanced editor, Add data destination, Choose columns, Remove columns, Keep rows, Remove rows, Filter rows, Sort, Transform, Combine, Map to entity, and Export temp.
- Queries pane:** Located on the left, showing a list of queries: orders and orders-2. The orders query is selected.
- Visual diagram:** Located in the center, showing a flow diagram with two queries: orders-2 (3 steps) and orders (9 steps). The orders query is highlighted.
- Data preview grid:** Located in the center, showing a table with 9 columns and 15 rows. The columns are SalesOrderID, OrderDate, MonthNo, CustomerID, Linetitem, ProductID, and OrderID. The data is filtered to show rows 1 through 15.
- Query settings pane:** Located on the right, showing the Name (orders), Entity type (Custom), and Applied steps (Source, Promoted h..., Changed co..., Appended ..., Filtered rows, Added cust..., Reordered c..., Changed co..., Changed co...). The Data destination is set to Lakehouse.

The bottom status bar indicates Columns: 9 Rows: 99+.



Copying data with a Copy job



Copy job in Fabric Data Factory

Generally Available

Ingest data from any source to any destination with Copy job

Choose your data delivery style

Full copy mode copies all data from the source to the destination **at once**

Incremental copy mode copies all data in the initial job run, and subsequent job runs only copies **changes since the last run**

Settings



Copy job mode

Choose how you want the data to be copied. This copy mode will be applied every time you run the job, which can be run once or multiple times. After this new copy job item is created, you can then schedule how often the job is run and monitor statuses.



Full copy

Copy all data at once



Incremental copy

Initially copies all data, subsequent runs copy only changes





Copy job in Fabric Data Factory

Generally Available

Personalize your data update method

By default, Copy job **appends** data to your destination

But you can also write behavior to **overwrite**

Copy job

- ✓ Choose data source
- ✓ Choose data
- ✓ Choose data destination
- Map to destination**
Select and map to folder path or table.
- Settings
- Review + save

Map to destination

Files

Tables

Settings > Update method

Update method

Existing data

New data

→

Append

Overwrite

Back

Next



Copy job

Exploring Copy job

Navigation

1. Command ribbon
2. Visual diagram
3. Copy job monitoring

HomeView

1

Run

Choose data source

Edit mapping

Advanced settings

View run history

2

Source

Lakehouse1
Lakehouse

Full copy

Destination

copyJobTestLake
Lakehouse

3

Results

Refresh

Status: ✓ Succeeded

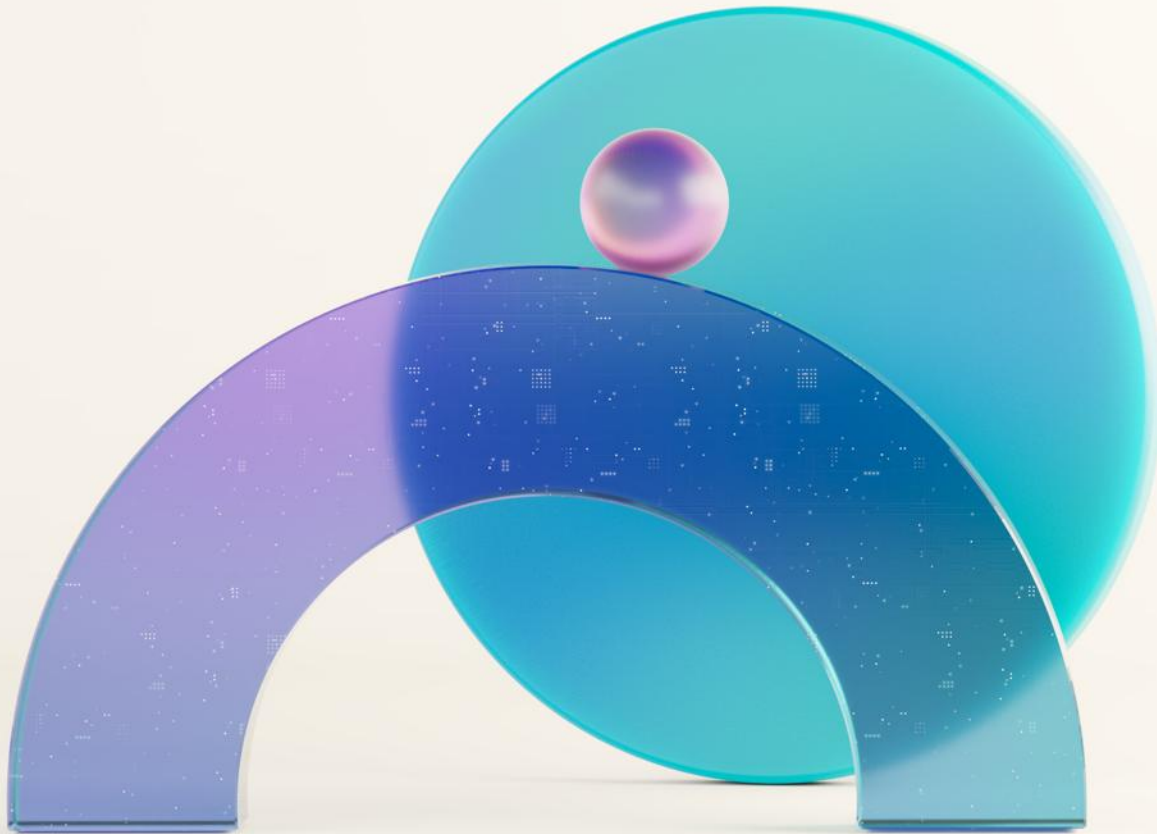
Rows read: 65436

Rows written: 65436

Throughput: 87.39 KB/s

More

Source → Destination	Status	Rows read	Rows written	Duration	Run start	Run end
sales_table → sales_table	✓ Succeeded	32718	32718	19 sec	3/6/2025, 6:49:51 PM	3/6/2025, 6:50:10 PM
sales → sales	✓ Succeeded	32718	32718	19 sec	3/6/2025, 6:49:51 PM	3/6/2025, 6:50:10 PM



Lab Demo: Transforming and copying data

- Create a **Dataflow Gen2** and prepare data using **Power Query Online**
- Utilize **AI-powered Copilot** to transform data
- **Output data** to a **Warehouse** destination
- Copy data into a Warehouse with a **Copy Job** item

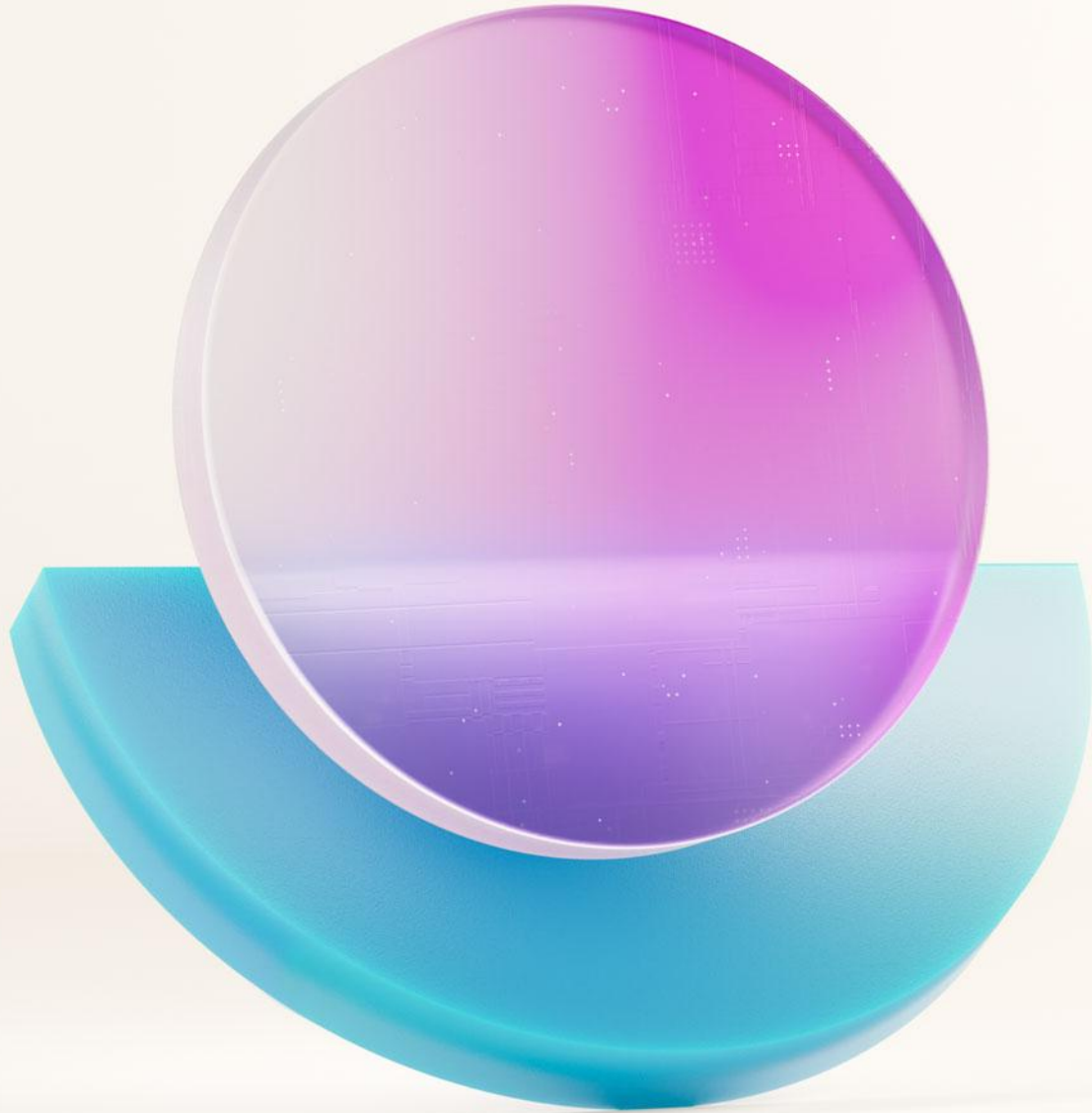
Reminder of how to access lab instructions

All materials are available at:

<https://aka.ms/dfiad>

Lab Instructions:

Transforming and copying data



Questions?

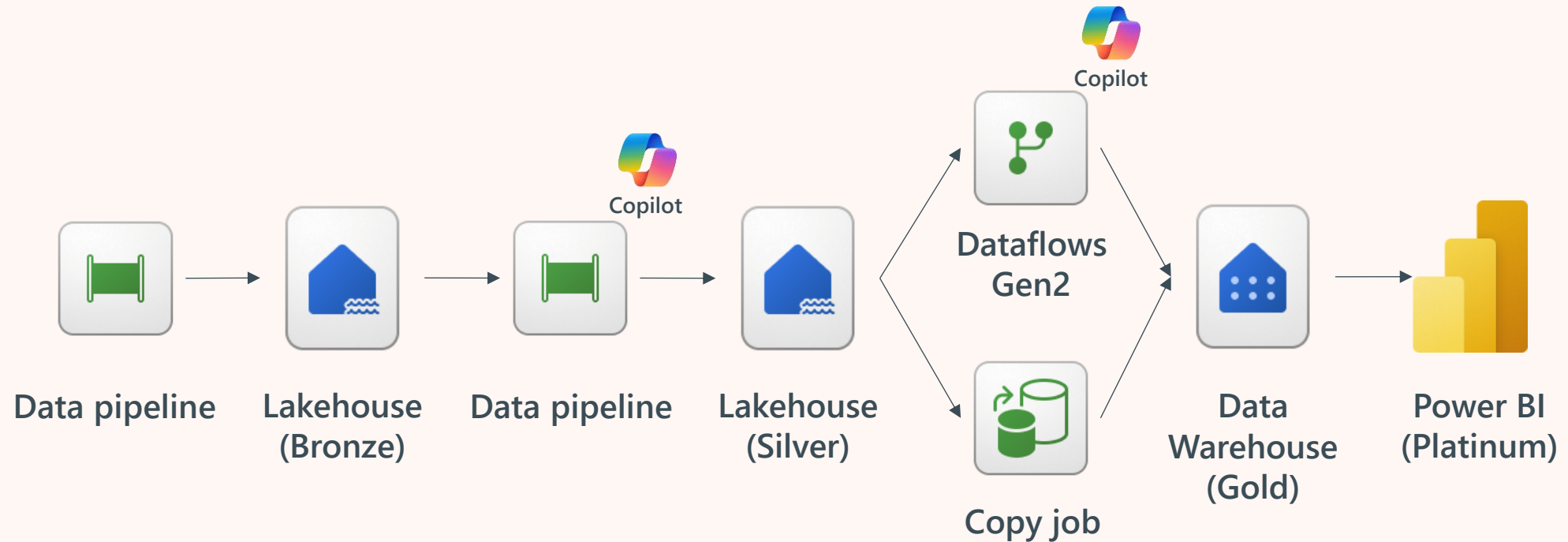


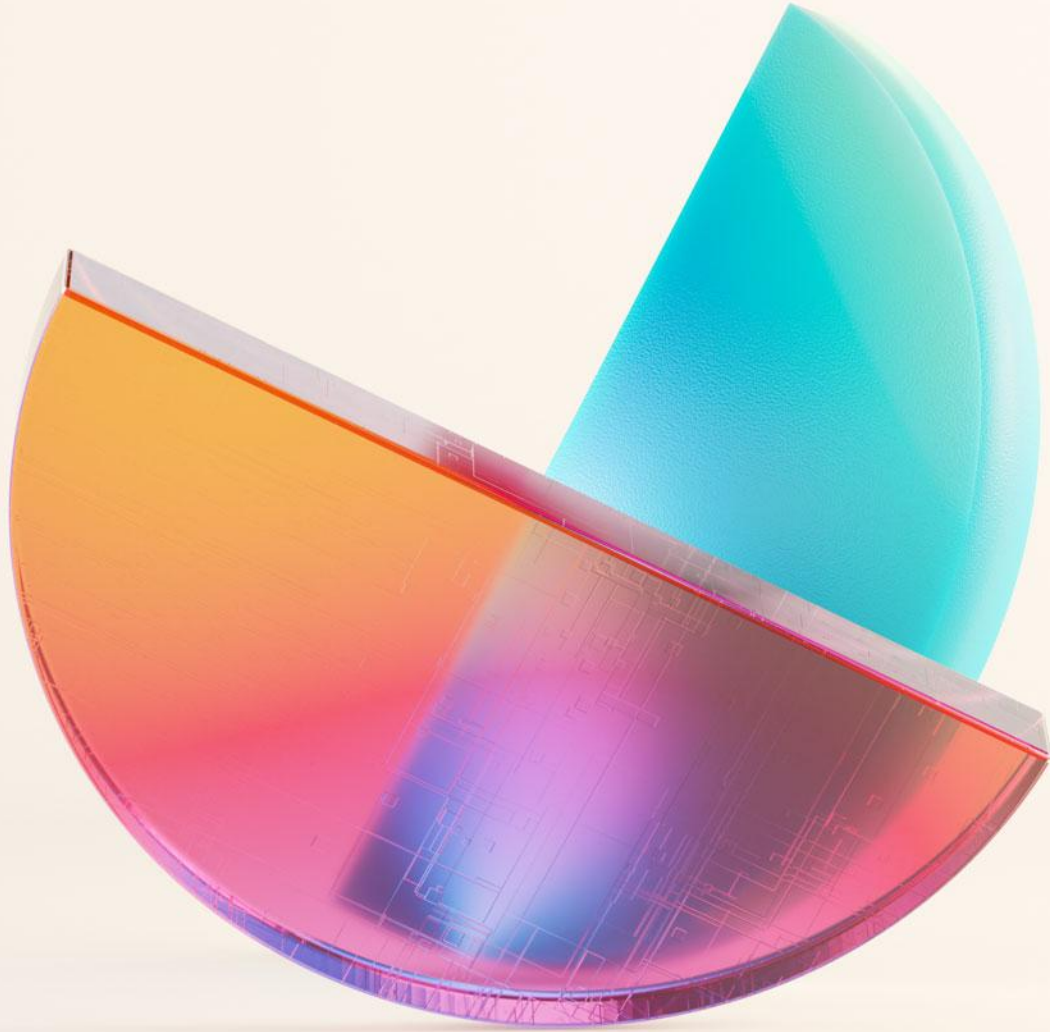
Let's take a short break...

Be back @ 3:45 PM

Influence our product roadmap
and ensure Fabric meets your real-life needs
<https://aka.ms/fabricideas>

A quick recap of what you did today!





One last thing...

Thank you!

Generally Available



Mirroring support for on-prem sources & sources behind firewall (Azure SQL MI, Snowflake, Azure SQL DB)



Mirroring for Azure SQL MI



Pipeline variable libraries



Save Dataflow Gen1 as Dataflow Gen2 (CI/CD)



Dataflow Gen2 Public APIs w/ SPN support



Copy job & Email & Teams pipeline activities



Dataflow Gen2 – Performance enhancements across Run time and Design Time



Azure Key Vault integration in Connections



Easily add SPN connections, Notebooks, and Pipelines to Airflow DAGs



Open Mirroring Support for Delimited Text



Copy job – more copy patterns including Merge, Reset, Truncate



Datra Pipelines are now "Pipelines"



Fast Copy supports Share point files & SQL in Fabric



Invoke Pipeline & Dataflow Activity w/param support



Multiple schedules per pipeline



Pipeline and Copy Job integration with VNet Data Gateway



Snowflake Key-Pair authentication support in Connections



Workspace Identity authentication support in Connections

Public Preview

Mirroring for Oracle/Google Big Query

Mirroring Support for Data Agent / Workspace PL

Dataflow Gen2 ALM: Fabric Variables

Copy job – Native CDC with SQL Server, SQL MI, SQL DB, Fabric Lakehouse

Pipeline expression evaluator

Copy job – Variable library for CICD

Dataflow Gen2 – SharePoint (XLSX), Snowflake & Lakehouse Files output destinations

Pipeline Copy Job Activity

Copilot in Modern Get Data

Workspace Private Link support for Data Factory

Microsoft Fabric Conference Vienna 2025



Fabric Data Factory:
What's New and Roadmap

Tues, 10:30 – 11:30



Performance Tuning Tips
for Fabric Data Factory

Tues, 14:15 – 15:15



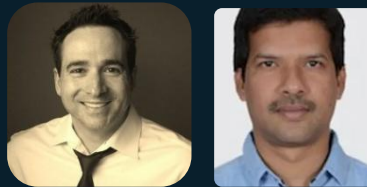
Bringing data to OneLake: a
deep dive into shortcuts and
mirroring

Wed, 10:30 – 11:30



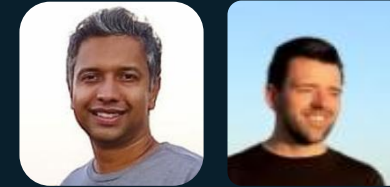
Fabric Data Factory–Maximize
Business Impact with Best-In-
Class Connectivity to SAP,
Oracle and more

Thurs, 12:00 – 13:00



Unlock the power of
Pipelines and Data
Movement with Fabric Data
Factory

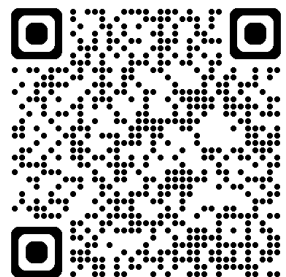
Thurs, 14:15 – 15:15



Unlock the power of
Dataflow Gen2 with
Fabric Data Factory

Thurs, 15:30 – 16:30

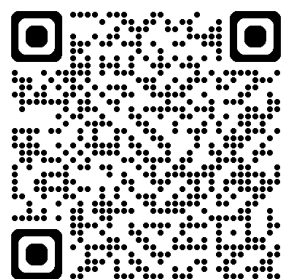
Get Fabric Certified for FREE



100% Discount For You

DP-700 or DP-600

<https://aka.ms/fabcon/cert100>



50% Discount To Share

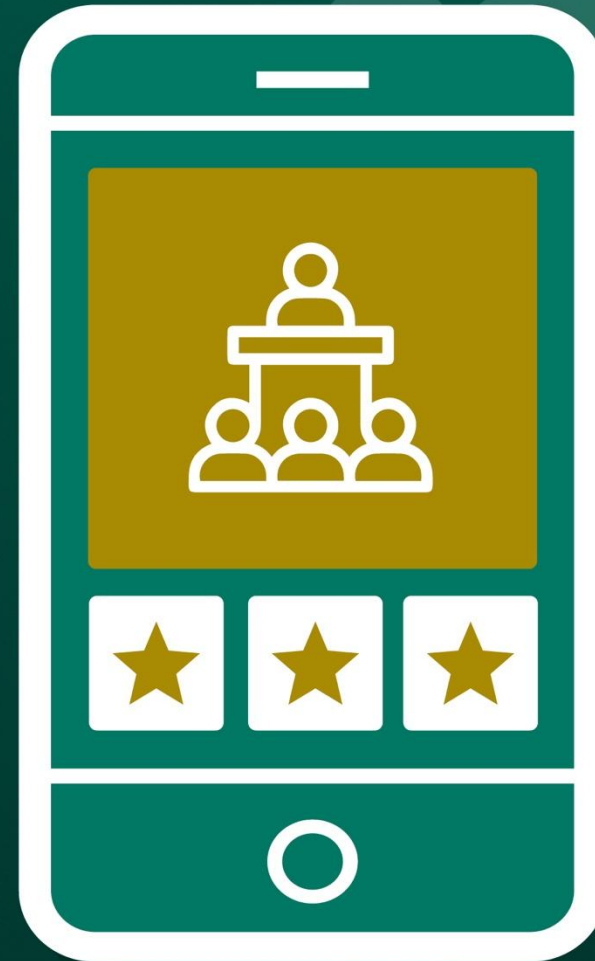
PL-300, DP-600, DP-700 or DP-900

<https://aka.ms/fabcon/cert50>





Please rate
this session
on the app



cvent

